

Linear Algebra

Lecture Notes before the Midterm

Chapter 1. Matrices

Matrices and products

Definition 1.1. Let m, n be positive integers. An $m \times n$ **matrix** $A = (a_{ij})$ is a rectangular array of real numbers, its **entries**; a_{ij} stands in row i and column j ($1 \leq i \leq m, 1 \leq j \leq n$). The set of all $m \times n$ matrices is $\mathbb{R}^{m \times n}$.

- $A = B$ means that A and B have the same size and $a_{ij} = b_{ij}$ for all i, j .
- $\mathbb{R}^n = \mathbb{R}^{n \times 1}$ is the set of **column vectors** $x = (x_i)$, and $\mathbb{R}^{1 \times n}$ the set of **row vectors**. A 1×1 matrix is identified with its entry.
- $\text{col}_j(A) \in \mathbb{R}^m$ is the j th column and $\text{row}_i(A) = (a_{i1}, \dots, a_{in}) \in \mathbb{R}^{1 \times n}$ the i th row of A . We write $a_j = \text{col}_j(A)$, so $A = [a_1 \cdots a_n]$.
- Sums and scalar multiples are taken entrywise: $(A + B)_{ij} = a_{ij} + b_{ij}$, $(cA)_{ij} = ca_{ij}$ ($c \in \mathbb{R}$); $-A = (-1)A$ and $A - B = A + (-B)$. They obey the laws of numbers, such as $A + B = B + A$ and $c(A + B) = cA + cB$, because these hold entry by entry.
- The **zero matrix** 0 has all entries 0 . The **identity matrix** is $I_n = (\delta_{ij}) \in \mathbb{R}^{n \times n}$, where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$; its columns $e_1, \dots, e_n$ are the **unit vectors** of $\mathbb{R}^n$.

In statistics, measurements of k variables on n units are stored in an $n \times k$ **data matrix** (x_{ij}) : row i holds the measurements on unit i , column j holds the n values of variable j , and x_{ij} is the value of variable j on unit i .

Definition 1.2. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$. The **product** $AB \in \mathbb{R}^{m \times p}$ is defined by

$$(AB)_{ij} = \sum_{r=1}^n a_{ir}b_{rj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj} \quad (1 \leq i \leq m, 1 \leq j \leq p)$$

It is defined only when A has as many columns as B has rows.

Theorem 1.3. Whenever the sizes allow the operations,

$$A(BC) = (AB)C, \quad A(B + C) = AB + AC, \quad (A + B)C = AC + BC, \quad A(cB) = c(AB) = (cA)B, \\ I_m A = A = A I_n, \quad 0A = 0, \quad A0 = 0 \quad (A \in \mathbb{R}^{m \times n})$$

Each rule is checked entry by entry; for instance $[A(BC)]_{ij} = \sum_r a_{ir} \sum_s b_{rs} c_{sj} = \sum_s \left(\sum_r a_{ir} b_{rs} \right) c_{sj} = [(AB)C]_{ij}$. So a product of several matrices needs no parentheses. For $A \in \mathbb{R}^{n \times n}$ we put $A^0 = I_n$ and $A^{k+1} = A^k A$ ($k \geq 0$).

Theorem 1.4. Let $A \in \mathbb{R}^{m \times n}$ have columns $a_1, \dots, a_n$ and $B \in \mathbb{R}^{n \times p}$ columns $b_1, \dots, b_p$.

- $AB = [Ab_1 \cdots Ab_p]$, that is, $\text{col}_j(AB) = Ab_j$.
- For $x \in \mathbb{R}^n$, $Ax = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n$. In particular $Ae_j = a_j$.
- $\text{row}_i(AB) = \text{row}_i(A)B = a_{i1} \text{row}_1(B) + \cdots + a_{in} \text{row}_n(B)$.
- $(AB)_{ij} = \text{row}_i(A) b_j$, and $AB = a_1 \text{row}_1(B) + \cdots + a_n \text{row}_n(B)$, a sum of n products of a column and a row (the **outer product expansion**).

Proof. In each statement the entry in position (i, j) (or i , or j) of both sides is $\sum_r a_{ir} b_{rj}$; in (b) the i th entry of Ax is $\sum_j a_{ij} x_j$, which is the i th entry of $\sum_j x_j a_j$. $\square$

Example 1.5. (a) $\begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 2 \\ 3 & 4 & 8 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 7 & 2 \end{pmatrix}$. By Theorem 1.4(c), its second row is $-3(1, -1, 2) + 1(3, 4, 8) = (0, 7, 2)$.

(b) $\begin{pmatrix} 2 & 1 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 5 \end{pmatrix} - 1 \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}$, by Theorem 1.4(b).

(c) For $N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ and $M = \begin{pmatrix} 1 & -5 & 2 \\ 2 & 3 & 4 \\ 9 & -1 & 3 \end{pmatrix}$,

$$NM = \begin{pmatrix} 2 & 3 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad MN = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 9 & 0 \end{pmatrix}$$

Row 1 of NM is row 2 of M , and column 2 of MN is column 1 of M . So $NM \neq MN$, although both are 3×3 .

Remark 1.6. Matrix products differ from products of numbers. In general $AB \neq BA$ (Example 1.5(c)); square matrices with $AB = BA$ are said to **commute**. A product can be 0 although no factor is 0: for

$$X = \begin{pmatrix} 1 & 1 & 3 \\ 2 & 2 & 6 \\ -1 & -1 & -3 \end{pmatrix}, \quad X^2 = 0,$$

since $(X^2)_{ij} = \text{row}_i(X) \text{col}_j(X)$ (Theorem 1.4(d)), rows 1, 2, 3 of X are 1, 2, -1 times $\text{row}_1(X)$, columns 1, 2, 3 are 1, 1, 3 times $\text{col}_1(X)$, and $\text{row}_1(X) \text{col}_1(X) = 1 + 2 - 3 = 0$. Hence $AB = 0$ does not force $A = 0$ or $B = 0$, and $AB = AC$ does not force $B = C$.

Transpose and trace

Definition 1.7. The **transpose** of $A \in \mathbb{R}^{m \times n}$ is $A' \in \mathbb{R}^{n \times m}$ with $(A')_{ij} = a_{ji}$: the rows of A' are the transposed columns of A . We write column vectors as $x = (x_1, \dots, x_n)'$. A square A is **symmetric** if $A' = A$.

Theorem 1.8. For matrices of sizes for which the operations are defined and $c \in \mathbb{R}$,

$$(A')' = A, \quad (A + B)' = A' + B', \quad (cA)' = cA', \quad (AB)' = B'A', \quad (ABC)' = C'B'A'$$

For $x, y \in \mathbb{R}^n$, $x'y = y'x = \sum_{i=1}^n x_i y_i$; in particular $x'x = \sum_i x_i^2$. For every $B \in \mathbb{R}^{m \times n}$, $B'B$ and BB' are symmetric.

Proof. The first three rules hold entrywise. For $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, the (k, i) entry of $B'A'$ is $\sum_j b_{jk} a_{ij} = (AB)_{ik} = ((AB)')_{ki}$; applying this twice gives $(ABC)'$. The 1×1 matrix $x'y = \sum_i x_i y_i$ equals its transpose $y'x$. Finally $(B'B)' = B'(B')' = B'B$, and likewise for BB' . $\square$

Definition 1.9. The **trace** of $A \in \mathbb{R}^{n \times n}$ is $\text{tr}(A) = a_{11} + a_{22} + \dots + a_{nn}$.

Theorem 1.10. (a) $\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$, $\text{tr}(cA) = c \text{tr}(A)$ and $\text{tr}(A') = \text{tr}(A)$ for $A, B \in \mathbb{R}^{n \times n}$.

(b) $\text{tr}(AB) = \text{tr}(BA)$ for $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$.

(c) $\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA)$ for $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, $C \in \mathbb{R}^{p \times m}$.

(d) $\text{tr}(A'A) = \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2$ for $A \in \mathbb{R}^{m \times n}$; hence $\text{tr}(A'A) = 0$ only if $A = 0$.

Proof. (a) holds entrywise on the diagonal. (b) Both traces equal $\sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ji}$. (c) Apply (b) to the pairs AB, C and A, BC . (d) The j th diagonal entry of $A'A$ is $\sum_i a_{ij}^2$; a sum of squares of real numbers is 0 only if every term is 0. $\square$

Special matrices; means and sums of squares

Diagonal and triangular matrices. $D = \text{diag}(d_1, \dots, d_n)$ is the $n \times n$ matrix with diagonal entries $d_1, \dots, d_n$ and all other entries 0. Left multiplication by a diagonal matrix scales the rows, right multiplication scales the columns:

$$\text{row}_i(DA) = d_i \text{row}_i(A), \quad \text{col}_j(AD) = d_j \text{col}_j(A),$$

since $(DA)_{ij} = d_i a_{ij}$ and $(AD)_{ij} = a_{ij} d_j$. In particular $\text{diag}(d_1, \dots, d_n) \text{diag}(f_1, \dots, f_n) = \text{diag}(d_1 f_1, \dots, d_n f_n)$. A square A is **upper triangular** if $a_{ij} = 0$ for $i > j$ and **lower triangular** if $a_{ij} = 0$ for $i < j$. A product of two upper (lower) triangular matrices is upper (lower) triangular, with diagonal entries $a_{ii} b_{ii}$: for upper triangular A, B , each term $a_{ik} b_{kj}$ of $(AB)_{ij}$ has $a_{ik} = 0$ if $k < i$ and $b_{kj} = 0$ if $k > j$, so $(AB)_{ij} = 0$ for $i > j$ and $(AB)_{ii} = a_{ii} b_{ii}$ (lower triangular: reverse the inequalities).

Definition 1.11. The **summing vector** $\mathbf{1}_n \in \mathbb{R}^n$ has all entries 1. Put

$$J_n = \mathbf{1}_n \mathbf{1}'_n \in \mathbb{R}^{n \times n}, \quad \bar{J}_n = \frac{1}{n} J_n, \quad C_n = I_n - \bar{J}_n$$

Every entry of J_n is 1, and C_n is the **centering matrix**. Subscripts are dropped when the size is clear. A square matrix A with $A^2 = A$ is **idempotent**.

Proposition 1.12. $\mathbf{1}'\mathbf{1} = n$, $J_n^2 = nJ_n$, and $\bar{J}_n$ and $C = C_n$ are symmetric and idempotent:

$$\bar{J}'_n = \bar{J}_n = \bar{J}_n^2, \quad C = C' = C^2, \quad C\mathbf{1} = 0$$

Proof. $\mathbf{1}'\mathbf{1} = 1 + \dots + 1 = n$, so $J_n^2 = \mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = nJ_n$ and $\bar{J}_n^2 = \frac{1}{n^2} nJ_n = \bar{J}_n$. Next $J'_n = (\mathbf{1}\mathbf{1}')' = \mathbf{1}\mathbf{1}' = J_n$ gives $\bar{J}'_n = \bar{J}_n$ and $C' = C$, and $C^2 = I - 2\bar{J}_n + \bar{J}_n^2 = I - \bar{J}_n = C$. Finally $C\mathbf{1} = \mathbf{1} - \frac{1}{n}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \mathbf{1} - \mathbf{1} = 0$. $\square$

Example 1.13. For data $x = (x_1, \dots, x_n)'$ the mean is $\bar{x} = \frac{1}{n}\mathbf{1}'x$. Then

$$Cx = x - \frac{1}{n}\mathbf{1}(\mathbf{1}'x) = x - \bar{x}\mathbf{1} = (x_1 - \bar{x}, \dots, x_n - \bar{x})',$$

the vector of **deviations** from the mean. Moreover $x'Cx = x'x - \bar{x}(\mathbf{1}'x) = \sum_i x_i^2 - n\bar{x}^2$, and, since $\sum_i x_i = n\bar{x}$, $\sum_i (x_i - \bar{x})^2 = \sum_i x_i^2 - 2\bar{x} \sum_i x_i + n\bar{x}^2 = \sum_i x_i^2 - n\bar{x}^2$. Hence the sum of squares about the mean is

$$\sum_{i=1}^n (x_i - \bar{x})^2 = x'x - n\bar{x}^2 = x'Cx$$

Example 1.14. Four units ($n = 4$) are measured on two variables ($k = 2$); x_{ij} is the value of variable j on unit i :

$$X_0 = (x_{ij}) = \begin{pmatrix} 2 & 3 \\ 0 & 3 \\ 2 & 1 \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{4 \times 2}, \quad \mathbf{1}'X_0 = (4, 8), \quad \bar{x} = \frac{1}{4}X'_0\mathbf{1} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

The vector $\bar{x} = (\bar{x}_1, \bar{x}_2)'$ of column means $\bar{x}_j = \frac{1}{n} \sum_i x_{ij}$ is the **mean vector**. By Example 1.13, applied to each column ($\text{col}_j(CX_0) = C \text{col}_j(X_0)$, Theorem 1.4(a)), the centered data matrix and the matrix of **corrected sums of squares and products** are

$$CX_0 = X_0 - \mathbf{1}\bar{x}' = \begin{pmatrix} 1 & 1 \\ -1 & 1 \\ 1 & -1 \\ -1 & -1 \end{pmatrix}, \quad X'_0CX_0 = (CX_0)'(CX_0) = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix},$$

using $C = C'C$ (Proposition 1.12). For instance $4 = 1 + 1 + 1 + 1$ and $0 = 1 - 1 - 1 + 1$. The same matrix comes from the uncorrected sums: in general $X'_0 C X_0 = X'_0 X_0 - \frac{1}{n}(X'_0 \mathbf{1})(\mathbf{1}' X_0) = X'_0 X_0 - n\bar{x}\bar{x}'$, and here

$$X'_0 X_0 - 4\bar{x}\bar{x}' = \begin{pmatrix} 8 & 8 \\ 8 & 20 \end{pmatrix} - \begin{pmatrix} 4 & 8 \\ 8 & 16 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$$

In regression one also uses the **design matrix** $X = [\mathbf{1}_4 \ X_0] \in \mathbb{R}^{4 \times 3}$, of size $n \times (k+1)$: its first column is $\mathbf{1}_4$, and its i th row is $x'_i = (1, x_{i1}, x_{i2})$. Write $X_{\cdot 1}$ and $X_{\cdot 2}$ for its other two columns, the columns of X_0 . By Theorems 1.4(d) and 1.8, the (j, l) entry of $X'X$ is $\text{col}_j(X)' \text{col}_l(X)$, so

$$X'X = \begin{pmatrix} \mathbf{1}'\mathbf{1} & \mathbf{1}'X_{\cdot 1} & \mathbf{1}'X_{\cdot 2} \\ X'_{\cdot 1}\mathbf{1} & X'_{\cdot 1}X_{\cdot 1} & X'_{\cdot 1}X_{\cdot 2} \\ X'_{\cdot 2}\mathbf{1} & X'_{\cdot 2}X_{\cdot 1} & X'_{\cdot 2}X_{\cdot 2} \end{pmatrix} = \begin{pmatrix} 4 & 4 & 8 \\ 4 & 8 & 8 \\ 8 & 8 & 20 \end{pmatrix} :$$

the entries are n , the column sums $\sum_i x_{i1}$, $\sum_i x_{i2}$, and the uncorrected sums of squares and products $\sum_i x_{i1}^2$, $\sum_i x_{i1}x_{i2}$, $\sum_i x_{i2}^2$.

Exercises and solutions

Exercise 1.1. Let $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix}$. Compute AB' , BA' , $A'B$ and $B'A$.

Solution.

$$AB' = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 6 \\ 5 & 4 \end{pmatrix}, \quad BA' = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 6 & 4 \end{pmatrix},$$

$$A'B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 4 & 5 \\ 2 & 5 & 4 \end{pmatrix}, \quad B'A = \begin{pmatrix} 0 & 1 \\ 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 4 & 5 \\ 2 & 5 & 4 \end{pmatrix}$$

For instance, $(AB')_{12} = 1 \cdot 1 + 2 \cdot 2 + 1 \cdot 1 = 6$. Here $A'B$ happens to be symmetric while AB' is not. The four products have the same trace 8; this is no coincidence: $\text{tr}(AB') = \text{tr}(B'A)$ and $\text{tr}(BA') = \text{tr}(A'B)$ by Theorem 1.10(b), and $\text{tr}(BA') = \text{tr}((AB')') = \text{tr}(AB')$ by Theorems 1.8 and 1.10(a).

Exercise 1.2. Let $A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 \\ 0 & -1 \\ 1 & 1 \end{pmatrix}$.

- (a) Compute AB and BA .
- (b) Write AB as the sum of the three products $\text{col}_j(A) \text{row}_j(B)$ of Theorem 1.4(d).
- (c) Compute $B'A'$, $\text{tr}(AB)$, $\text{tr}(BA)$ and $\text{tr}(A'A)$.

Solution. (a) By Theorem 1.4(c), $\text{row}_i(AB) = \text{row}_i(A)B$ is a combination of the rows of B :

$$\begin{aligned} \text{row}_1(AB) &= 1(2, 1) + 0(0, -1) + 2(1, 1) = (4, 3), \\ \text{row}_2(AB) &= -1(2, 1) + 3(0, -1) + 1(1, 1) = (-1, -3) \end{aligned}$$

Likewise each row of $BA \in \mathbb{R}^{3 \times 3}$ is a combination of the rows $(1, 0, 2)$ and $(-1, 3, 1)$ of A :

$$\begin{aligned} \text{row}_1(BA) &= 2(1, 0, 2) + 1(-1, 3, 1) = (1, 3, 5), \\ \text{row}_2(BA) &= 0(1, 0, 2) - 1(-1, 3, 1) = (1, -3, -1), \\ \text{row}_3(BA) &= 1(1, 0, 2) + 1(-1, 3, 1) = (0, 3, 3) \end{aligned}$$

Hence

$$AB = \begin{pmatrix} 4 & 3 \\ -1 & -3 \end{pmatrix}, \quad BA = \begin{pmatrix} 1 & 3 & 5 \\ 1 & -3 & -1 \\ 0 & 3 & 3 \end{pmatrix}$$

The two products do not even have the same size.

(b)

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix} (2, 1) + \begin{pmatrix} 0 \\ 3 \end{pmatrix} (0, -1) + \begin{pmatrix} 2 \\ 1 \end{pmatrix} (1, 1) = \begin{pmatrix} 2 & 1 \\ -2 & -1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & -3 \end{pmatrix} + \begin{pmatrix} 2 & 2 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 3 \\ -1 & -3 \end{pmatrix} = AB$$

(c) Here $B' = \begin{pmatrix} 2 & 0 & 1 \\ 1 & -1 & 1 \end{pmatrix}$ and $A' = \begin{pmatrix} 1 & -1 \\ 0 & 3 \\ 2 & 1 \end{pmatrix}$, so

$$B'A' = \begin{pmatrix} 2 \cdot 1 + 0 \cdot 0 + 1 \cdot 2 & 2 \cdot (-1) + 0 \cdot 3 + 1 \cdot 1 \\ 1 \cdot 1 + (-1) \cdot 0 + 1 \cdot 2 & 1 \cdot (-1) + (-1) \cdot 3 + 1 \cdot 1 \end{pmatrix} = \begin{pmatrix} 4 & -1 \\ 3 & -3 \end{pmatrix} = (AB)',$$

as Theorem 1.8 says. Next $\text{tr}(AB) = 4 - 3 = 1$ and $\text{tr}(BA) = 1 - 3 + 3 = 1$, equal by Theorem 1.10(b); and by Theorem 1.10(d), $\text{tr}(A'A) = 1^2 + 0^2 + 2^2 + (-1)^2 + 3^2 + 1^2 = 16$.

Exercise 1.3. Give examples of real 2×2 matrices such that: (a) $A^2 = -I$; (b) $B^2 = 0$ ($B \neq 0$); (c) $CD = -DC$ ($CD \neq 0$); (d) $EF = 0$ (no entries of E or F are zero).

Solution. (a) No real number a has $a^2 = -1$, but $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ gives $A^2 = \begin{pmatrix} 0 \cdot 0 + 1 \cdot (-1) & 0 \cdot 1 + 1 \cdot 0 \\ (-1) \cdot 0 + 0 \cdot (-1) & (-1) \cdot 1 + 0 \cdot 0 \end{pmatrix} = -I$.

(b) $B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0$ satisfies $B^2 = 0$.

(c) For $C = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ and $D = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$, $CD = \begin{pmatrix} 0 & -2 \\ 2 & 0 \end{pmatrix}$ and $DC = \begin{pmatrix} 0 & 2 \\ -2 & 0 \end{pmatrix} = -CD \neq 0$.

(d) For $E = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and $F = \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix}$, every entry of EF is $1 \cdot 1 + 1 \cdot (-1) = 0$, so $EF = 0$.

Exercise 1.4. Let $A, B \in \mathbb{R}^{n \times n}$. Show that $(A + B)(A - B) = A^2 - B^2$ if and only if A and B commute.

Solution. By Theorem 1.3, $(A + B)(A - B) = A(A - B) + B(A - B) = A^2 - AB + BA - B^2$. Thus $(A + B)(A - B) = A^2 - B^2$ if and only if $-AB + BA = 0$, that is, if and only if $AB = BA$.

Exercise 1.5. Let $A = \begin{pmatrix} 1 & 1 \\ 3 & 4 \end{pmatrix}$. Show that the matrices B with $AB = BA$ are exactly $B = \alpha \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \beta \begin{pmatrix} 0 & 1 \\ 3 & 3 \end{pmatrix}$ ($\alpha, \beta \in \mathbb{R}$).

Solution. Let $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Then $AB = BA$ reads

$$\begin{pmatrix} a + c & b + d \\ 3a + 4c & 3b + 4d \end{pmatrix} = \begin{pmatrix} a + 3b & a + 4b \\ c + 3d & c + 4d \end{pmatrix}$$

Comparing entries, the (1, 1) and (2, 2) entries give the same equation, and the conditions are

$$3b - c = 0, \quad a + 3b - d = 0, \quad a + c - d = 0$$

(the (2, 1) entries give $3a + 3c - 3d = 0$). The first gives $c = 3b$, and then the third gives $d = a + 3b$; with these values the second holds ($a + 3b - a - 3b = 0$). So a and b are free, and the solutions are

$$B = \begin{pmatrix} a & b \\ 3b & a + 3b \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 3 & 3 \end{pmatrix},$$

the stated form with $\alpha = a$, $\beta = b$. Conversely, every α, β arise (take $a = \alpha$, $b = \beta$), so all these matrices commute with A .

Exercise 1.6. (a) Show that the product AB of two symmetric matrices $A, B \in \mathbb{R}^{n \times n}$ is symmetric if and only if A and B commute.

(b) Give an example of two symmetric matrices of the same size whose product is not symmetric.

Solution. (a) Since A and B are symmetric, $(AB)' = B'A' = BA$ (Theorem 1.8). So $AB = (AB)'$ holds if and only if $AB = BA$.

(b) Take $A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Then $AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 \\ 1 & 0 \end{pmatrix} = BA$, so by (a) AB is not symmetric.

Exercise 1.7. Show that for any $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$ and $C \in \mathbb{R}^{p \times m}$, $\text{tr}(ABC) = \text{tr}(B'A'C') = \text{tr}(A'C'B')$.

Solution. By Theorem 1.10(c), then (a), then Theorem 1.8, and finally (c) again,

$$\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}((CAB)') = \text{tr}(B'A'C') = \text{tr}(A'C'B')$$

In the last step the factors $B' \in \mathbb{R}^{p \times n}$, $A' \in \mathbb{R}^{n \times m}$, $C' \in \mathbb{R}^{m \times p}$ are moved cyclically.

Exercise 1.8. Let $A \in \mathbb{R}^{n \times n}$ and let $D = \text{diag}(a_{11}, \dots, a_{nn})$. Show that the sum of all entries of A is

$$\mathbf{1}'_n A \mathbf{1}_n = \mathbf{1}'_n D \mathbf{1}_n + \text{tr}((\mathbf{1}_n \mathbf{1}'_n - I_n)A)$$

Solution. By Theorem 1.10(a),(b),

$$\text{tr}((\mathbf{1}\mathbf{1}' - I_n)A) = \text{tr}(\mathbf{1}\mathbf{1}'A) - \text{tr}(A) = \text{tr}(\mathbf{1}'A\mathbf{1}) - \text{tr}(A) = \mathbf{1}'A\mathbf{1} - \mathbf{1}'D\mathbf{1}$$

Here $\text{tr}(\mathbf{1}\mathbf{1}'A) = \text{tr}(\mathbf{1}'A\mathbf{1})$ moves the $n \times 1$ factor $\mathbf{1}$ past the $1 \times n$ factor $\mathbf{1}'A$; the trace of the 1×1 matrix $\mathbf{1}'A\mathbf{1}$ is that number; and $\mathbf{1}'D\mathbf{1} = \sum_i a_{ii} = \text{tr}(A)$, because $D\mathbf{1} = (a_{11}, \dots, a_{nn})'$. Rearranging gives the identity. Also $\mathbf{1}'A\mathbf{1} = \sum_i \sum_j a_{ij}$ by Theorem 1.4(b),(c).

Exercise 1.9. (a) Show that the only symmetric idempotent 2×2 matrices are I_2 , 0 , and uu' with $u \in \mathbb{R}^2$, $u'u = 1$.

(b) Give an example of an $n \times n$ idempotent matrix ($n \geq 2$) that is not symmetric.

Solution. (a) The symmetric matrix $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$ is idempotent if and only if $\begin{pmatrix} a^2+b^2 & b(a+d) \\ b(a+d) & b^2+d^2 \end{pmatrix} = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$, that is, $a^2 + b^2 = a$, $b(a + d) = b$, $d^2 + b^2 = d$.

- If $a + d \neq 1$, the second equation gives $b = 0$; then $a^2 = a$ and $d^2 = d$, so $a, d \in \{0, 1\}$, and $a + d \neq 1$ leaves $a = d = 0$ or $a = d = 1$: $A = 0$ or $A = I_2$.
- If $a + d = 1$, then $d = 1 - a$ and the first equation gives $b^2 = a(1 - a)$, so $0 \leq a \leq 1$ and $b = \pm\sqrt{a(1 - a)}$; the third equation then holds, since $(1 - a)^2 + a(1 - a) = 1 - a$. Hence

$$A = \begin{pmatrix} a & \pm\sqrt{a(1-a)} \\ \pm\sqrt{a(1-a)} & 1-a \end{pmatrix} = \begin{pmatrix} \sqrt{a} \\ \pm\sqrt{1-a} \end{pmatrix} \begin{pmatrix} \sqrt{a} & \pm\sqrt{1-a} \end{pmatrix} = uu',$$

with the same sign throughout, and $u'u = a + (1 - a) = 1$.

Conversely, I_2 and 0 are symmetric idempotent, and if $u'u = 1$, then uu' is symmetric and $(uu')(uu') = u(u'u)u' = uu'$.

(b) Let $a, b \in \mathbb{R}^n$ with $b'a = 1$, and $A = ab'$. Then $A^2 = a(b'a)b' = ab' = A$. This A is symmetric only when b is a multiple of a : if $ab' = ba'$, multiplying on the right by a gives $a(b'a) = b(a'a)$, that is, $a = (a'a)b$; here $a \neq 0$ because $b'a = 1$, so $a'a > 0$ and $b = a/(a'a)$. Conversely, if $b = ca$, then $ab' = caa'$ is symmetric. For a concrete example take $a = e_1$ and $b = e_1 + e_2$: $b'a = 1$, and $A = e_1e'_1 + e_1e'_2$ has $(1, 1)$ and $(1, 2)$ entries 1 and all other entries 0; for $n = 2$, $A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$, with $A^2 = A \neq A'$.

Exercise 1.10. Four units are measured on two variables ($n = 4$, $k = 2$), with data matrix

$$X_0 = \begin{pmatrix} 1 & 0 \\ 3 & 2 \\ 2 & 0 \\ 2 & 2 \end{pmatrix} \in \mathbb{R}^{4 \times 2}$$

Find

- (a) the mean vector $\bar{x}$ and the centered data matrix $C_4 X_0$;
- (b) the matrix $X_0' C_4 X_0$ of corrected sums of squares and products, both as $(C_4 X_0)'(C_4 X_0)$ and as $X_0' X_0 - 4\bar{x}\bar{x}'$;
- (c) $X'X$ for the design matrix $X = [\mathbf{1}_4 \ X_0]$.

Solution. (a) $\mathbf{1}'X_0 = (1+3+2+2, 0+2+0+2) = (8, 4)$, so $\bar{x} = \frac{1}{4}X_0'\mathbf{1} = (2, 1)'$. As in Example 1.14, $C_4 X_0 = X_0 - \mathbf{1}\bar{x}'$:

$$C_4 X_0 = \begin{pmatrix} 1-2 & 0-1 \\ 3-2 & 2-1 \\ 2-2 & 0-1 \\ 2-2 & 2-1 \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ 1 & 1 \\ 0 & -1 \\ 0 & 1 \end{pmatrix}$$

(b) The (j, l) entry of $(C_4 X_0)'(C_4 X_0)$ is $\text{col}_j(C_4 X_0)' \text{col}_l(C_4 X_0)$ (Theorems 1.4(d) and 1.8): $1+1+0+0 = 2$ for $(1, 1)$, $1+1+0+0 = 2$ for $(1, 2)$ and $(2, 1)$, and $1+1+1+1 = 4$ for $(2, 2)$. On the other hand

$$X_0' X_0 - 4\bar{x}\bar{x}' = \begin{pmatrix} 18 & 10 \\ 10 & 8 \end{pmatrix} - 4 \begin{pmatrix} 4 & 2 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 18 & 10 \\ 10 & 8 \end{pmatrix} - \begin{pmatrix} 16 & 8 \\ 8 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}$$

(for instance $18 = 1 + 9 + 4 + 4$ and $10 = 0 + 6 + 0 + 4$). So $X_0' C_4 X_0 = \begin{pmatrix} 2 & 2 \\ 2 & 4 \end{pmatrix}$: the sums of squares about the means are $\sum_i (x_{i1} - \bar{x}_1)^2 = 2$ and $\sum_i (x_{i2} - \bar{x}_2)^2 = 4$, and the corrected sum of products is $\sum_i (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2) = 2$.

(c) As in Example 1.14, the entries of $X'X$ are $n = 4$, the column sums 8, 4 of X_0 and the entries of $X_0' X_0$:

$$X'X = \begin{pmatrix} 4 & 8 & 4 \\ 8 & 18 & 10 \\ 4 & 10 & 8 \end{pmatrix}$$

Chapter 2. Gauss elimination, RREF, and inverses

Linear systems and augmented matrices

Definition 2.1. Let $A = (a_{ij}) \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$. The system of m linear equations in n unknowns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

is the matrix equation $Ax = b$, since $(Ax)_i = \sum_j a_{ij}x_j$. A is the **coefficient matrix** and $[A \mid b] \in \mathbb{R}^{m \times (n+1)}$, the matrix A with the column b appended, is the **augmented matrix**. A **solution** is an $x \in \mathbb{R}^n$ with $Ax = b$. The system is **consistent** if it has a solution and **inconsistent** otherwise. It is **homogeneous** if $b = 0$; then $x = 0$ is a solution, the **trivial solution**.

Elementary row operations

Definition 2.2. The **elementary row operations** on a matrix with m rows are, for row indices $r \neq s$:

- (1) multiply row r by a number $c \neq 0$, written $R_r \leftarrow cR_r$;
- (2) add c times row s to row r ($c \in \mathbb{R}$), written $R_r \leftarrow R_r + cR_s$;
- (3) interchange rows r and s , written $R_r \leftrightarrow R_s$.

When several operations label one arrow, they are performed one after another, in the order written (upper label first). B is **row-equivalent** to A if B is obtained from A by finitely many elementary row operations.

Theorem 2.3. (a) Each elementary row operation is undone by an operation of the same type:

$R_r \leftarrow cR_r$ by $R_r \leftarrow c^{-1}R_r$, $R_r \leftarrow R_r + cR_s$ by $R_r \leftarrow R_r - cR_s$, and $R_r \leftrightarrow R_s$ by itself.

- (b) If row operations take $[A \mid b]$ to $[B \mid z]$, then the systems $Ax = b$ and $Bx = z$ have exactly the same solutions.

Proof. (a) is checked row by row. (b) It suffices to consider one operation. Each equation of the new system is a multiple of an equation of the old one, or a sum of an old equation and a multiple of another, so every solution of $Ax = b$ solves $Bx = z$. By (a), the old system is obtained back from the new one by a row operation, so every solution of $Bx = z$ solves $Ax = b$. $\square$

So we solve a system by simplifying its augmented matrix with row operations; the next definition says how far to simplify.

Echelon form and RREF

Definition 2.4. A matrix R is in **reduced row echelon form (RREF)** if

- (a) the first nonzero entry of each nonzero row is 1, its **pivot**;
- (b) each column that contains a pivot has all its other entries 0;
- (c) every zero row lies below every nonzero row;
- (d) if rows $1, \dots, r$ are the nonzero rows and the pivot of row i lies in column k_i , then $k_1 < k_2 < \cdots < k_r$.

The columns $k_1, \dots, k_r$ are the **pivot columns**; column k_i of R is the unit vector e_i . A matrix that satisfies (a), (c), (d), but not necessarily (b), is in **echelon form**; then every entry below a pivot is 0.

For example, I_n and 0 are in RREF, and so is

$$R = \begin{pmatrix} 0 & 1 & -3 & 0 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad (r = 2, \quad k_1 = 2, \quad k_2 = 4)$$

The matrix $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ is in echelon form but not in RREF (column 3 violates (b)), and $\begin{pmatrix} 0 & 2 & 1 \\ 1 & 0 & -3 \\ 0 & 0 & 0 \end{pmatrix}$ violates (a) and (d).

Definition 2.5 (Gauss–Jordan elimination). Let M be any matrix (for instance an augmented matrix). Start with $i = 1$.

Step 1. Find the first column that has a nonzero entry in rows $i, i+1, \dots$; if there is none, go to Step 5. This is the next pivot column.

Step 2. If the entry of row i in this column is 0, interchange row i with a lower row that has a nonzero entry there.

Step 3. Multiply row i by the reciprocal of this entry, so that the pivot is 1.

Step 4. Add suitable multiples of row i to the rows below it, so that every entry below the pivot is 0. Increase i by 1 and return to Step 1.

Step 5. (Backward elimination.) For each pivot, from the last one to the first, add suitable multiples of its row to the rows above it, so that every entry above the pivot is 0.

Steps 1–4 (**forward elimination**) produce an echelon form; Step 5 produces the RREF. The scaling in Step 3 may also be postponed, to the end of forward elimination or even to the very end (the pivots become 1 only then); this often avoids fractions. Every matrix is thus row-equivalent to a matrix in RREF; this RREF is unique (proved after Theorem 4.3), and is called the RREF of M .

Example 2.6. Solve the homogeneous system

$$x_1 + x_2 + x_3 - 2x_4 = 0, \quad x_1 + x_2 + 2x_3 - 3x_4 = 0, \quad 2x_1 + x_2 + 5x_3 - 4x_4 = 0$$

Solution. Row-reduce the coefficient matrix A (the column $b = 0$ never changes, so it is not written).

$$\begin{aligned} & \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & 2 & -3 \\ 2 & 1 & 5 & -4 \end{pmatrix} \xrightarrow[\substack{R_2 \leftarrow R_2 - R_1 \\ R_3 \leftarrow R_3 - 2R_1}]{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & 0 & 1 & -1 \\ 0 & -1 & 3 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} \\ & \xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & 1 & 1 & -2 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow[\substack{R_1 \leftarrow R_1 - R_3}]{R_2 \leftarrow R_2 + 3R_3} \begin{pmatrix} 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & -1 \end{pmatrix} \end{aligned}$$

After the first step the entry of row 2 in column 2 is 0, so Step 2 interchanges rows 2 and 3, and Step 3 multiplies the new row 2 by -1 . The first matrix of the second line is an echelon form (end of forward elimination); the last matrix is the RREF R , with pivot columns 1, 2, 3. The system $Rx = 0$ reads $x_1 + 2x_4 = 0$, $x_2 - 3x_4 = 0$, $x_3 - x_4 = 0$, so $x_4 = t$ is free and all solutions are

$$x = t(-2, 3, 1, 1)' \quad (t \in \mathbb{R})$$

Check: $-2 + 3 + 1 - 2 = 0$, $-2 + 3 + 2 - 3 = 0$, $-4 + 3 + 5 - 4 = 0$.

Reading off the solutions

Proposition 2.7. Let row operations take $[A \mid b]$ ($A \in \mathbb{R}^{m \times n}$) to $[R \mid z]$ with R in RREF, nonzero rows $1, \dots, r$ and pivot columns $k_1 < \dots < k_r$. Call $x_{k_1}, \dots, x_{k_r}$ the **pivot unknowns** and the other $n - r$ unknowns the **free unknowns**.

- (a) $Ax = b$ is consistent if and only if $z_i = 0$ for every $i > r$, that is, no row of $[R \mid z]$ reads $(0 \ \dots \ 0 \mid z_i)$ with $z_i \neq 0$.
- (b) If it is consistent, all solutions are obtained by choosing arbitrary values for the free unknowns and computing each pivot unknown from its row: $x_{k_i} = z_i - \sum_{j \text{ free}} R_{ij} x_j$ ($1 \leq i \leq r$).

Proof. By Theorem 2.3, $Ax = b$ and $Rx = z$ have the same solutions. (a) Row $i > r$ of $Rx = z$ reads $0 = z_i$. So there is no solution if one of these z_i is not 0; if all of them are 0, these rows hold for every x , and the argument for (b) gives a solution. (b) For $i \leq r$, column k_i of R is e_i , so the pivot unknown x_{k_i} occurs only in equation i , with coefficient 1; each choice of the free unknowns therefore gives exactly one solution. $\square$

Example 2.8. For the RREF R displayed after Definition 2.4, the system $Rx = 0$ is $x_2 - 3x_3 + 2x_5 = 0$, $x_4 + 2x_5 = 0$; the pivot unknowns are x_2, x_4 and the free unknowns x_1, x_3, x_5 . With $x_1 = a$, $x_3 = b$, $x_5 = c$, all solutions are

$$x = \begin{pmatrix} a \\ 3b - 2c \\ b \\ -2c \\ c \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + b \begin{pmatrix} 0 \\ 3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + c \begin{pmatrix} 0 \\ -2 \\ 0 \\ -2 \\ 1 \end{pmatrix} \quad (a, b, c \in \mathbb{R})$$

Example 2.9. For which $y = (y_1, y_2, y_3)'$ is $Ax = y$ consistent, where $A = \begin{pmatrix} 1 & -2 & 1 \\ 2 & -3 & 1 \\ 0 & 1 & -1 \end{pmatrix}$? Solve it for $y = (1, 2, 0)'$.

Solution. Row-reduce the augmented matrix, keeping y_1, y_2, y_3 as letters:

$$\begin{aligned} & \left(\begin{array}{ccc|c} 1 & -2 & 1 & y_1 \\ 2 & -3 & 1 & y_2 \\ 0 & 1 & -1 & y_3 \end{array} \right) \xrightarrow{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & -2 & 1 & y_1 \\ 0 & 1 & -1 & y_2 - 2y_1 \\ 0 & 1 & -1 & y_3 \end{array} \right) \\ & \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & -2 & 1 & y_1 \\ 0 & 1 & -1 & y_2 - 2y_1 \\ 0 & 0 & 0 & y_3 - y_2 + 2y_1 \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \left(\begin{array}{ccc|c} 1 & 0 & -1 & 2y_2 - 3y_1 \\ 0 & 1 & -1 & y_2 - 2y_1 \\ 0 & 0 & 0 & y_3 - y_2 + 2y_1 \end{array} \right) \end{aligned}$$

Here $r = 2$ and $z_3 = y_3 - y_2 + 2y_1$. By Proposition 2.7, the system is consistent if and only if $2y_1 - y_2 + y_3 = 0$; then $x_3 = c$ is free and

$$x_1 = 2y_2 - 3y_1 + c, \quad x_2 = y_2 - 2y_1 + c, \quad x_3 = c \quad (c \in \mathbb{R})$$

For $y = (1, 2, 0)'$, $2 - 2 + 0 = 0$, and the solutions are $x = (1, 0, 0)' + c(1, 1, 1)'$: the solution $(1, 0, 0)'$ plus the multiples of $(1, 1, 1)'$, which solve $Ax = 0$. For $y = (1, 0, 0)'$, $2y_1 - y_2 + y_3 = 2 \neq 0$, so $Ax = y$ has no solution.

Homogeneous systems

Theorem 2.10. (a) If $A \in \mathbb{R}^{m \times n}$ with $m < n$ (fewer equations than unknowns), then $Ax = 0$ has a nontrivial solution.

- (b) For $A \in \mathbb{R}^{n \times n}$, $Ax = 0$ has only the trivial solution if and only if the RREF of A is I_n .

Proof. Let R be the RREF of A , with r nonzero rows. (a) $r \leq m < n$, so there is a free unknown; giving it the value 1 and the other free unknowns the value 0 gives a solution $x \neq 0$ (Proposition 2.7(b)). (b) If $R = I$, then $Rx = x$, so $x = 0$ is the only solution. If $Ax = 0$ has only the trivial solution, there is no free unknown, so $r = n$; then $1 \leq k_1 < \dots < k_n \leq n$ forces $k_i = i$, so column i of R is e_i and $R = I$. $\square$

Inverses

Definition 2.11. $A \in \mathbb{R}^{n \times n}$ is **invertible** if $AB = BA = I_n$ for some $B \in \mathbb{R}^{n \times n}$. Such a B is unique: if $BA = I$ and $AC = I$, then $B = B(AC) = (BA)C = C$. It is called the **inverse** of A , written A^{-1} . For example, $\text{diag}(d_1, \dots, d_n)$ with all $d_i \neq 0$ has inverse $\text{diag}(d_1^{-1}, \dots, d_n^{-1})$, since the products in both orders are $\text{diag}(1, \dots, 1) = I$.

Theorem 2.12. Let $A, B \in \mathbb{R}^{n \times n}$ be invertible.

- (a) A^{-1} is invertible and $(A^{-1})^{-1} = A$.
- (b) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$; more generally $(A_1 \cdots A_k)^{-1} = A_k^{-1} \cdots A_1^{-1}$ for invertible $A_1, \dots, A_k$.
- (c) A' is invertible and $(A')^{-1} = (A^{-1})'$.

Proof. (a) $AA^{-1} = A^{-1}A = I$ says that A is the inverse of A^{-1} . (b) $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$, and similarly $(B^{-1}A^{-1})(AB) = I$; for k factors repeat this. (c) $A'(A^{-1})' = (A^{-1}A)' = I$ and $(A^{-1})'A' = (AA^{-1})' = I$ by Theorem 1.8. $\square$

Elementary matrices

Definition 2.13. An **elementary matrix** is a matrix $E = \omega(I_m)$ obtained from I_m by one elementary row operation ω . For $r \neq s$ we write:

- $D_r(c)$ ($c \neq 0$): I_m with its (r, r) entry replaced by c (from $R_r \leftarrow cR_r$);
- $T_{rs}(c)$: I_m with its (r, s) entry replaced by c (from $R_r \leftarrow R_r + cR_s$);
- P_{rs} : I_m with rows r and s interchanged (from $R_r \leftrightarrow R_s$).

For example, for $m = 3$:

$$P_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad T_{31}(c) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ c & 0 & 1 \end{pmatrix}, \quad D_3(c) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & c \end{pmatrix}$$

Theorem 2.14. Let ω be an elementary row operation on matrices with m rows and $E = \omega(I_m)$.

- (a) $\omega(A) = EA$ for every $A \in \mathbb{R}^{m \times n}$: a row operation is a left multiplication.
- (b) E is invertible, and E^{-1} is the elementary matrix of the inverse operation: $D_r(c)^{-1} = D_r(c^{-1})$, $T_{rs}(c)^{-1} = T_{rs}(-c)$, $P_{rs}^{-1} = P_{rs}$.
- (c) If row operations $\omega_1, \dots, \omega_k$ (in this order) take A to B , then $B = PA$ with $P = E_k \cdots E_2 E_1$, $E_j = \omega_j(I_m)$; and P is invertible.

Proof. (a) By Theorem 1.4(c), $\text{row}_i(EA) = \text{row}_i(E)A$. An unchanged row of E is e'_i , and $e'_i A = \text{row}_i(A)$. For type (2), $\text{row}_r(E) = e'_r + ce'_s$, so $\text{row}_r(EA) = \text{row}_r(A) + c \text{row}_s(A)$; types (1) and (3) are the same computation. (b) Let $\tilde{\omega}$ be the inverse operation (Theorem 2.3(a)) and $\tilde{E} = \tilde{\omega}(I)$. By (a), $E\tilde{E} = \omega(\tilde{\omega}(I)) = I$ and $\tilde{E}E = \tilde{\omega}(\omega(I)) = I$. (c) By (a), $B = E_k(\cdots(E_1A)) = (E_k \cdots E_1)A$, and a product of invertible matrices is invertible (Theorem 2.12(b)). $\square$

Deciding invertibility and computing inverses

Theorem 2.15. For $A \in \mathbb{R}^{n \times n}$ the following are equivalent:

- (i) A is invertible;
- (ii) the RREF of A is I_n ;
- (iii) $Ax = 0$ has only the trivial solution;
- (iv) $Ax = b$ has a solution for every $b \in \mathbb{R}^n$;
- (v) A is a product of elementary matrices.

Moreover, if $BA = I$ or $AB = I$ for some $B \in \mathbb{R}^{n \times n}$, then A is invertible and $B = A^{-1}$.

Proof. Let $R = PA$ be the RREF of A , with $P = E_k \cdots E_1$ (Theorem 2.14(c)).

- If $R = I$, then $A = P^{-1} = E_1^{-1} \cdots E_k^{-1}$ is a product of elementary matrices, hence invertible (Theorem 2.12(b)); $Ax = 0$ has only the trivial solution (Theorem 2.10(b)); and $x = A^{-1}b$ solves $Ax = b$.
- If $R \neq I$, then R has a zero row, hence a zero last row (by (c) of Definition 2.4); otherwise $r = n$ and $R = I$ as in the proof of Theorem 2.10(b). Then $Ax = 0$ has a solution $x \neq 0$ (Theorem 2.10(b)), so A has no inverse (else $x = A^{-1}Ax = 0$ ✗), and A is not a product of elementary matrices. For $b = P^{-1}e_n$, the system $Ax = b$ has the same solutions as $Rx = Pb = e_n$, whose last equation reads $0 = 1$; so (iv) fails.

Thus each of (i)–(v) holds exactly when $R = I$. If $BA = I$, then $Ax = 0$ gives $x = BAx = 0$, so A is invertible by (iii), and $B = BAA^{-1} = A^{-1}$. If $AB = I$, then B is invertible with $B^{-1} = A$ by the case just shown, so $A = B^{-1}$ is invertible with $A^{-1} = B$. $\square$

Computing A^{-1} . Row-reduce the $n \times 2n$ matrix $[A \mid I_n]$ by Gauss–Jordan elimination.

- If the left half becomes I_n , the right half is A^{-1} : $[A \mid I] \rightarrow [I \mid B]$ means $E[A \mid I] = [EA \mid E] = [I \mid B]$ for the product E of the elementary matrices used (Theorem 2.14(c); E multiplies each column separately, Theorem 1.4(a)), so $EA = I$ and $B = E = A^{-1}$ by Theorem 2.15.
- If a zero row appears in the left half, A is not invertible. The left half is then EA , with E the (invertible) product of the elementary matrices used so far (Theorem 2.14(c)). For every $C \in \mathbb{R}^{n \times n}$, EAC has the same zero row (Theorem 1.4(c)), so $EAC \neq I_n$ and EA is not invertible; if A were invertible, so would be EA (Theorem 2.12(b)).

Example 2.16. Find the inverse of $A = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 1 & -2 \\ 2 & -1 & 2 \end{pmatrix}$ and write A and A^{-1} as products of elementary matrices.

Solution.

$$\begin{aligned}
 & \left(\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ -1 & 1 & -2 & 0 & 1 & 0 \\ 2 & -1 & 2 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 + R_1} \left(\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 1 & 1 & 0 \\ 0 & 1 & 0 & -2 & 0 & 1 \end{array} \right) \\
 & \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 0 & 1 \\ 0 & 0 & -1 & 1 & 1 & 0 \end{array} \right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{ccc|ccc} 1 & -1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 1 & -1 & -1 & 0 \end{array} \right) \\
 & \xrightarrow{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{ccc|ccc} 1 & -1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 1 & -1 & -1 & 0 \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 + R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & -2 & 0 & 1 \\ 0 & 0 & 1 & -1 & -1 & 0 \end{array} \right)
 \end{aligned}$$

After the first step the entry of row 2 in column 2 is 0, so Step 2 interchanges rows 2 and 3. The left half has become I_3 , so $A^{-1} = \begin{pmatrix} 0 & 1 & 1 \\ -2 & 0 & 1 \\ -1 & -1 & 0 \end{pmatrix}$. The six operations have the elementary matrices $E_1 = T_{21}(1)$, $E_2 = T_{31}(-2)$, $E_3 = P_{23}$, $E_4 = D_3(-1)$, $E_5 = T_{13}(-1)$, $E_6 = T_{12}(1)$, and by Theorem 2.14(c)

$$A^{-1} = E_6 E_5 E_4 E_3 E_2 E_1 = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

By Theorem 2.12(b) and Theorem 2.14(b),

$$A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1} E_5^{-1} E_6^{-1} = T_{21}(-1) T_{31}(2) P_{23} D_3(-1) T_{13}(1) T_{12}(-1)$$

Example 2.17. For the matrix $X'X$ of Example 1.14, find $(X'X)^{-1}$.

Solution. We postpone the scaling to the end (Definition 2.5), so that fractions appear only in the last step.

$$\begin{aligned} & \left(\begin{array}{ccc|ccc} 4 & 4 & 8 & 1 & 0 & 0 \\ 4 & 8 & 8 & 0 & 1 & 0 \\ 8 & 8 & 20 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{ccc|ccc} 4 & 4 & 8 & 1 & 0 & 0 \\ 0 & 4 & 0 & -1 & 1 & 0 \\ 0 & 0 & 4 & -2 & 0 & 1 \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 - 2R_3} \left(\begin{array}{ccc|ccc} 4 & 4 & 0 & 5 & 0 & -2 \\ 0 & 4 & 0 & -1 & 1 & 0 \\ 0 & 0 & 4 & -2 & 0 & 1 \end{array} \right) \\ & \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{ccc|ccc} 4 & 0 & 0 & 6 & -1 & -2 \\ 0 & 4 & 0 & -1 & 1 & 0 \\ 0 & 0 & 4 & -2 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \leftarrow \frac{1}{4}R_3]{R_1 \leftarrow \frac{1}{4}R_1, R_2 \leftarrow \frac{1}{4}R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{2} & -\frac{1}{4} & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{4} & \frac{1}{4} & 0 \\ 0 & 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{4} \end{array} \right) \end{aligned}$$

So $(X'X)^{-1} = \frac{1}{4} \begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$. The first step produces the corrected sums of Example 1.14: row 2 of $X'X$ is $(n\bar{x}_1, X'_{\cdot 1}X_{\cdot 1}, X'_{\cdot 1}X_{\cdot 2}) = (4, 8, 8)$, and subtracting $\bar{x}_1 = 1$ times row 1, $(n, n\bar{x}_1, n\bar{x}_2) = (4, 4, 8)$, leaves $(0, X'_{\cdot 1}X_{\cdot 1} - n\bar{x}_1^2, X'_{\cdot 1}X_{\cdot 2} - n\bar{x}_1\bar{x}_2) = (0, 4, 0)$; likewise row 3 minus $\bar{x}_2 = 2$ times row 1 is $(0, 0, 4)$. These are the rows of X'_0CX_0 .

Example 2.18. The matrix A of Example 2.9 is not invertible: its RREF is $\begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \neq I_3$ (the left part of the last matrix there). Accordingly $Ax = 0$ has the nontrivial solution $x = (1, 1, 1)'$ (take $c = 1$), and $Ax = y$ has no solution for $y = (1, 0, 0)'$.

Solving $Ax = b$

In practice $Ax = b$ is solved by forward elimination of $[A \mid b]$ alone: an echelon form already shows whether there is a solution, and how many.

Theorem 2.19. Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$, and let row operations take $[A \mid b]$ to a matrix $[E \mid d]$ in echelon form ($E \in \mathbb{R}^{m \times n}$, $d \in \mathbb{R}^m$). Then $Ax = b$ is consistent if and only if $[E \mid d]$ has no row $(0 \cdots 0 \mid c)$ with $c \neq 0$, that is, if and only if the last column of $[E \mid d]$ is not a pivot column.

Proof. • By Theorem 2.3, $Ax = b$ and $Ex = d$ have the same solutions. In an echelon form, a row $(0 \cdots 0 \mid c)$ with $c \neq 0$ is exactly a row whose pivot lies in the last column.

- If $[E \mid d]$ has such a row, the corresponding equation reads $0 = c$, and no x satisfies it. The same holds if such a row appears at any stage of the elimination, so one may stop there.
- Suppose there is no such row. Then every zero row of E is a zero row of $[E \mid d]$, and its equation reads $0 = 0$. Let rows $1, \dots, r$ be the nonzero rows of E , with pivots in columns $k_1 < \dots < k_r$; call $x_{k_1}, \dots, x_{k_r}$ the pivot unknowns and the others free, as in Proposition 2.7.
- Give the free unknowns any values. Row i of $E = (e_{ij})$ is 0 before column k_i and 1 there, so for $i = r, r-1, \dots, 1$ equation i reads $x_{k_i} + \sum_{j > k_i} e_{ij}x_j = d_i$. Each x_j with $j > k_i$ is free or the pivot unknown of a lower row, already found; so the equation determines x_{k_i} (**back substitution**). The resulting x solves $Ex = d$, hence $Ax = b$.

- Every solution arises in this way from the values of its free unknowns, since the equations force the pivot unknowns. $\square$

Theorem 2.20. Let $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.

- If x_p is one solution of $Ax = b$ (a **particular solution**), then the solutions of $Ax = b$ are exactly the vectors $x_p + z$ with $Az = 0$.
- $Ax = b$ has no solution, exactly one solution, or infinitely many solutions. If it is consistent, it has exactly one solution if and only if there are no free unknowns, that is, every column of E in Theorem 2.19 is a pivot column.
- If A is square, $Ax = b$ has exactly one solution if and only if A is invertible; this solution is $x = A^{-1}b$.

Proof. (a) If $Az = 0$, then $A(x_p + z) = Ax_p + Az = b$. Conversely, if $Ax = b$, then $z = x - x_p$ satisfies $Az = b - b = 0$, and $x = x_p + z$.

(b) Suppose $Ax = b$ is consistent, and take $[E \mid d]$ as in Theorem 2.19.

- By the proof of that theorem, each choice of values of the free unknowns gives exactly one solution, every solution arises so, and different choices give different solutions (they differ in a free unknown, an entry of x).
 - Without free unknowns there is therefore exactly one solution; with a free unknown, each of its values gives a different solution, so there are infinitely many.
- (c) • If A is invertible, $x = A^{-1}b$ is a solution, since $A(A^{-1}b) = b$; and every solution satisfies $x = A^{-1}(Ax) = A^{-1}b$.
- If $Ax = b$ has exactly one solution x_p , then $Az = 0$ only for $z = 0$, since otherwise $x_p + z$ would be a second solution by (a). So A is invertible by Theorem 2.15 ((iii) $\Rightarrow$ (i)). $\square$

So, to solve $Ax = b$: reduce $[A \mid b]$ by forward elimination to an echelon form; if a row $(0 \cdots 0 \mid c)$ with $c \neq 0$ appears, there is no solution; otherwise choose the free unknowns arbitrarily and find the pivot unknowns by back substitution (or continue to the RREF and use Proposition 2.7).

Example 2.21. Solve the system $x_1 - x_2 + x_3 = 1$, $-x_1 + x_2 - 2x_3 = 0$, $2x_1 - x_2 + 2x_3 = 3$.

Solution. The coefficient matrix is the matrix A of Example 2.16, and $b = (1, 0, 3)'$. Forward elimination of $[A \mid b]$ uses the operations of the forward elimination there:

$$\begin{aligned} & \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ -1 & 1 & -2 & 0 \\ 2 & -1 & 2 & 3 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 + R_1} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 1 & 0 & 1 \end{array} \right) \\ & \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & -1 & 1 \end{array} \right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{ccc|c} 1 & -1 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & -1 \end{array} \right) \end{aligned}$$

- The last matrix is an echelon form with no row $(0 \ 0 \ 0 \mid c)$, $c \neq 0$, so the system is consistent (Theorem 2.19).
- Every column of the left part is a pivot column, so there are no free unknowns and the solution is unique (Theorem 2.20(b)).
- Back substitution: $x_3 = -1$; $x_2 = 1$; $x_1 = 1 + x_2 - x_3 = 1 + 1 + 1 = 3$. So $x = (3, 1, -1)'$.
- Check, with A^{-1} from Example 2.16 (Theorem 2.20(c)): $A^{-1}b = \begin{pmatrix} 0 & 1 & 1 \\ -2 & 0 & 1 \\ -1 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -1 \end{pmatrix}$.

Example 2.22. Solve the system $x_1 + x_2 + x_3 - 2x_4 = 2$, $x_1 + x_2 + 2x_3 - 3x_4 = 3$, $2x_1 + x_2 + 5x_3 - 4x_4 = 7$.

Solution. The coefficient matrix is the matrix A of Example 2.6, and $b = (2, 3, 7)'$. Forward elimination of $[A \mid b]$ uses the same operations as there:

$$\begin{aligned} & \left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 1 & 1 & 2 & -3 & 3 \\ 2 & 1 & 5 & -4 & 7 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 0 & 0 & 1 & -1 & 1 \\ 0 & -1 & 3 & 0 & 3 \end{array} \right) \\ & \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 0 & -1 & 3 & 0 & 3 \\ 0 & 0 & 1 & -1 & 1 \end{array} \right) \xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 2 \\ 0 & 1 & -3 & 0 & -3 \\ 0 & 0 & 1 & -1 & 1 \end{array} \right) \end{aligned}$$

- The last matrix is an echelon form with no row $(0 \ 0 \ 0 \ 0 \mid c)$, $c \neq 0$: the system is consistent (Theorem 2.19).
- The pivot columns are 1, 2, 3, so x_4 is free and there are infinitely many solutions (Theorem 2.20(b)).
- Back substitution with $x_4 = t$: $x_3 = 1 + t$; $x_2 = -3 + 3x_3 = 3t$; $x_1 = 2 - x_2 - x_3 + 2x_4 = 2 - 3t - (1 + t) + 2t = 1 - 2t$. So

$$x = \begin{pmatrix} 1 - 2t \\ 3t \\ 1 + t \\ t \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} -2 \\ 3 \\ 1 \\ 1 \end{pmatrix} \quad (t \in \mathbb{R})$$

- This is the particular solution $x_p = (1, 0, 1, 0)'$ plus the solutions $t(-2, 3, 1, 1)'$ of $Ax = 0$ found in Example 2.6, as Theorem 2.20(a) says.

Example 2.23. Show that the system $x_1 - 2x_2 + x_3 + 2x_4 = 1$, $x_1 - x_2 - x_3 + x_4 = 2$, $x_1 + x_2 - 5x_3 - x_4 = 3$ has no solution.

Solution. Forward elimination of the augmented matrix $[A \mid b]$, $b = (1, 2, 3)'$:

$$\begin{aligned} & \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & 1 \\ 1 & -1 & -1 & 1 & 2 \\ 1 & 1 & -5 & -1 & 3 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & 1 \\ 0 & 1 & -2 & -1 & 1 \\ 0 & 3 & -6 & -3 & 2 \end{array} \right) \\ & \xrightarrow{R_3 \leftarrow R_3 - 3R_2} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & 1 \\ 0 & 1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 0 & -1 \end{array} \right) \xrightarrow{R_3 \leftarrow -R_3} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & 1 \\ 0 & 1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right) \end{aligned}$$

The last matrix is an echelon form whose last column is a pivot column: row 3 reads $0 = 1$. So the system has no solution (Theorem 2.19). Indeed, the coefficient matrix A has $\text{row}_3(A) = 3\text{row}_2(A) - 2\text{row}_1(A)$, while $3b_2 - 2b_1 = 4 \neq 3 = b_3$: the third equation contradicts the first two.

Exercises and solutions

Exercise 2.1. Reduce the matrix $A = \begin{pmatrix} 0 & 0 & 1 & 2 & 3 \\ 2 & 4 & -2 & 0 & 4 \\ 2 & 4 & -1 & 2 & 7 \end{pmatrix}$ to echelon form by elementary row operations, and record the elementary matrices used.

Solution. Follow Definition 2.5, postponing the scaling (Step 3) to the end. Column 1 is the first pivot column; since $a_{11} = 0$, interchange rows 1 and 2; then clear column 1 below the pivot, and then column 3, the next pivot column:

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 2 & 4 & -1 & 2 & 7 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 & 3 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 2 & 4 & -2 & 0 & 4 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Finally divide row 1 by 2:

$$H = \begin{pmatrix} 1 & 2 & -1 & 0 & 2 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} = D_1\left(\frac{1}{2}\right) T_{32}(-1) T_{31}(-1) P_{12} A$$

(Theorem 2.14(c)). H is in echelon form with pivot columns 1 and 3. One more operation, $R_1 \leftarrow R_1 + R_2$, clears the entry above the second pivot and gives the RREF $\begin{pmatrix} 1 & 2 & 0 & 2 & 5 \\ 0 & 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$.

Exercise 2.2. Find the solutions of the equations $x_1 + 3x_2 + 2x_3 = 0$, $2x_1 + 7x_2 + 3x_3 = 0$, $x_1 + 4x_2 + x_3 = 0$.

Solution. Row-reduce the coefficient matrix:

$$\begin{pmatrix} 1 & 3 & 2 \\ 2 & 7 & 3 \\ 1 & 4 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 3R_2} \begin{pmatrix} 1 & 0 & 5 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

Column 3 has no pivot, so x_3 is a free unknown (Proposition 2.7): with $x_3 = \lambda$, the nonzero rows give $x_1 = -5\lambda$, $x_2 = \lambda$. All solutions are $x = \lambda(-5, 1, 1)'$, $\lambda \in \mathbb{R}$; in particular the system has nontrivial solutions, although it has as many equations as unknowns (the third equation is the second minus the first).

Exercise 2.3. Solve the homogeneous system

$$\begin{aligned} x_1 - x_2 + 2x_3 + x_4 &= 0 \\ x_1 - x_3 + 3x_4 &= 0 \\ 3x_1 - 2x_2 + 3x_3 + 5x_4 &= 0 \\ x_1 + x_2 - 4x_3 + 5x_4 &= 0 \end{aligned}$$

Solution. Transform the coefficient matrix to echelon form:

$$\begin{pmatrix} 1 & -1 & 2 & 1 \\ 1 & 0 & -1 & 3 \\ 3 & -2 & 3 & 5 \\ 1 & 1 & -4 & 5 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1, R_3 \leftarrow R_3 - 3R_1} \begin{pmatrix} 1 & -1 & 2 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 1 & -3 & 2 \\ 0 & 2 & -6 & 4 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 2R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & -1 & 2 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

By Theorem 2.3, the system has the same solutions as the two equations $x_1 - x_2 + 2x_3 + x_4 = 0$ and $x_2 - 3x_3 + 2x_4 = 0$. For any values of x_3 and x_4 , the second equation determines x_2 and then the first determines x_1 ; so x_3 and x_4 are free. Let $x_3 = \lambda$ and $x_4 = \mu$. Then $x_2 = 3x_3 - 2x_4 = 3\lambda - 2\mu$ and $x_1 = x_2 - 2x_3 - x_4 = \lambda - 3\mu$. Hence

$$x = \lambda(1, 3, 1, 0)' + \mu(-3, -2, 0, 1)' \quad (\lambda, \mu \in \mathbb{R})$$

(One more step, $R_1 \leftarrow R_1 + R_2$, gives the RREF, with rows $(1, 0, -1, 3)$ and $(0, 1, -3, 2)$, from which the same solutions are read off.)

Exercise 2.4. Solve the equations

$$\begin{pmatrix} 1 & -3 & -2 \\ -1 & 2 & 1 \\ -1 & 2 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 1 & -2 & 0 \\ 1 & -2 & -1 \\ 1 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}$$

by Gaussian elimination.

Solution. First system. Forward elimination:

$$\begin{pmatrix} 1 & -3 & -2 & | & 1 \\ -1 & 2 & 1 & | & -1 \\ -1 & 2 & 2 & | & -2 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} 1 & -3 & -2 & | & 1 \\ 0 & -1 & -1 & | & 0 \\ 0 & -1 & 0 & | & -1 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & -3 & -2 & | & 1 \\ 0 & 1 & 1 & | & 0 \\ 0 & -1 & 0 & | & -1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & -3 & -2 & | & 1 \\ 0 & 1 & 1 & | & 0 \\ 0 & 0 & 1 & | & -1 \end{pmatrix}$$

Backward elimination:

$$\xrightarrow[R_1 \leftarrow R_1 + 2R_3]{R_2 \leftarrow R_2 - R_3} \begin{pmatrix} 1 & -3 & 0 & | & -1 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & -1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + 3R_2} \begin{pmatrix} 1 & 0 & 0 & | & 2 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & -1 \end{pmatrix}$$

By Proposition 2.7, the unique solution is $(x_1, x_2, x_3) = (2, 1, -1)$. Check: $2 - 3 + 2 = 1$, $-2 + 2 - 1 = -1$, $-2 + 2 - 2 = -2$.

Second system. Forward elimination; after the first step the entry of row 2 in column 2 is 0, so rows 2 and 3 are interchanged (Step 2):

$$\begin{pmatrix} 1 & -2 & 0 & | & 3 \\ 1 & -2 & -1 & | & 1 \\ 1 & -1 & 1 & | & 4 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & -2 & 0 & | & 3 \\ 0 & 0 & -1 & | & -2 \\ 0 & 1 & 1 & | & 1 \end{pmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & -2 & 0 & | & 3 \\ 0 & 1 & 1 & | & 1 \\ 0 & 0 & -1 & | & -2 \end{pmatrix} \xrightarrow{R_3 \leftarrow -R_3} \begin{pmatrix} 1 & -2 & 0 & | & 3 \\ 0 & 1 & 1 & | & 1 \\ 0 & 0 & 1 & | & 2 \end{pmatrix}$$

Backward elimination:

$$\xrightarrow{R_2 \leftarrow R_2 - R_3} \begin{pmatrix} 1 & -2 & 0 & | & 3 \\ 0 & 1 & 0 & | & -1 \\ 0 & 0 & 1 & | & 2 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \begin{pmatrix} 1 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & | & -1 \\ 0 & 0 & 1 & | & 2 \end{pmatrix}$$

The unique solution is $(x_1, x_2, x_3) = (1, -1, 2)$. Check: $1 + 2 + 0 = 3$, $1 + 2 - 2 = 1$, $1 + 1 + 2 = 4$.

Exercise 2.5. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}$. Find an elementary matrix E with $B = EA$ when: (a) B is obtained

from A by interchanging the first and third rows; (b) B is obtained from A by multiplying the second row by 7; (c) B is obtained from A by adding five times the second row to the third row.

Solution. Perform each operation on I_3 (Theorem 2.14(a)):

$$P_{13} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad D_2(7) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad T_{32}(5) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & 1 \end{pmatrix}$$

and

$$P_{13}A = \begin{pmatrix} 5 & 6 \\ 3 & 4 \\ 1 & 2 \end{pmatrix}, \quad D_2(7)A = \begin{pmatrix} 1 & 2 \\ 21 & 28 \\ 5 & 6 \end{pmatrix}, \quad T_{32}(5)A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 20 & 26 \end{pmatrix}$$

Exercise 2.6. Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 1 & 0 & 1 \end{pmatrix}$.

- (a) Find the matrix E such that EA is obtained from A by first interchanging rows 1 and 2 and then multiplying row 2 by 2. Does the order of the two operations matter?
- (b) Find the matrix E such that EA is obtained from A by first interchanging rows 1 and 2 and then adding 3 times row 1 to row 2. Does the order matter?
- (c) Do elementary matrices commute?

Solution. By Theorem 2.14, the operation done first is the factor next to A .

$$(a) E = D_2(2)P_{12} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \text{ and}$$

$$D_2(2)P_{12}A = \begin{pmatrix} 0 & 1 & 2 \\ 2 & 4 & 6 \\ 1 & 0 & 1 \end{pmatrix} \neq \begin{pmatrix} 0 & 2 & 4 \\ 1 & 2 & 3 \\ 1 & 0 & 1 \end{pmatrix} = P_{12}D_2(2)A,$$

since $P_{12}D_2(2) = \begin{pmatrix} 0 & 2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. The order matters.

(b) $E = T_{21}(3)P_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, while $P_{12}T_{21}(3) = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. Accordingly $EA = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 5 & 9 \\ 1 & 0 & 1 \end{pmatrix}$ but $P_{12}T_{21}(3)A = \begin{pmatrix} 3 & 7 & 11 \\ 1 & 2 & 3 \\ 1 & 0 & 1 \end{pmatrix}$: the order matters.

(c) No: (a) and (b) give pairs of elementary matrices that do not commute.

Exercise 2.7. Show that the product of two elementary matrices is, in general, not elementary.

Solution. For example,

$$P_{13}D_2(7) = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 7 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

This is not elementary: a matrix $D_r(c)$ or $T_{rs}(c)$ differs from I_3 in one entry only, while this product differs from I_3 in five entries; and every P_{rs} has all entries 0 or 1, while this product has the entry 7.

Exercise 2.8. Use row reduction of $[A \mid I_3]$ to find the inverse of $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 3 & 4 & 3 \end{pmatrix}$.

Solution.

$$\begin{aligned} & \left(\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 3 & 4 & 3 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 3 & 4 & 3 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 - 3R_1} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -3 & 1 \end{array} \right) \\ & \xrightarrow[\begin{smallmatrix} R_3 \leftarrow R_3 - R_2 \\ R_3 \leftarrow -R_3 \end{smallmatrix}]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 3 & -1 \end{array} \right) \xrightarrow[\begin{smallmatrix} R_2 \leftarrow R_2 - R_3 \\ R_1 \leftarrow R_1 - R_3 \end{smallmatrix}]{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & -1 & -2 & 1 \\ 0 & 1 & 0 & 0 & -3 & 1 \\ 0 & 0 & 1 & 1 & 3 & -1 \end{array} \right) \\ & \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 1 & 0 & 0 & -3 & 1 \\ 0 & 0 & 1 & 1 & 3 & -1 \end{array} \right) \end{aligned}$$

Hence $A^{-1} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -3 & 1 \\ 1 & 3 & -1 \end{pmatrix}$. Check: $\text{row}_3(A)A^{-1} = (3, 4, 3)A^{-1} = (-3 + 3, 3 - 12 + 9, 4 - 3) = (0, 0, 1)$, and similarly for the other rows.

Exercise 2.9. Do $A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 3 & 1 \\ 2 & 5 & 1 \\ 1 & 2 & 1 \end{pmatrix}$ have inverses? Find the inverse when it exists.

Solution. (A) Row-reduce A :

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 3 \\ 3 & 6 & 4 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - 3R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

A zero row appears, so the RREF of A is not I_3 , and A has no inverse (Theorem 2.15). (Indeed row 3 of A is the sum of rows 1 and 2.)

(B) Row-reduce $[B \mid I_3]$:

$$\begin{aligned} & \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 2 & 5 & 1 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & -1 & 0 & -1 & 0 & 1 \end{array} \right) \\ & \xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & -1 & 0 & -1 & 0 & 1 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{ccc|ccc} 1 & 3 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right) \\ & \xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 - R_3} \left(\begin{array}{ccc|ccc} 1 & 3 & 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 - 3R_2} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 1 & 2 \\ 0 & 1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array} \right) \end{aligned}$$

The left block is I_3 , so B is invertible and $B^{-1} = \begin{pmatrix} -3 & 1 & 2 \\ 1 & 0 & -1 \\ 1 & -1 & 1 \end{pmatrix}$ (Theorem 2.15). Check: the first row of BB^{-1} is $(-3 + 3 + 1, 1 + 0 - 1, 2 - 3 + 1) = (1, 0, 0)$, and similarly for the other rows.

Exercise 2.10. Let $A \in \mathbb{R}^{n \times n}$ be invertible. Show that:

- (a) $(\lambda A)^{-1} = \frac{1}{\lambda} A^{-1}$ ($\lambda \neq 0$);
- (b) if A and B commute and B is invertible, then A and B^{-1} commute.

Solution. (a) We check both products (Definition 2.11): $(\lambda A)(\frac{1}{\lambda} A^{-1}) = AA^{-1} = I$ and $(\frac{1}{\lambda} A^{-1})(\lambda A) = A^{-1}A = I$. (b) Multiply $AB = BA$ on the left and on the right by B^{-1} : $B^{-1}ABB^{-1} = B^{-1}BAB^{-1}$, that is, $B^{-1}A = AB^{-1}$.

Exercise 2.11. Let $A \in \mathbb{R}^{n \times n}$.

- (a) If $A^2 = 0$, show that $I - A$ is invertible.
- (b) If $A^3 = 0$, show that $I - A$ is invertible.
- (c) In general, if $A^k = 0$ for some positive integer k , show that $I - A$ is invertible.
- (d) Suppose that $A^2 + 2A + I = 0$. Show that A is invertible.
- (e) Suppose that $A^3 - A + I = 0$. Show that A is invertible.

Solution. (a)–(c) For every positive integer k , by Theorem 1.3,

$$(I - A)(I + A + A^2 + \cdots + A^{k-1}) = (I + A + \cdots + A^{k-1}) - (A + A^2 + \cdots + A^k) = I - A^k,$$

and in the same way $(I + A + \cdots + A^{k-1})(I - A) = I - A^k$. If $A^k = 0$, both products equal I , so $I - A$ is invertible with inverse $I + A + \cdots + A^{k-1}$. Parts (a) and (b) are the cases $k = 2$ and $k = 3$: $(I - A)^{-1} = I + A$ and $I + A + A^2$, respectively.

(d) $A(-A - 2I) = (-A - 2I)A = -A^2 - 2A = I$, since $A^2 + 2A + I = 0$. So $A^{-1} = -A - 2I$.

(e) $A(-A^2 + I) = (-A^2 + I)A = -A^3 + A = I$, since $A^3 - A + I = 0$. So $A^{-1} = I - A^2$.

Exercise 2.12. Find the inverses of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ ($ad - bc \neq 0$) and of $\begin{pmatrix} 1 & 1 & 2 \\ 3 & 2 & 3 \\ 3 & 2 & 4 \end{pmatrix}$.

Solution. (2×2) Put $\Delta = ad - bc$. If $a \neq 0$,

$$\begin{pmatrix} a & b & | & 1 & 0 \\ c & d & | & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \leftarrow \frac{1}{a} R_1} \begin{pmatrix} 1 & \frac{b}{a} & | & \frac{1}{a} & 0 \\ c & d & | & 0 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - c R_1} \begin{pmatrix} 1 & \frac{b}{a} & | & \frac{1}{a} & 0 \\ 0 & \frac{d}{a} & | & -\frac{c}{a} & 1 \end{pmatrix} \\ \xrightarrow{R_2 \leftarrow \frac{a}{\Delta} R_2} \begin{pmatrix} 1 & \frac{b}{a} & | & \frac{1}{a} & 0 \\ 0 & 1 & | & -\frac{c}{\Delta} & \frac{a}{\Delta} \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - \frac{b}{a} R_2} \begin{pmatrix} 1 & 0 & | & \frac{d}{\Delta} & -\frac{b}{\Delta} \\ 0 & 1 & | & -\frac{c}{\Delta} & \frac{a}{\Delta} \end{pmatrix}$$

using $\frac{1}{a} + \frac{bc}{a\Delta} = \frac{\Delta + bc}{a\Delta} = \frac{d}{\Delta}$. The result

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

also holds when $a = 0$: in all cases $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} ad - bc & 0 \\ 0 & ad - bc \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

(3×3)

$$\begin{pmatrix} 1 & 1 & 2 & | & 1 & 0 & 0 \\ 3 & 2 & 3 & | & 0 & 1 & 0 \\ 3 & 2 & 4 & | & 0 & 0 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - 3R_1]{R_2 \leftarrow R_2 - 3R_1} \begin{pmatrix} 1 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & -1 & -3 & | & -3 & 1 & 0 \\ 0 & -1 & -2 & | & -3 & 0 & 1 \end{pmatrix} \\ \xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 1 & 3 & | & 3 & -1 & 0 \\ 0 & -1 & -2 & | & -3 & 0 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 1 & 2 & | & 1 & 0 & 0 \\ 0 & 1 & 3 & | & 3 & -1 & 0 \\ 0 & 0 & 1 & | & 0 & -1 & 1 \end{pmatrix} \\ \xrightarrow[R_1 \leftarrow R_1 - 2R_3]{R_2 \leftarrow R_2 - 3R_3} \begin{pmatrix} 1 & 1 & 0 & | & 1 & 2 & -2 \\ 0 & 1 & 0 & | & 3 & 2 & -3 \\ 0 & 0 & 1 & | & 0 & -1 & 1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 & | & -2 & 0 & 1 \\ 0 & 1 & 0 & | & 3 & 2 & -3 \\ 0 & 0 & 1 & | & 0 & -1 & 1 \end{pmatrix}$$

So the inverse is $\begin{pmatrix} -2 & 0 & 1 \\ 3 & 2 & -3 \\ 0 & -1 & 1 \end{pmatrix}$. Check: the first row $(1, 1, 2)$ of the matrix times the inverse is $(-2 + 3 + 0, 0 + 2 - 2, 1 - 3 + 2) = (1, 0, 0)$, and similarly for the other rows.

Exercise 2.13. For the four units of Example 1.14 a response y_i is also measured: $y = (9, 3, 5, 3)'$. With $X'X$ as in Example 2.17, compute $X'y$ and solve the normal equations $X'X\hat{\beta} = X'y$ for the least-squares estimate $\hat{\beta} = (\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2)'$ by Gauss–Jordan elimination; check the answer with $\hat{\beta} = (X'X)^{-1}X'y$.

Solution. The j th entry of $X'y$ is $\text{col}_j(X)'y$ (Theorem 1.4(d)):

$$X'y = \begin{pmatrix} \sum_i y_i \\ \sum_i x_{i1} y_i \\ \sum_i x_{i2} y_i \end{pmatrix} = \begin{pmatrix} 9 + 3 + 5 + 3 \\ 2 \cdot 9 + 0 \cdot 3 + 2 \cdot 5 + 0 \cdot 3 \\ 3 \cdot 9 + 3 \cdot 3 + 1 \cdot 5 + 1 \cdot 3 \end{pmatrix} = \begin{pmatrix} 20 \\ 28 \\ 44 \end{pmatrix}$$

As in Example 2.17, postpone the scaling to the end:

$$\begin{pmatrix} 4 & 4 & 8 & | & 20 \\ 4 & 8 & 8 & | & 28 \\ 8 & 8 & 20 & | & 44 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 4 & 4 & 8 & | & 20 \\ 0 & 4 & 0 & | & 8 \\ 0 & 0 & 4 & | & 4 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - 2R_3} \begin{pmatrix} 4 & 4 & 0 & | & 12 \\ 0 & 4 & 0 & | & 8 \\ 0 & 0 & 4 & | & 4 \end{pmatrix} \\ \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 4 & 0 & 0 & | & 4 \\ 0 & 4 & 0 & | & 8 \\ 0 & 0 & 4 & | & 4 \end{pmatrix} \xrightarrow[R_3 \leftarrow \frac{1}{4} R_3]{R_1 \leftarrow \frac{1}{4} R_1, R_2 \leftarrow \frac{1}{4} R_2} \begin{pmatrix} 1 & 0 & 0 & | & 1 \\ 0 & 1 & 0 & | & 2 \\ 0 & 0 & 1 & | & 1 \end{pmatrix}$$

So $\hat{\beta} = (1, 2, 1)'$ is the unique solution. Check:

$$(X'X)^{-1}X'y = \frac{1}{4} \begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 20 \\ 28 \\ 44 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 120 - 28 - 88 \\ -20 + 28 \\ -40 + 44 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 4 \\ 8 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

Exercise 2.14. Let A be the coefficient matrix of Example 2.23.

- (a) For which $b = (b_1, b_2, b_3)'$ is $Ax = b$ consistent?
 (b) Find all solutions of $Ax = b$ for $b = (1, 2, 4)'$.

Solution. (a) Row-reduce $[A \mid b]$, keeping b_1, b_2, b_3 as letters:

$$\begin{aligned} & \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & b_1 \\ 1 & -1 & -1 & 1 & b_2 \\ 1 & 1 & -5 & -1 & b_3 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & b_1 \\ 0 & 1 & -2 & -1 & b_2 - b_1 \\ 0 & 3 & -6 & -3 & b_3 - b_1 \end{array} \right) \\ & \xrightarrow{R_3 \leftarrow R_3 - 3R_2} \left(\begin{array}{cccc|c} 1 & -2 & 1 & 2 & b_1 \\ 0 & 1 & -2 & -1 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 - 3b_2 + 2b_1 \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 + 2R_2} \left(\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 2b_2 - b_1 \\ 0 & 1 & -2 & -1 & b_2 - b_1 \\ 0 & 0 & 0 & 0 & b_3 - 3b_2 + 2b_1 \end{array} \right) \end{aligned}$$

The left part of the last matrix is the RREF of A , with $r = 2$. By Proposition 2.7(a), $Ax = b$ is consistent if and only if $2b_1 - 3b_2 + b_3 = 0$. (In Example 2.23, $2 - 6 + 3 = -1 \neq 0$.)

(b) For $b = (1, 2, 4)'$, $2 - 6 + 4 = 0$, so the system is consistent, and the last matrix becomes

$$\left(\begin{array}{cccc|c} 1 & 0 & -3 & 0 & 3 \\ 0 & 1 & -2 & -1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

The free unknowns are $x_3 = s$, $x_4 = t$, and by Proposition 2.7(b) $x_1 = 3 + 3s$, $x_2 = 1 + 2s + t$. So

$$x = \begin{pmatrix} 3 \\ 1 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} 3 \\ 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \quad (s, t \in \mathbb{R}),$$

infinitely many solutions (Theorem 2.20(b)): the particular solution $(3, 1, 0, 0)'$ plus the solutions of $Ax = 0$ (Theorem 2.20(a)). Check: $A(3, 1, 0, 0)' = (3 - 2, 3 - 1, 3 + 1)' = (1, 2, 4)'$.

Exercise 2.15. Let

$$A = \begin{pmatrix} 1 & 3 & 5 & -1 \\ 1 & 2 & 2 & 0 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 1 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 6 \\ 4 \\ 2 \\ 0 \end{pmatrix}, \quad c = \begin{pmatrix} 6 \\ 4 \\ 3 \\ 0 \end{pmatrix}$$

- (a) Find all solutions of $Ax = b$.
 (b) Show that $Ax = c$ has no solution.

Solution. (a) Row-reduce $[A \mid b]$:

$$\begin{aligned} & \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 1 & 2 & 2 & 0 & 4 \\ 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 1 & -1 & 0 \end{array} \right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 0 & -1 & -3 & 1 & -2 \\ 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 1 & -1 & 0 \end{array} \right) \xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 0 & 1 & 3 & -1 & 2 \\ 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 1 & -1 & 0 \end{array} \right) \\ & \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 \end{array} \right) \xrightarrow{R_3 \leftrightarrow R_4} \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \\ & \xrightarrow[R_1 \leftarrow R_1 - 5R_3]{R_2 \leftarrow R_2 - 3R_3} \left(\begin{array}{cccc|c} 1 & 3 & 0 & 4 & 6 \\ 0 & 1 & 0 & 2 & 2 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 - 3R_2} \left(\begin{array}{cccc|c} 1 & 0 & 0 & -2 & 0 \\ 0 & 1 & 0 & 2 & 2 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \end{aligned}$$

After the third step the entry of row 3 in column 3 is 0, so Step 2 interchanges rows 3 and 4. The left part of the last matrix is the RREF R of A , with $r = 3$ and pivot columns 1, 2, 3; the unknown x_4 is free. The last matrix is $[R \mid z]$ with $z = (0, 2, 0, 0)'$. As $z_4 = 0$, the system is consistent (Proposition 2.7(a)), and with $x_4 = t$, Proposition 2.7(b) gives $x_1 = 2t$, $x_2 = 2 - 2t$, $x_3 = t$:

$$x = \begin{pmatrix} 0 \\ 2 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} 2 \\ -2 \\ 1 \\ 1 \end{pmatrix} \quad (t \in \mathbb{R}),$$

infinitely many solutions (Theorem 2.20(a),(b)). Check: $A(0, 2, 0, 0)' = 2 \operatorname{col}_2(A) = (6, 4, 2, 0)' = b$ and $A(2, -2, 1, 1)' = (2 - 6 + 5 - 1, 2 - 4 + 2, -2 + 3 - 1, 1 - 1)' = 0$.

(b) Row-reduce $[A \mid c]$ by the same first three operations:

$$\begin{aligned} & \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 1 & 2 & 2 & 0 & 4 \\ 0 & 1 & 3 & -1 & 3 \\ 0 & 0 & 1 & -1 & 0 \end{array} \right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 0 & -1 & -3 & 1 & -2 \\ 0 & 1 & 3 & -1 & 3 \\ 0 & 0 & 1 & -1 & 0 \end{array} \right) \\ & \xrightarrow{R_2 \leftarrow -R_2} \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 0 & 1 & 3 & -1 & 2 \\ 0 & 1 & 3 & -1 & 3 \\ 0 & 0 & 1 & -1 & 0 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|c} 1 & 3 & 5 & -1 & 6 \\ 0 & 1 & 3 & -1 & 2 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & -1 & 0 \end{array} \right) \end{aligned}$$

The third row of the last matrix reads $(0 \ 0 \ 0 \ 0 \mid 1)$, that is, $0 = 1$; so $Ax = c$ has no solution (Theorem 2.19). Indeed $\operatorname{row}_3(A) = \operatorname{row}_1(A) - \operatorname{row}_2(A)$, while $c_3 = 3 \neq 2 = c_1 - c_2$.

Exercise 2.16. For which values of k and h does the system

$$x_1 + x_2 + x_3 = 1, \quad x_1 + 2x_2 + 3x_3 = 3, \quad x_1 + 3x_2 + kx_3 = h$$

have no solution, exactly one solution, or infinitely many solutions? Give the solutions when they exist.

Solution. Forward elimination:

$$\left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 3 \\ 1 & 3 & k & h \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 2 & k-1 & h-1 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & k-5 & h-5 \end{array} \right)$$

- $k \neq 5$: $R_3 \leftarrow \frac{1}{k-5}R_3$ gives an echelon form with pivots in columns 1, 2, 3 and none in the last column. So there is exactly one solution (Theorems 2.19 and 2.20(b)): back substitution gives $x_3 = \frac{h-5}{k-5}$, $x_2 = 2 - 2x_3$, $x_1 = 1 - x_2 - x_3 = -1 + x_3$, that is, $x = (-1, 2, 0)' + \frac{h-5}{k-5}(1, -2, 1)'$.
- $k = 5$, $h \neq 5$: row 3 reads $0 = h - 5 \neq 0$, so there is no solution.
- $k = 5$, $h = 5$: row 3 is 0, and the first two rows form an echelon form with pivots in columns 1, 2; $x_3 = t$ is free, so there are infinitely many solutions: $x_2 = 2 - 2t$, $x_1 = 1 - x_2 - x_3 = -1 + t$, that is, $x = (-1, 2, 0)' + t(1, -2, 1)'$ ($t \in \mathbb{R}$).

Chapter 3. Vector spaces, subspaces, span, and bases

Vector spaces

Definition 3.1. A (real) **vector space** is a set V , whose elements are called **vectors**, with an **addition** that gives $u + v \in V$ for $u, v \in V$ and a **scalar multiplication** that gives $cv \in V$ for $c \in \mathbb{R}$ and $v \in V$, such that for all $u, v, w \in V$ and $c, d \in \mathbb{R}$:

- (V1) $u + v = v + u$ and (V2) $u + (v + w) = (u + v) + w$;
- (V3) there is a **zero vector** $0 \in V$ with $v + 0 = v$ for every $v \in V$;
- (V4) for each $v \in V$ there is $-v \in V$ with $v + (-v) = 0$;
- (V5) $1v = v$ and (V6) $(cd)v = c(dv)$;
- (V7) $c(u + v) = cu + cv$ and (V8) $(c + d)v = cv + dv$.

A vector $c_1v_1 + \cdots + c_kv_k$ ($c_i \in \mathbb{R}$) is a **linear combination** of $v_1, \dots, v_k$.

From these rules, $0v = 0$ and $(-1)v = -v$: indeed $0v = (0 + 0)v = 0v + 0v$, and adding $-(0v)$ to both sides gives $0 = 0v$; then $v + (-1)v = (1 + (-1))v = 0v = 0$, and $-v$ is the only vector w with $v + w = 0$, since such a w equals $w + (v + (-v)) = (w + v) + (-v) = 0 + (-v) = -v$. Likewise $c0 = 0$, and $cv = 0$ only if $c = 0$ or $v = 0$ (if $c \neq 0$, then $v = c^{-1}(cv) = c^{-1}0 = 0$).

Example 3.2. $\mathbb{R}^n$, $\mathbb{R}^{1 \times n}$ and $\mathbb{R}^{m \times n}$, with the entrywise sum and scalar multiple of Definition 1.1, are vector spaces: (V1)–(V8) hold entry by entry, the zero vector is the zero matrix, and $-A = (-1)A$. In this course every vector space is one of these or a subspace of one of them.

Subspaces

Definition 3.3. A **subspace** of a vector space V is a subset $W \subseteq V$ that is itself a vector space with the addition and scalar multiplication of V .

Theorem 3.4. For a subset W of a vector space V the following are equivalent:

- (a) W is a subspace of V ;
- (b) $0 \in W$, and $u + v \in W$ and $cu \in W$ for all $u, v \in W$ and $c \in \mathbb{R}$ (W is **closed under addition and scalar multiplication**);
- (c) $0 \in W$, and $cu + v \in W$ for all $u, v \in W$ and $c \in \mathbb{R}$.

Proof. (a) $\Rightarrow$ (b): the operations of the vector space W are those of V and give vectors of W ; W has a zero vector, so it contains some w , and then $0 = 0w \in W$. (b) $\Rightarrow$ (c): $cu \in W$, so $cu + v \in W$. (c) $\Rightarrow$ (a): taking $v = 0$ gives $cu \in W$, and $c = 1$ gives $u + v \in W$; so the operations of V give operations on W . The rules (V1), (V2), (V5)–(V8) hold in W because they hold in V ; (V3) holds since $0 \in W$, and (V4) since $-w = (-1)w \in W$. $\square$

So, to show that W is a subspace, check that it contains 0 and is closed under the two operations; to show that it is not, exhibit one failure.

Example 3.5. (a) $\{0\}$ and V itself are subspaces of V .

- (b) In $\mathbb{R}^n$, $W = \{x : x_1 = 0\}$ is a subspace: $0 \in W$, and $(cx + y)_1 = cx_1 + y_1 = 0$ for $x, y \in W$. For $n \geq 2$, $\{x : x_1 = 1 + x_2\}$ is not, since it does not contain 0 .
- (c) In $\mathbb{R}^2$, $\{(x, y)' : 2x - 5y = 0\}$ is a subspace, since $2(cx_1 + x_2) - 5(cy_1 + y_2) = c(2x_1 - 5y_1) + (2x_2 - 5y_2)$ for $(x_1, y_1)'$ and $(x_2, y_2)'$ in it; but for $k \neq 0$ the set $\{(x, y)' : ax + by = k\}$ is not, since it does not contain 0 .

- (d) The symmetric $n \times n$ matrices form a subspace of $\mathbb{R}^{n \times n}$: $0' = 0$, and $(cA+B)' = cA' + B' = cA+B$ for symmetric A, B . So do the **skew-symmetric** matrices, those with $A' = -A$: $0' = -0$, and $(cA+B)' = cA' + B' = -(cA+B)$ if $A' = -A$ and $B' = -B$.
- (e) For $A \in \mathbb{R}^{m \times n}$, the solution set $\{x \in \mathbb{R}^n : Ax = 0\}$ of a homogeneous system is a subspace of $\mathbb{R}^n$: $A0 = 0$, and $A(cu+v) = cAu + Av = 0$ if $Au = Av = 0$. For $b \neq 0$, the solutions of $Ax = b$ do not form a subspace ($A0 = 0 \neq b$).
- (f) In statistics, the **centered** vectors $W_0 = \{x \in \mathbb{R}^n : \mathbf{1}'x = 0\}$, whose entries add up to 0, form a subspace: this is (e) with the $1 \times n$ matrix $A = \mathbf{1}'$. For every x , the deviation vector Cx of Example 1.13 lies in W_0 , since $\mathbf{1}'Cx = (C\mathbf{1})'x = 0$.
- (g) The set $\{(x, y, z)' \in \mathbb{R}^3 : xy = 0\}$ is not a subspace: $(1, 0, 0)'$ and $(0, 1, 0)'$ lie in it, but their sum $(1, 1, 0)'$ does not. It is the union of the subspaces $\{x = 0\}$ and $\{y = 0\}$; a union of subspaces need not be a subspace.

Span

Definition 3.6. Let $v_1, \dots, v_k$ be vectors of a vector space V . Their **span** is the set of all their linear combinations,

$$\text{span}\{v_1, \dots, v_k\} = \{c_1v_1 + \dots + c_kv_k : c_1, \dots, c_k \in \mathbb{R}\},$$

and $\text{span } \emptyset = \{0\}$. If $\text{span}\{v_1, \dots, v_k\} = W$, we say that $v_1, \dots, v_k$ **span** W .

Theorem 3.7. $\text{span}\{v_1, \dots, v_k\}$ is a subspace of V that contains $v_1, \dots, v_k$, and it is contained in every subspace that contains $v_1, \dots, v_k$: it is the smallest subspace containing them.

Proof. $0 = 0v_1 + \dots + 0v_k$ lies in the span, and $c \sum_i a_i v_i + \sum_i b_i v_i = \sum_i (ca_i + b_i)v_i$; so the span is a subspace (Theorem 3.4(c)). It contains $v_j = 1v_j + \sum_{i \neq j} 0v_i$. A subspace that contains $v_1, \dots, v_k$ contains every $c_i v_i$ and every sum of these, by closure. $\square$

For vectors $v_1, \dots, v_k \in \mathbb{R}^m$, put $V = [v_1 \ \dots \ v_k] \in \mathbb{R}^{m \times k}$. Since $Vc = c_1v_1 + \dots + c_kv_k$ (Theorem 1.4(b)),

$$\text{span}\{v_1, \dots, v_k\} = \{Vc : c \in \mathbb{R}^k\}, \quad b \in \text{span}\{v_1, \dots, v_k\} \iff \text{the system } Vc = b \text{ is consistent}$$

So membership in a span is decided by row-reducing $[V \mid b]$ (Proposition 2.7, Theorem 2.19). For instance, by Example 2.9, a vector $y \in \mathbb{R}^3$ lies in the span of the columns of the matrix A there if and only if $2y_1 - y_2 + y_3 = 0$.

Example 3.8. Let $W = \text{span}\{v_1, v_2, v_3\} \subseteq \mathbb{R}^5$, where $v_1 = (1, 2, 0, 3, 0)'$, $v_2 = (0, 0, 1, 4, 0)'$ and $v_3 = (0, 0, 0, 0, 1)'$. Which $w = (w_1, \dots, w_5)'$ lie in W ?

Solution. Row-reduce $[v_1 \ v_2 \ v_3 \mid w]$, keeping the w_i as letters:

$$\begin{aligned} \left(\begin{array}{ccc|c} 1 & 0 & 0 & w_1 \\ 2 & 0 & 0 & w_2 \\ 0 & 1 & 0 & w_3 \\ 3 & 4 & 0 & w_4 \\ 0 & 0 & 1 & w_5 \end{array} \right) & \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & w_1 \\ 0 & 0 & 0 & w_2 - 2w_1 \\ 0 & 1 & 0 & w_3 \\ 0 & 4 & 0 & w_4 - 3w_1 \\ 0 & 0 & 1 & w_5 \end{array} \right) & \xrightarrow{R_4 \leftarrow R_4 - 4R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & w_1 \\ 0 & 0 & 0 & w_2 - 2w_1 \\ 0 & 1 & 0 & w_3 \\ 0 & 0 & 0 & w_4 - 3w_1 - 4w_3 \\ 0 & 0 & 1 & w_5 \end{array} \right) \\ & \xrightarrow[R_3 \leftrightarrow R_5]{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc|c} 1 & 0 & 0 & w_1 \\ 0 & 1 & 0 & w_3 \\ 0 & 0 & 1 & w_5 \\ 0 & 0 & 0 & w_4 - 3w_1 - 4w_3 \\ 0 & 0 & 0 & w_2 - 2w_1 \end{array} \right) \end{aligned}$$

By Proposition 2.7, $w \in W$ if and only if

$$w_2 = 2w_1, \quad w_4 = 3w_1 + 4w_3,$$

and then $w = w_1v_1 + w_3v_2 + w_5v_3$. For example, $(-3, -6, 1, -5, 2)' \in W$, since $-6 = 2(-3)$ and $-5 = 3(-3) + 4 \cdot 1$; indeed it equals $-3v_1 + v_2 + 2v_3$. But $(2, 4, 6, 7, 8)' \notin W$, since $3 \cdot 2 + 4 \cdot 6 = 30 \neq 7$.

Example 3.9. $\text{span}\{\mathbf{1}_n\} = \{c\mathbf{1}_n : c \in \mathbb{R}\}$ is the subspace of constant vectors. Every $x \in \mathbb{R}^n$ is the sum of a constant vector and a centered vector:

$$x = \bar{x}\mathbf{1} + Cx, \quad \bar{x}\mathbf{1} \in \text{span}\{\mathbf{1}\}, \quad Cx \in W_0,$$

since $Cx = x - \bar{x}\mathbf{1}$ (Example 1.13) and $Cx \in W_0$ (Example 3.5(f)). For $x = (1, 3, 5, 7, 9)'$: $x = 5\mathbf{1} + (-4, -2, 0, 2, 4)'$.

Linear independence

Definition 3.10. Vectors $v_1, \dots, v_k$ of a vector space are **linearly dependent** if $c_1v_1 + \dots + c_kv_k = 0$ for some scalars $c_1, \dots, c_k$, not all 0 (a **linear relation**). Otherwise they are **linearly independent**: $c_1v_1 + \dots + c_kv_k = 0$ only for $c_1 = \dots = c_k = 0$. The empty list is independent.

Proposition 3.11. Let $v_1, \dots, v_k \in \mathbb{R}^m$ and $V = [v_1 \ \dots \ v_k] \in \mathbb{R}^{m \times k}$. Then $v_1, \dots, v_k$ are linearly independent if and only if $Vc = 0$ has only the solution $c = 0$, that is, if and only if every column of the RREF of V is a pivot column (no free unknowns). In particular, more than m vectors of $\mathbb{R}^m$ are linearly dependent.

Proof. $Vc = c_1v_1 + \dots + c_kv_k$, so the linear relations among the v_i are the solutions of $Vc = 0$. This system is consistent ($c = 0$), so by Theorem 2.20(b) $c = 0$ is its only solution exactly when there is no free unknown. If $k > m$, there is a free unknown, by Theorem 2.10(a). $\square$

Proposition 3.12. Let $v_1, \dots, v_k$ be vectors of a vector space.

- (a) If some $v_i = 0$, or $v_i = v_j$ for some $i \neq j$, then $v_1, \dots, v_k$ are dependent. A single vector v is independent if and only if $v \neq 0$.
- (b) A sublist of an independent list is independent.
- (c) For $k \geq 2$, $v_1, \dots, v_k$ are dependent if and only if some v_j is a linear combination of the others. In particular, two vectors are dependent if and only if one of them is a multiple of the other.

Proof. (a) $1 \cdot 0 = 0$ and $v_i - v_j = 0$ are linear relations with a coefficient 1; and if $v \neq 0$, then $cv = 0$ forces $c = 0$. (b) A relation among some of the vectors is a relation among all of them, with coefficient 0 for the others. (c) If $v_j = \sum_{i \neq j} x_i v_i$, then $v_j - \sum_{i \neq j} x_i v_i = 0$ is a relation with coefficient 1 on v_j . Conversely, if $\sum_i x_i v_i = 0$ with $x_j \neq 0$, then $v_j = \sum_{i \neq j} (-x_i/x_j)v_i$. $\square$

Example 3.13. Let a_1, a_2, a_3, a_4 be the columns of the matrix A of Example 2.6:

$$a_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad a_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad a_3 = \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix}, \quad a_4 = \begin{pmatrix} -2 \\ -3 \\ -4 \end{pmatrix}$$

Four vectors of $\mathbb{R}^3$ are dependent (Proposition 3.11). A relation comes from the solutions $x = t(-2, 3, 1, 1)'$ of $Ax = 0$ found there:

$$-2a_1 + 3a_2 + a_3 + a_4 = 0, \quad \text{so} \quad a_4 = 2a_1 - 3a_2 - a_3$$

On the other hand a_1, a_2, a_3 are independent: the same row operations take $[a_1 \ a_2 \ a_3]$ to the first three columns of the RREF found there, which form I_3 , so every column is a pivot column.

Bases

Definition 3.14. Vectors $v_1, \dots, v_k$ of a vector space V form a **basis** of a subspace W if they lie in W , are linearly independent, and span W .

- The unit vectors $e_1, \dots, e_n$ form a basis of $\mathbb{R}^n$, the **standard basis**: $x = x_1e_1 + \dots + x_ne_n$ for every $x \in \mathbb{R}^n$, and this is 0 only for $x = 0$.
- Let $E_{ij} \in \mathbb{R}^{m \times n}$ have (i, j) entry 1 and all other entries 0 (two indices; not the elementary matrices $E_1, E_2, \dots$ of Theorem 2.14(c)). The mn matrices E_{ij} form a basis of $\mathbb{R}^{m \times n}$: $A = \sum_i \sum_j a_{ij} E_{ij}$, and $\sum_i \sum_j c_{ij} E_{ij} = (c_{ij})$ is 0 only if every $c_{ij} = 0$.
- The empty list is a basis of $\{0\}$.

Theorem 3.15. If $v_1, \dots, v_k$ form a basis of W , every $w \in W$ can be written as $w = c_1v_1 + \dots + c_kv_k$ in exactly one way; the numbers $c_1, \dots, c_k$ are the **coordinates** of w in this basis.

Proof. Such an expression exists because the v_i span W . If $\sum_i x_i v_i = \sum_i y_i v_i$, then $\sum_i (x_i - y_i) v_i = 0$, and independence gives $x_i = y_i$ for every i . $\square$

Example 3.16. If $P \in \mathbb{R}^{n \times n}$ is invertible, its columns $p_1, \dots, p_n$ form a basis of $\mathbb{R}^n$, and the coordinates of $y \in \mathbb{R}^n$ are $x = P^{-1}y$. Indeed $Px = \sum_j x_j p_j$; $Px = 0$ only for $x = 0$ (Theorem 2.15), so the columns are independent; and $y = P(P^{-1}y)$, so they span. For example, the columns $(1, -1, 2)'$, $(-1, 1, -1)'$, $(1, -2, 2)'$ of the matrix A of Example 2.16 form a basis of $\mathbb{R}^3$. The coordinates of $y = (1, 0, 3)'$ are $A^{-1}y = (3, 1, -1)'$, the solution of $Ax = y$ found in Example 2.21: $y = 3(1, -1, 2)' + (-1, 1, -1)' - (1, -2, 2)'$.

Example 3.17. Let $u_1 = (1, 2, 0)'$, $u_2 = (1, 3, 0)'$, $u_3 = (2, 5, 0)'$, and $W = \{(a, b, 0)' : a, b \in \mathbb{R}\}$, a subspace of $\mathbb{R}^3$. Any two of u_1, u_2, u_3 form a basis of W . For $w = (a, b, 0)' \in W$, the equation $cu_i + du_j = w$ has third entries $0 = 0$, and its first two entries give

- u_1, u_2 : $c + d = a$, $2c + 3d = b$, so $d = b - 2a$, $c = 3a - b$;
- u_1, u_3 : $c + 2d = a$, $2c + 5d = b$, so $d = b - 2a$, $c = 5a - 2b$;
- u_2, u_3 : $c + 2d = a$, $3c + 5d = b$, so $d = 3a - b$, $c = 2b - 5a$.

In each case a solution exists for every w , so the pair spans W , and it is unique, so for $w = 0$ only $c = d = 0$: the pair is independent. All three vectors together also span W , but they are dependent ($u_1 + u_2 - u_3 = 0$), so they do not form a basis. A subspace has many bases.

Example 3.18. The symmetric 2×2 matrices (for instance 2×2 covariance matrices) form a subspace of $\mathbb{R}^{2 \times 2}$ (Example 3.5(d)). The matrices E_{11} , $E_{12} + E_{21}$, E_{22} form a basis of it: they are symmetric, every symmetric matrix is

$$\begin{pmatrix} a & b \\ b & d \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix},$$

and this combination is 0 only if $a = b = d = 0$.

Example 3.19. In $W_0 = \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$, the vectors $d_1 = (1, -1, 0)'$ and $d_2 = (0, 1, -1)'$ form a basis. They lie in W_0 . They span W_0 : for $x \in W_0$, $x_3 = -x_1 - x_2$, so

$$x = x_1 d_1 + (x_1 + x_2) d_2 = \begin{pmatrix} x_1 & -x_1 + x_1 + x_2 & -x_1 - x_2 \end{pmatrix}'$$

They are independent: $c_1 d_1 + c_2 d_2 = (c_1, c_2 - c_1, -c_2) = 0$ forces $c_1 = c_2 = 0$.

Sums and direct sums of subspaces

Definition 3.20. Let U and W be subspaces of a vector space V . Their **intersection** $U \cap W$ is the set of vectors that lie in both U and W , and their **sum** is

$$U + W = \{u + w : u \in U, w \in W\}$$

The sum is **direct**, written $U \oplus W$, if $U \cap W = \{0\}$, that is, if 0 is the only vector that U and W have in common. Thus $V = U \oplus W$ means that $V = U + W$ and $U \cap W = \{0\}$.

Theorem 3.21. Let U and W be subspaces of a vector space V .

- (a) $U \cap W$ and $U + W$ are subspaces of V .
- (b) $U + W$ contains U and W , and it is contained in every subspace of V that contains $U \cup W$: it is the smallest subspace of V containing $U \cup W$. Moreover $U + W = \text{span}(U \cup W)$, the set of all linear combinations $c_1x_1 + \cdots + c_kx_k$ of finitely many vectors $x_1, \dots, x_k \in U \cup W$.
- (c) If $U = \text{span}\{u_1, \dots, u_p\}$ and $W = \text{span}\{w_1, \dots, w_q\}$, then $U + W = \text{span}\{u_1, \dots, u_p, w_1, \dots, w_q\}$.

Proof. (a)

- Both sets contain 0 : $0 \in U$, $0 \in W$, and $0 = 0 + 0$.
- Let $x, y \in U \cap W$ and $c \in \mathbb{R}$. Then $cx + y \in U$ and $cx + y \in W$, since U and W are subspaces; so $cx + y \in U \cap W$.
- Let $x = u_1 + w_1$ and $y = u_2 + w_2$ with $u_1, u_2 \in U$, $w_1, w_2 \in W$, and let $c \in \mathbb{R}$. Then $cx + y = (cu_1 + u_2) + (cw_1 + w_2)$ with $cu_1 + u_2 \in U$ and $cw_1 + w_2 \in W$; so $cx + y \in U + W$.
- By Theorem 3.4(c), $U \cap W$ and $U + W$ are subspaces.

(b)

- Every $u \in U$ is $u = u + 0$, and every $w \in W$ is $w = 0 + w$; so U and W lie in $U + W$.
- If a subspace X of V contains $U \cup W$, then it contains u and w for all $u \in U$, $w \in W$, hence also $u + w$; so $U + W \subseteq X$.
- In a combination $c_1x_1 + \cdots + c_kx_k$ with all $x_i \in U \cup W$, the terms with $x_i \in U$ add up to a vector $u \in U$, and the other terms, whose x_i lie in W , add up to a vector $w \in W$ (an empty sum is 0). So the combination is $u + w \in U + W$. Conversely, $u + w = 1u + 1w$ is such a combination.

(c)

- If $u = \sum_i a_i u_i \in U$ and $w = \sum_j b_j w_j \in W$, then $u + w = \sum_i a_i u_i + \sum_j b_j w_j$ is a combination of $u_1, \dots, u_p, w_1, \dots, w_q$.
- Conversely, $\sum_i a_i u_i + \sum_j b_j w_j = u + w$ with $u = \sum_i a_i u_i \in U$ and $w = \sum_j b_j w_j \in W$. □

For matrices: let $A \in \mathbb{R}^{m \times p}$ have columns $a_1, \dots, a_p$, let $B \in \mathbb{R}^{m \times q}$ have columns $b_1, \dots, b_q$, and let $[A \ B] \in \mathbb{R}^{m \times (p+q)}$ be the matrix with columns $a_1, \dots, a_p, b_1, \dots, b_q$. By (c), the columns of $[A \ B]$ span the sum of the subspaces spanned by the columns of A and by the columns of B ,

$$\text{span}\{a_1, \dots, a_p, b_1, \dots, b_q\} = \text{span}\{a_1, \dots, a_p\} + \text{span}\{b_1, \dots, b_q\}$$

and, by Theorem 1.4(b), the vectors of this subspace are the vectors $Ax + By$ ($x \in \mathbb{R}^p$, $y \in \mathbb{R}^q$).

Theorem 3.22. Let U and W be subspaces of a vector space V . Then $U \cap W = \{0\}$ if and only if every $x \in U + W$ can be written in exactly one way as $x = u + w$ with $u \in U$ and $w \in W$, that is: if $u + w = u_1 + w_1$ with $u, u_1 \in U$ and $w, w_1 \in W$, then $u = u_1$ and $w = w_1$.

Proof. • Let $U \cap W = \{0\}$, and let $u + w = u_1 + w_1$ as above. Then $u - u_1 = w_1 - w$. The left side lies in U and the right side in W , so this vector lies in $U \cap W = \{0\}$. Hence $u = u_1$ and $w = w_1$.

• Let $U \cap W \neq \{0\}$, and take $x \neq 0$ in $U \cap W$. Then $x = x + 0$ (with $x \in U$, $0 \in W$) and $x = 0 + x$ (with $0 \in U$, $x \in W$) are two different ways of writing $x \in U + W$. $\square$

So, if $V = U \oplus W$, each $v \in V$ is $v = u + w$ for exactly one $u \in U$ and one $w \in W$ (the parts of v in U and in W).

Theorem 3.23. Let U and W be subspaces of a vector space V , let $u_1, \dots, u_p$ form a basis of U , and let $w_1, \dots, w_q$ form a basis of W . If $U \cap W = \{0\}$, then $u_1, \dots, u_p, w_1, \dots, w_q$ form a basis of $U \oplus W$. If $U \cap W \neq \{0\}$, then $u_1, \dots, u_p, w_1, \dots, w_q$ are linearly dependent. Hence the sum $U + W$ is direct if and only if $u_1, \dots, u_p, w_1, \dots, w_q$ are linearly independent.

Proof. • The vectors $u_1, \dots, u_p, w_1, \dots, w_q$ lie in $U + W$ and span it, by Theorem 3.21(b),(c).

- Let $U \cap W = \{0\}$ and $\sum_i a_i u_i + \sum_j b_j w_j = 0$. Then $x = \sum_i a_i u_i = \sum_j (-b_j) w_j$ lies in U and in W , so $x = 0$. Hence $\sum_i a_i u_i = 0$ and $\sum_j b_j w_j = 0$, and since each list is independent, every a_i and every b_j is 0. So the vectors form a basis of $U + W = U \oplus W$.
- Let $U \cap W \neq \{0\}$, take $x \neq 0$ in $U \cap W$, and write $x = \sum_i a_i u_i = \sum_j b_j w_j$. Then $\sum_i a_i u_i + \sum_j (-b_j) w_j = 0$, and not every a_i is 0, since $x \neq 0$: a linear relation among $u_1, \dots, u_p, w_1, \dots, w_q$. $\square$

Example 3.24. In $\mathbb{R}^3$ let $U = \{x : x_3 = 0\} = \text{span}\{e_1, e_2\}$ (the subspace W of Example 3.17) and $W_0 = \{x : x_1 + x_2 + x_3 = 0\} = \text{span}\{d_1, d_2\}$, with $d_1 = (1, -1, 0)'$ and $d_2 = (0, 1, -1)'$ (Example 3.19); both are planes through 0.

(a) $U + W_0 = \mathbb{R}^3$, but the sum is not direct.

- Every $x \in \mathbb{R}^3$ is $x = (x_1, x_2 + x_3, 0)'+(0, -x_3, x_3)'$, with the first vector in U and the second, $-x_3 d_2$, in W_0 . So $U + W_0 = \mathbb{R}^3$.
- $x \in U \cap W_0$ means $x_3 = 0$ and $x_1 + x_2 + x_3 = 0$, that is, $x = (t, -t, 0)' = t d_1$ for some $t \in \mathbb{R}$. So $U \cap W_0 = \text{span}\{d_1\}$, a line, and the sum is not direct.
- Accordingly (Theorem 3.22) the parts are not unique: for example $(1, 2, 3)' = (1, 5, 0)' + (0, -3, 3)' = (4, 2, 0)' + (-3, 0, 3)'$, and the two parts in U differ by $3d_1 \in U \cap W_0$. Likewise e_1, e_2, d_1, d_2 are dependent, since $d_1 = e_1 - e_2$ (Theorem 3.23).

(b) $\mathbb{R}^3 = W_0 \oplus \text{span}\{e_3\}$.

- $ce_3 = (0, 0, c)'$ lies in W_0 only for $c = 0$; so $W_0 \cap \text{span}\{e_3\} = \{0\}$.
- For $x \in \mathbb{R}^3$ and $c \in \mathbb{R}$, $x - ce_3 = (x_1, x_2, x_3 - c)'$ lies in W_0 exactly when $c = x_1 + x_2 + x_3$. So every x is $x = (x_1, x_2, -x_1 - x_2)' + (x_1 + x_2 + x_3)e_3$ with the first vector in W_0 , and in no other way. For example, $(1, 2, 3)' = (1, 2, -3)' + 6e_3$.
- By Theorem 3.23, d_1, d_2, e_3 form a basis of $\mathbb{R}^3$.

Example 3.25. The constant vectors and the centered vectors $W_0 = \{x \in \mathbb{R}^n : \mathbf{1}'x = 0\}$ (Example 3.5(f)) give a direct sum:

$$\mathbb{R}^n = \text{span}\{\mathbf{1}_n\} \oplus W_0, \quad x = \bar{x}\mathbf{1} + Cx$$

- Sum: every $x \in \mathbb{R}^n$ is $x = \bar{x}\mathbf{1} + Cx$ with $\bar{x}\mathbf{1} \in \text{span}\{\mathbf{1}\}$ and $Cx \in W_0$ (Example 3.9).
- Intersection: if $c\mathbf{1} \in W_0$, then $0 = \mathbf{1}'(c\mathbf{1}) = cn$ (Proposition 1.12), so $c = 0$.
- Hence $\bar{x}\mathbf{1}$ is the only constant vector whose difference from x is centered (Theorem 3.22). In matrix form, $x = \bar{J}_n x + C_n x$ with $\bar{J}_n x = \frac{1}{n}\mathbf{1}(\mathbf{1}'x) = \bar{x}\mathbf{1}$, and $\bar{J}_n + C_n = I_n$ (Definition 1.11). For $x = (1, 3, 5, 7, 9)'$ the two parts are $\bar{x}\mathbf{1} = 5\mathbf{1}$ and $Cx = (-4, -2, 0, 2, 4)'$.

- For $n = 3$, $\mathbf{1}_3$ together with the basis d_1, d_2 of W_0 (Example 3.19) is a basis of $\mathbb{R}^3$ (Theorem 3.23).
- The second subspace matters: also $\mathbb{R}^3 = W_0 \oplus \text{span}\{e_3\}$ (Example 3.24(b)), and the parts of $x = (1, 2, 3)'$ in W_0 differ: here $\bar{x} = 2$ and $Cx = (-1, 0, 1)'$, there $(1, 2, -3)'$.

Exercises and solutions

Exercise 3.1. Is the set of vectors $(x, y)' \in \mathbb{R}^2$ with $x \geq 0$ and $y \geq 0$ a subspace of $\mathbb{R}^2$?

Solution. No: $(1, 1)'$ lies in the set, but $(-1)(1, 1)' = (-1, -1)'$ does not; so the set is not closed under scalar multiplication (Theorem 3.4(b)).

Exercise 3.2. In the vector space $\mathbb{R}^{3 \times 3}$, show that the set of lower triangular matrices is a subspace.

Solution. Let $A, B \in \mathbb{R}^{3 \times 3}$ be lower triangular, that is, $a_{ij} = b_{ij} = 0$ for $i < j$, and let $c \in \mathbb{R}$. For $i < j$, $(A + B)_{ij} = a_{ij} + b_{ij} = 0$ and $(cA)_{ij} = ca_{ij} = 0$; so $A + B$ and cA are lower triangular, and so is 0. By Theorem 3.4(b), the lower triangular matrices form a subspace.

Exercise 3.3. Consider $e_1 = (1, 0, 0)'$, $e_2 = (0, 1, 0)'$ and $v = (0, -2, 0)'$ in $\mathbb{R}^3$. Show that e_1, e_2, v span $W = \{(x, y, 0)' : x, y \in \mathbb{R}\}$, that e_1 and e_2 alone span the same subspace W , and that e_2 and v span only $\{(0, y, 0)' : y \in \mathbb{R}\}$.

Solution. For scalars a_1, a_2, a_3 put $\lambda = a_2 - 2a_3$. Since $v = -2e_2$,

$$a_1 e_1 + a_2 e_2 + a_3 v = a_1 e_1 + \lambda e_2 = (a_1, \lambda, 0)', \quad a_2 e_2 + a_3 v = \lambda e_2 = (0, \lambda, 0)'$$

So $\text{span}\{e_1, e_2, v\}$ consists of the vectors $(a_1, \lambda, 0)'$. Each lies in W , and every $(x, y, 0)' \in W$ is obtained with $a_1 = x$, $a_2 = y$, $a_3 = 0$; so $\text{span}\{e_1, e_2, v\} = W$. The same computation with $a_3 = 0$ gives $\text{span}\{e_1, e_2\} = W$, and $\text{span}\{e_2, v\} = \{\lambda e_2 : \lambda \in \mathbb{R}\} = \{(0, y, 0)' : y \in \mathbb{R}\}$.

Exercise 3.4. Let $W = \text{span}\{v_1, v_2, v_3\} \subseteq \mathbb{R}^4$, where $v_1 = (1, 2, 1, 0)'$, $v_2 = (0, 1, 1, 1)'$ and $v_3 = (1, 3, 3, 2)'$.

- Which $w = (w_1, w_2, w_3, w_4)'$ lie in W ?
- Show that $(3, 6, 4, 1)' \in W$, and write it as a combination of v_1, v_2, v_3 .
- Does $(1, 1, 1, 1)'$ lie in W ?

Solution. (a) With $V = [v_1 \ v_2 \ v_3]$, $w \in W$ if and only if $Vc = w$ is consistent (the paragraph after Theorem 3.7). Row-reduce $[V \mid w]$, keeping the w_i as letters:

$$\begin{aligned} & \left(\begin{array}{ccc|c} 1 & 0 & 1 & w_1 \\ 2 & 1 & 3 & w_2 \\ 1 & 1 & 3 & w_3 \\ 0 & 1 & 2 & w_4 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 0 & 1 & w_1 \\ 0 & 1 & 1 & w_2 - 2w_1 \\ 0 & 1 & 2 & w_3 - w_1 \\ 0 & 1 & 2 & w_4 \end{array} \right) \\ & \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & w_1 \\ 0 & 1 & 1 & w_2 - 2w_1 \\ 0 & 0 & 1 & w_1 - w_2 + w_3 \\ 0 & 0 & 1 & 2w_1 - w_2 + w_4 \end{array} \right) \xrightarrow{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{ccc|c} 1 & 0 & 1 & w_1 \\ 0 & 1 & 1 & w_2 - 2w_1 \\ 0 & 0 & 1 & w_1 - w_2 + w_3 \\ 0 & 0 & 0 & w_1 - w_3 + w_4 \end{array} \right) \end{aligned}$$

The last matrix is an echelon form. By Theorem 2.19, $Vc = w$ is consistent if and only if its last row is not $(0 \ 0 \ 0 \mid d)$ with $d \neq 0$:

$$w \in W \iff w_1 - w_3 + w_4 = 0$$

Then every column of the left part is a pivot column, so the coefficients are unique (Theorem 2.20(b)); back substitution gives $c_3 = w_1 - w_2 + w_3$, $c_2 = (w_2 - 2w_1) - c_3 = -3w_1 + 2w_2 - w_3$ and $c_1 = w_1 - c_3 = w_2 - w_3$.

(b) For $w = (3, 6, 4, 1)'$, $3 - 4 + 1 = 0$, so $w \in W$, with $c_3 = 3 - 6 + 4 = 1$, $c_2 = -9 + 12 - 4 = -1$ and $c_1 = 6 - 4 = 2$: $w = 2v_1 - v_2 + v_3$. Check: $(2, 4, 2, 0)' - (0, 1, 1, 1)' + (1, 3, 3, 2)' = (3, 6, 4, 1)'$.

(c) No: $w_1 - w_3 + w_4 = 1 - 1 + 1 = 1 \neq 0$.

Exercise 3.5. Show that the following vectors are linearly independent. (a) $(1, 2)'$ and $(1, 3)'$. (b) $(1, 1, 0)'$, $(1, 1, 1)'$ and $(0, 1, -1)'$. (c) $(0, 1, 1)'$, $(0, 2, 1)'$ and $(1, 5, 3)'$.

Solution. (a) If $a(1, 2)' + b(1, 3)' = 0$, then $a + b = 0$ and $2a + 3b = 0$. The second equation minus twice the first gives $b = 0$, and then $a = 0$.

(b) If $a(1, 1, 0)' + b(1, 1, 1)' + c(0, 1, -1)' = 0$, then $a + b = 0$, $a + b + c = 0$, $b - c = 0$. The second equation minus the first gives $c = 0$; then the third gives $b = 0$, and the first $a = 0$.

(c) If $a(0, 1, 1)' + b(0, 2, 1)' + c(1, 5, 3)' = 0$, then $c = 0$, $a + 2b + 5c = 0$, $a + b + 3c = 0$. Subtracting the third equation from the second gives $b + 2c = 0$, so $b = 0$ (as $c = 0$), and then $a = 0$.

Exercise 3.6. Let $u_1 = (1, 2, 3, 1)'$, $u_2 = (0, 1, 1, 1)'$, $u_3 = (1, 2, 3, 0)'$ and $u_4 = (0, 2, 2, 1)'$.

(a) Are u_1, u_2, u_3 linearly independent?

(b) Are u_1, u_2, u_3, u_4 linearly independent? If not, find a linear relation among them, and write u_4 as a combination of u_1, u_2, u_3 .

Solution. Row-reduce $V = [u_1 \ u_2 \ u_3 \ u_4]$:

$$\begin{aligned} & \begin{pmatrix} 1 & 0 & 1 & 0 \\ 2 & 1 & 2 & 2 \\ 3 & 1 & 3 & 2 \\ 1 & 1 & 0 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1, \ R_3 \leftarrow R_3 - 3R_1} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & -1 & 1 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & -1 \end{pmatrix} \\ & \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow -R_3} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_3} \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

After the second step the entry of row 3 in column 3 is 0, so Step 2 interchanges rows 3 and 4. The last matrix is the RREF of V , with pivot columns 1, 2, 3.

(a) Yes. Row operations act on each column separately (Theorem 2.14(c) and Theorem 1.4(a)), so the same operations take $[u_1 \ u_2 \ u_3]$ to the first three columns of the last matrix, whose columns are the unit vectors e_1, e_2, e_3 of $\mathbb{R}^4$. This is the RREF of $[u_1 \ u_2 \ u_3]$, and every column is a pivot column; so u_1, u_2, u_3 are independent (Proposition 3.11).

(b) No. Column 4 is not a pivot column, so $Vc = 0$ has a free unknown and a solution $c \neq 0$ (Proposition 3.11). The RREF gives $c_1 - c_4 = 0$, $c_2 + 2c_4 = 0$, $c_3 + c_4 = 0$; with $c_4 = t$, $c = t(1, -2, -1, 1)'$. For $t = 1$,

$$u_1 - 2u_2 - u_3 + u_4 = 0, \quad \text{so} \quad u_4 = -u_1 + 2u_2 + u_3$$

Check: $-(1, 2, 3, 1)' + 2(0, 1, 1, 1)' + (1, 2, 3, 0)' = (0, 2, 2, 1)' = u_4$.

Exercise 3.7. For what values of the scalar k are the three row vectors $(k, 1, 0)$, $(1, k, 1)$ and $(0, 1, k)$ linearly dependent, and for what values are they linearly independent?

Solution. Let x_1, x_2, x_3 be scalars with $x_1(k, 1, 0) + x_2(1, k, 1) + x_3(0, 1, k) = 0$, that is,

$$kx_1 + x_2 = 0, \quad x_1 + kx_2 + x_3 = 0, \quad x_2 + kx_3 = 0,$$

or equivalently (the first and third equations give (1), the second gives (2))

$$x_2 = -kx_3 = -kx_1 \quad (1), \quad kx_2 = -x_1 - x_3 \quad (2)$$

- $k = 0$: (1) and (2) become $x_2 = 0$ and $x_3 = -x_1$; for instance $(x_1, x_2, x_3) = (1, 0, -1)$ is a solution other than 0.
- $k \neq 0$: (1) gives $kx_3 = kx_1$, so $x_3 = x_1$; then (1) and (2) give $-k^2x_1 = kx_2 = -2x_1$, so $(2 - k^2)x_1 = 0$. Hence $k^2 = 2$, or else $x_1 = 0$ and then $x_3 = x_2 = 0$.
- If $k = \sqrt{2}$, conditions (1), (2) are equivalent to $x_3 = x_1$ and $x_2 = -\sqrt{2}x_1$; if $k = -\sqrt{2}$, to $x_3 = x_1$ and $x_2 = \sqrt{2}x_1$. In both cases $x_1 = 1$ gives a solution other than 0.

Thus there are solutions other than $x_1 = x_2 = x_3 = 0$ if and only if $k = 0$ or $k = \pm\sqrt{2}$. The three vectors are linearly dependent if $k \in \{0, \sqrt{2}, -\sqrt{2}\}$ and linearly independent otherwise.

Exercise 3.8. Show that the columns of the matrix $A = \begin{pmatrix} 1 & 3 & 2 & 0 \\ 1 & -5 & -3 & 1 \\ 5 & -1 & 0 & 2 \end{pmatrix}$ are linearly dependent.

How about the rows?

Solution. The columns satisfy, for example, $3\text{col}_1(A) - \text{col}_2(A) - 8\text{col}_4(A) = 0$:

$$3 \begin{pmatrix} 1 \\ 1 \\ 5 \end{pmatrix} - \begin{pmatrix} 3 \\ -5 \\ -1 \end{pmatrix} - 8 \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 - 3 - 0 \\ 3 + 5 - 8 \\ 15 + 1 - 16 \end{pmatrix} = 0$$

The coefficients 3, -1, 0, -8 are not all 0, so the columns are dependent; equivalently, $Ax = 0$ for $x = (3, -1, 0, -8)' \neq 0$ (Theorem 1.4(b)). The rows satisfy $3\text{row}_1(A) + 2\text{row}_2(A) - \text{row}_3(A) = 0$: $(3, 9, 6, 0) + (2, -10, -6, 2) - (5, -1, 0, 2) = (0, 0, 0, 0)$, that is, $y'A = 0$ for $y = (3, 2, -1)'$ (Theorem 1.4(c)). So the rows are dependent too.

Exercise 3.9. Suppose that $v_1, \dots, v_k$ span a vector space V but do not form a basis of V . Show that, for any $v \in V$, the representation of v as a linear combination of $v_1, \dots, v_k$ is not unique.

Solution. Let $v \in V$. Since $v_1, \dots, v_k$ span V , there are scalars $x_1, \dots, x_k$ with $v = \sum_i x_i v_i$. Since $v_1, \dots, v_k$ span V but do not form a basis, they are linearly dependent (Definition 3.14): there are scalars $z_1, \dots, z_k$, not all 0, with $\sum_i z_i v_i = 0$. Let $y_i = x_i + z_i$. Then $\sum_i y_i v_i = \sum_i x_i v_i + \sum_i z_i v_i = v + 0 = v$, and $(y_1, \dots, y_k) \neq (x_1, \dots, x_k)$ because some $z_i \neq 0$. So v has two different representations. (Compare Theorem 3.15.)

Exercise 3.10. Let U and W be the subspaces of $\mathbb{R}^3$ spanned by the columns of

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 2 \\ 1 & 1 \\ 2 & 3 \end{pmatrix},$$

respectively. Find (a) a basis of $U + W$; (b) a basis of $U \cap W$ (is the sum $U + W$ direct?); (c) a vector of $U + W$ that is not in $U \cup W$.

Solution. Let $b_1 = (0, 1, 2)'$ and $b_2 = (2, 1, 3)'$ be the columns of B ; the columns of A are e_1, e_2 .

(a)

- By Theorem 3.21(c), $U + W = \text{span}\{e_1, e_2, b_1, b_2\}$.
- e_1, e_2, b_1 are independent: $c_1e_1 + c_2e_2 + c_3b_1 = (c_1, c_2 + c_3, 2c_3)' = 0$ gives $c_3 = 0$, then $c_2 = 0$ and $c_1 = 0$.
- So $M = [e_1 \ e_2 \ b_1] \in \mathbb{R}^{3 \times 3}$ has $Mc = 0$ only for $c = 0$. By Theorem 2.15, M is invertible, and by Example 3.16 its columns form a basis of $\mathbb{R}^3$.
- These columns lie in the subspace $U + W$, so every vector of $\mathbb{R}^3$, a combination of them, lies in $U + W$. Hence $U + W = \mathbb{R}^3$, and e_1, e_2, b_1 form a basis of $U + W$.

(b)

- U consists of the vectors $(x_1, x_2, 0)'$, and W of the vectors $y_1b_1 + y_2b_2 = (2y_2, y_1 + y_2, 2y_1 + 3y_2)'$ ($x_1, x_2, y_1, y_2 \in \mathbb{R}$).
- So $U \cap W$ consists of the vectors $y_1b_1 + y_2b_2$ with $2y_1 + 3y_2 = 0$. This equation holds exactly when $y_1 = 3t$ and $y_2 = -2t$ for some $t \in \mathbb{R}$: these values give $6t - 6t = 0$; conversely, if $2y_1 + 3y_2 = 0$, then $t = y_1 + y_2$ has $3t = 3y_1 - 2y_1 = y_1$ and $-2t = 3y_2 - 2y_2 = y_2$.
- For $y_1 = 3t, y_2 = -2t$: $y_1b_1 + y_2b_2 = (-4t, t, 0)' = t(-4, 1, 0)'$. Hence $U \cap W = \text{span}\{(-4, 1, 0)'\}$, and the single nonzero vector $(-4, 1, 0)'$ is a basis of it (Proposition 3.12(a)).
- Since $U \cap W \neq \{0\}$, the sum $U + W$ is not direct (Definition 3.20).

(c) Take $x = (0, 2, 2)'$; it lies in $U + W = \mathbb{R}^3$.

- $x \notin U$, since its third entry is not 0.
- If $x = (2y_2, y_1 + y_2, 2y_1 + 3y_2)'$, the first two entries give $y_2 = 0$ and $y_1 = 2$, and then the third entry would be $4 \neq 2$. So $x \notin W$.

Exercise 3.11. In $\mathbb{R}^3$ let $W = \text{span}\{e_1\}$ and $U = \text{span}\{u_1, u_2\}$, where $u_1 = (1, 1, 0)'$ and $u_2 = (0, 1, 1)'$. Show that $\mathbb{R}^3 = W \oplus U$, and write $x = (1, 2, 3)'$ as $x = w + u$ with $w \in W$ and $u \in U$.

Solution. The vectors e_1, u_1, u_2 are linearly independent: if $ae_1 + bu_1 + cu_2 = 0$, then

$$a + b = 0, \quad b + c = 0, \quad c = 0,$$

so $c = 0$, then $b = 0$ and $a = 0$.

- $W \cap U = \{0\}$: e_1 is a basis of W , and u_1, u_2 (independent by Proposition 3.12(b)) form a basis of U . Their combined list e_1, u_1, u_2 is independent, so the sum $W + U$ is direct (Theorem 3.23).
- $W + U = \mathbb{R}^3$: the matrix $M = [e_1 \ u_1 \ u_2] \in \mathbb{R}^{3 \times 3}$ has $Mc = 0$ only for $c = 0$, so it is invertible (Theorem 2.15), and its columns span $\mathbb{R}^3$ (Example 3.16). By Theorem 3.21(c), $W + U = \text{span}\{e_1, u_1, u_2\} = \mathbb{R}^3$.

Hence $\mathbb{R}^3 = W \oplus U$. To write $x = (1, 2, 3)'$ as $ae_1 + bu_1 + cu_2$, solve the system whose augmented matrix is already in echelon form,

$$\left(\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right)$$

Back substitution (Theorem 2.19) gives $c = 3$, $b = 2 - 3 = -1$, $a = 1 - (-1) = 2$. So $w = 2e_1 = (2, 0, 0)'$ and $u = -u_1 + 3u_2 = (-1, 2, 3)'$; indeed $w + u = (1, 2, 3)'$. By Theorem 3.22, this is the only such decomposition.

Exercise 3.12. Show that every $A \in \mathbb{R}^{n \times n}$ can be written in exactly one way as $A = B + C$ with B symmetric and C skew-symmetric, and conclude that $\mathbb{R}^{n \times n}$ is the direct sum of the subspace of symmetric matrices and the subspace of skew-symmetric matrices.

Solution. Both sets are subspaces of $\mathbb{R}^{n \times n}$ (Example 3.5(d)).

- Existence: let $B = \frac{1}{2}(A + A')$ and $C = \frac{1}{2}(A - A')$. Then $B + C = A$, and, since $(A')' = A$ (Theorem 1.8), $B' = \frac{1}{2}(A' + A) = B$ and $C' = \frac{1}{2}(A' - A) = -C$.
- Uniqueness: let also $A = B_1 + C_1$ with B_1 symmetric and C_1 skew-symmetric. Then $B_1 - B = C - C_1$, where $B_1 - B$ is symmetric and $C - C_1$ is skew-symmetric. So $B_1 - B = (B_1 - B)' = (C - C_1)' = -(C - C_1) = -(B_1 - B)$, hence $2(B_1 - B) = 0$. Thus $B_1 = B$, and then $C_1 = A - B_1 = C$.
- So the sum of the two subspaces is all of $\mathbb{R}^{n \times n}$, and by Theorem 3.22 their intersection is $\{0\}$: $\mathbb{R}^{n \times n}$ is their direct sum.

For example, $B = \frac{1}{2}(A + A')$ and $C = \frac{1}{2}(A - A')$ give

$$A = \begin{pmatrix} 1 & 3 & 0 \\ 1 & 2 & 4 \\ 2 & -2 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 5 \end{pmatrix} + \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & 3 \\ 1 & -3 & 0 \end{pmatrix} = B + C$$

Chapter 4. Bases, dimension, and rank

Bases and dimension of subspaces of $\mathbb{R}^n$

Theorem 4.1. Let R be the RREF of $A \in \mathbb{R}^{m \times n}$, with r nonzero rows. The following are equivalent:

- (i) the columns $a_1, \dots, a_n$ of A are linearly independent;
- (ii) the system $Ax = 0$ has only the solution $x = 0$;
- (iii) $r = n$, that is, every column of R is a pivot column.

In particular, if $n > m$, then any n vectors of $\mathbb{R}^m$ are linearly dependent.

We call this the **pivot test**.

Proof. • (i) $\Leftrightarrow$ (ii): since $Ax = x_1a_1 + \dots + x_na_n$ (Theorem 1.4(b)), the statement “ $x_1a_1 + \dots + x_na_n = 0$ only for $x = 0$ ” says exactly that $Ax = 0$ only for $x = 0$.

- (ii) $\Leftrightarrow$ (iii): $Ax = 0$ and $Rx = 0$ have the same solutions (Theorem 2.3(b)). If $r < n$, there is a free unknown; giving it the value 1 and the other free unknowns the value 0 yields a solution $x \neq 0$ (Proposition 2.7(b)). If $r = n$, every unknown is a pivot unknown, there are no free unknowns, and the equations of $Rx = 0$ read $x_{k_i} = 0$ ($1 \leq i \leq n$); so $x = 0$.
- If $n > m$, put the n vectors into a matrix $A \in \mathbb{R}^{m \times n}$. Its RREF has $r \leq m < n$ nonzero rows, so (iii) fails and the vectors are dependent. $\square$

Remark 4.2. For the pivot test one may stop at an echelon form E of A (Definitions 2.4 and 2.5). Backward elimination adds multiples of a row i , whose leading 1 is in column k_i , to rows $l < i$, whose leading 1 is in a column $k_l < k_i$. Row i is 0 before column k_i , so row l changes only in columns $\geq k_i$ and keeps its leading 1 in column k_l . Hence the RREF of A reached from E has the same leading columns as E : the pivot columns of A , and their number r , can be read off from any echelon form of A . In particular, (iii) of Theorem 4.1 says that every column of E contains a leading 1. The same holds before the leading entries are scaled to 1: scaling a row does not move its leading entry.

Theorem 4.3. Let R be the RREF of $A \in \mathbb{R}^{m \times n}$, with r nonzero rows, pivot columns $k_1 < \dots < k_r$ and entries R_{ij} .

- (a) The columns of A and those of R satisfy the same linear relations: for $x \in \mathbb{R}^n$, $x_1a_1 + \dots + x_na_n = 0$ if and only if $x_1 \text{col}_1(R) + \dots + x_n \text{col}_n(R) = 0$.
- (b) If $r \geq 1$, the pivot columns $a_{k_1}, \dots, a_{k_r}$ of A form a basis of $\text{span}\{a_1, \dots, a_n\}$, and for every $j = 1, \dots, n$

$$a_j = R_{1j}a_{k_1} + R_{2j}a_{k_2} + \dots + R_{rj}a_{k_r}$$

If $r = 0$, then $A = 0$ and $\text{span}\{a_1, \dots, a_n\} = \{0\}$.

In particular, every finite list that spans a subspace W contains a basis of W .

Proof. (a) The two sums are Ax and Rx , and $Ax = 0$ if and only if $Rx = 0$ (Theorem 2.3(b)).

(b) If $r = 0$, then $R = 0$, and $A = P^{-1}R = 0$, where $R = PA$ with $P \in \mathbb{R}^{m \times m}$ invertible (Theorem 2.14(c)). Let $r \geq 1$.

- Columns of R : $\text{col}_{k_i}(R) = e_i$, and rows $r+1, \dots, m$ of R are zero, so $\text{col}_j(R) = \sum_{i=1}^r R_{ij}e_i = \sum_{i=1}^r R_{ij} \text{col}_{k_i}(R)$ for every j .
- Relations: for a column $j \notin \{k_1, \dots, k_r\}$ let $x \in \mathbb{R}^n$ have $x_j = 1$, $x_{k_i} = -R_{ij}$ ($1 \leq i \leq r$) and all other entries 0. By the first item, $Rx = \text{col}_j(R) - \sum_i R_{ij} \text{col}_{k_i}(R) = 0$; by (a), $Ax = a_j - \sum_i R_{ij}a_{k_i} = 0$. For $j = k_l$ the formula reads $a_{k_l} = a_{k_l}$, since R_{ik_l} is 1 for $i = l$ and 0 otherwise.

- Spanning: each a_j lies in the subspace $U = \text{span}\{a_{k_1}, \dots, a_{k_r}\}$, so $\text{span}\{a_1, \dots, a_n\} \subseteq U$ (Theorem 3.7); the reverse inclusion is clear.
- Independence: if $c_1 a_{k_1} + \dots + c_r a_{k_r} = 0$, then by (a) $c_1 \text{col}_{k_1}(R) + \dots + c_r \text{col}_{k_r}(R) = c_1 e_1 + \dots + c_r e_r = 0$, so every $c_i = 0$.

For the last sentence, write the spanning vectors as the columns of a matrix and apply (b). $\square$

By Theorem 4.3(a), a_j is a linear combination of $a_1, \dots, a_{j-1}$ if and only if $\text{col}_j(R)$ is a linear combination of $\text{col}_1(R), \dots, \text{col}_{j-1}(R)$ (take $x_j = -1$ and $x_l = 0$ for $l > j$). In R this happens exactly when j is not a pivot column: $\text{col}_{k_i}(R) = e_i$, while the columns before k_i are 0 in rows $i, \dots, m$; and a non-pivot column j equals $\sum_{k_i < j} R_{ij} \text{col}_{k_i}(R)$. So the pivot columns of A are the columns that are not combinations of the earlier ones; they do not depend on the row operations used. Hence the RREF of A is unique, as stated in Definition 2.5: two RREFs R, S of A have the same pivot columns $k_1 < \dots < k_r$, hence the same zero rows $r+1, \dots, m$; and for each j , both $R_{1j}, \dots, R_{rj}$ and $S_{1j}, \dots, S_{rj}$ are coefficients of a_j in the basis $a_{k_1}, \dots, a_{k_r}$ (Theorem 4.3(b)), so they are equal (Theorem 3.15). Thus $R = S$.

Theorem 4.4. Every subspace W of $\mathbb{R}^n$ has a basis, and any two bases of W have the same number of elements. This number is the **dimension** of W , written $\dim W$. For example, $\dim \mathbb{R}^n = n$ (standard basis) and $\dim\{0\} = 0$ (the empty list is a basis of $\{0\}$).

Proof. • If W is spanned by $w_1, \dots, w_m$, then any $k > m$ vectors $v_1, \dots, v_k$ of W are linearly dependent. Indeed, write $v_j = a_{1j}w_1 + \dots + a_{mj}w_m$ ($1 \leq j \leq k$) and $A = (a_{ij}) \in \mathbb{R}^{m \times k}$. As $k > m$, the columns of A are dependent (Theorem 4.1): $Ax = 0$ for some $x \neq 0$. Then

$$x_1 v_1 + \dots + x_k v_k = \sum_{j=1}^k x_j \sum_{i=1}^m a_{ij} w_i = \sum_{i=1}^m (Ax)_i w_i = 0$$

- Same size: if $v_1, \dots, v_k$ and $w_1, \dots, w_m$ are bases of W , each list spans W and is independent, so the first item gives $k \leq m$ and $m \leq k$.
- Existence: start with the empty list and, as long as the vectors chosen do not span W , adjoin a vector $v \in W$ outside their span. The list stays independent: in a relation $c_1 v_1 + \dots + c_k v_k + cv = 0$, $c \neq 0$ would put v in the span of $v_1, \dots, v_k$, so $c = 0$, and then every $c_i = 0$. More than n vectors of $\mathbb{R}^n$ are dependent (Theorem 4.1), so the process stops after at most n steps, at a basis of W . $\square$

By Theorem 4.3, $\dim \text{span}\{a_1, \dots, a_n\} = r$, the number of pivots of the RREF of $[a_1 \ \dots \ a_n]$.

Theorem 4.5. Let W be a subspace of $\mathbb{R}^n$ with $\dim W = d$.

- If U is a subspace of W , then $\dim U \leq d$, and $\dim U = d$ only for $U = W$. In particular $d \leq n$, and $d = n$ only for $W = \mathbb{R}^n$.
- Every linearly independent list of vectors of W can be extended to a basis of W .
- More than d vectors of W are linearly dependent; fewer than d vectors do not span W .
- If $d \geq 1$, then d vectors of W are linearly independent $\Leftrightarrow$ they span $W \Leftrightarrow$ they form a basis of W .

Proof. If $d = 0$, then $W = \{0\}$, and (a)–(c) are immediate: every subspace U of W is $\{0\} = W$; every vector of W is 0, so a nonempty list of vectors of W is dependent (Proposition 3.12(a)); and the empty list, the only independent one, is a basis of W (Definition 3.14). Let $d \geq 1$, and let $w_1, \dots, w_d$ be a basis of W (Theorem 4.4). We prove (c) first.

(c)

- The vectors $w_1, \dots, w_d$ span W , so any $k > d$ vectors of W are linearly dependent, by the first step of the proof of Theorem 4.4 (with $m = d$).
- If $k < d$ vectors spanned W , the same step, applied to them, would make the $d > k$ vectors $w_1, \dots, w_d$ of W dependent ✗

(b)

- Let $v_1, \dots, v_k$ be linearly independent vectors of W . If they span W , they form a basis of W .
- Otherwise take $v \in W$ outside $\text{span}\{v_1, \dots, v_k\}$. Then $v_1, \dots, v_k, v$ are independent: in a relation $c_1 v_1 + \dots + c_k v_k + cv = 0$, $c \neq 0$ would put v in the span, so $c = 0$, and then every $c_i = 0$.
- Repeat this. Each list obtained consists of independent vectors of W , so it has at most d vectors by (c); hence the process stops, at a basis of W that contains $v_1, \dots, v_k$.

(a)

- A basis of U (Theorem 4.4) consists of $\dim U$ independent vectors of W , so $\dim U \leq d$ by (c).
- If $U \neq W$, take $v \in W$ outside U . Adjoining v to a basis of U gives, as in (b), $\dim U + 1$ independent vectors of W , so $\dim U + 1 \leq d$ by (c). Hence $\dim U = d$ only for $U = W$.
- For the last sentence, apply this to W as a subspace of $\mathbb{R}^n$, which has dimension n (Theorem 4.4).

(d) Let $v_1, \dots, v_d \in W$. A basis of W is independent and spans W (Definition 3.14).

- Let $v_1, \dots, v_d$ be independent, and let $w \in W$. By (c), the $d+1$ vectors $w, v_1, \dots, v_d$ are dependent: $x_0 w + x_1 v_1 + \dots + x_d v_d = 0$ with not all $x_i = 0$. Here $x_0 \neq 0$, since $v_1, \dots, v_d$ are independent; so $w = -(x_1/x_0)v_1 - \dots - (x_d/x_0)v_d$. Thus $v_1, \dots, v_d$ span W , and they form a basis.
- Let $v_1, \dots, v_d$ span W . By Theorem 4.3, some of them form a basis of W , which has d vectors (Theorem 4.4); so it consists of all of $v_1, \dots, v_d$, and they form a basis. $\square$

Transposition carries a subspace of $\mathbb{R}^{1 \times n}$ onto a subspace of $\mathbb{R}^n$ and preserves linear combinations, hence independence, spanning and bases; so Theorems 4.4 and 4.5 also hold for subspaces of $\mathbb{R}^{1 \times n}$, such as the row space $\mathcal{R}(A)$ below.

Example 4.6. Let $W = \text{span}\{u_1, u_2, u_3, u_4\} \subseteq \mathbb{R}^3$, where

$$u_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad u_2 = \begin{pmatrix} -2 \\ -4 \\ 2 \end{pmatrix}, \quad u_3 = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \quad u_4 = \begin{pmatrix} 3 \\ 7 \\ -2 \end{pmatrix}$$

(a) Are $u_1, \dots, u_4$ linearly independent? (b) Find a basis of W among $u_1, \dots, u_4$, and $\dim W$. (c) Express the remaining vectors in this basis.

Solution. Row-reduce $A = [u_1 \ u_2 \ u_3 \ u_4]$:

$$\begin{aligned} \begin{pmatrix} 1 & -2 & 1 & 3 \\ 2 & -4 & 3 & 7 \\ -1 & 2 & 0 & -2 \end{pmatrix} &\xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & -2 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & -2 & 1 & 3 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \\ &\xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & -2 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = R \end{aligned}$$

So $r = 2$, with pivot columns $k_1 = 1$, $k_2 = 3$.

- (a) No: $r = 2 < 4$ (Theorem 4.1); indeed any four vectors of $\mathbb{R}^3$ are dependent.
- (b) By Theorem 4.3, u_1, u_3 form a basis of W , and $\dim W = 2$. As $2 < 3$, $W \neq \mathbb{R}^3$ (Theorem 4.5(a)).
- (c) Columns 2 and 4 of R give $u_2 = -2u_1$ and $u_4 = 2u_1 + u_3$ (Theorem 4.3(b)). Check: $2u_1 + u_3 = (2, 4, -2)' + (1, 3, 0)' = (3, 7, -2)' = u_4$. Equivalently, $2u_1 + u_2 = 0$ and $2u_1 + u_3 - u_4 = 0$.

The four subspaces of a matrix

Definition 4.7. For $A \in \mathbb{R}^{m \times n}$,

$$\begin{aligned}\mathcal{C}(A) &= \{Ax : x \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, & \mathcal{R}(A) &= \{y'A : y \in \mathbb{R}^m\} \subseteq \mathbb{R}^{1 \times n}, \\ \mathcal{N}(A) &= \{x \in \mathbb{R}^n : Ax = 0\} \subseteq \mathbb{R}^n, & \mathcal{N}(A') &= \{y \in \mathbb{R}^m : A'y = 0\} \subseteq \mathbb{R}^m\end{aligned}$$

are the **column space**, **row space**, **null space** and **left null space** of A . The columns of A span $\mathcal{C}(A)$, the rows of A span $\mathcal{R}(A)$ (a space of row vectors), $\mathcal{N}(A)$ is the set of solutions of $Ax = 0$, and $y \in \mathcal{N}(A')$ if and only if $y'A = 0$ (transpose both sides). All four are subspaces (Theorem 3.7 for $\mathcal{C}(A)$ and $\mathcal{R}(A)$, Example 3.5(e) for $\mathcal{N}(A)$ and $\mathcal{N}(A')$).

Let R be the RREF of $A \in \mathbb{R}^{m \times n}$, with r nonzero rows, pivot columns $k_1 < \dots < k_r$ and other columns $f_1 < \dots < f_{n-r}$, so that $x_{f_1}, \dots, x_{f_{n-r}}$ are the free unknowns of $Ax = 0$ (Proposition 2.7(b)). For $j = 1, \dots, n - r$, the **special solution** $s_j \in \mathbb{R}^n$ is the solution of $Ax = 0$ with free unknowns $x_{f_j} = 1$ and $x_{f_l} = 0$ ($l \neq j$). By Proposition 2.7(b), its entries are

$$(s_j)_{f_l} = \delta_{jl} \quad (1 \leq l \leq n - r), \quad (s_j)_{k_i} = -R_{if_j} \quad (1 \leq i \leq r)$$

Theorem 4.8. Let $A \in \mathbb{R}^{m \times n}$, and let row operations take $[A \mid I_m]$ to $[R \mid P]$, where R is the RREF of A , with nonzero rows $\rho_1, \dots, \rho_r$, pivot columns $k_1 < \dots < k_r$ and free columns $f_1 < \dots < f_{n-r}$. Then $P \in \mathbb{R}^{m \times m}$ is invertible, $PA = R$, and:

- (a) if $r \geq 1$, the pivot columns $a_{k_1}, \dots, a_{k_r}$ of A form a basis of $\mathcal{C}(A)$;
- (b) if $r \geq 1$, the nonzero rows $\rho_1, \dots, \rho_r$ of R form a basis of $\mathcal{R}(A)$;
- (c) the special solutions $s_1, \dots, s_{n-r}$ form a basis of $\mathcal{N}(A)$ (if $r = n$, then $\mathcal{N}(A) = \{0\}$, with the empty basis);
- (d) the transposed rows $\text{row}_{r+1}(P)', \dots, \text{row}_m(P)'$ form a basis of $\mathcal{N}(A')$ (if $r = m$, then $\mathcal{N}(A') = \{0\}$, with the empty basis).

Hence $\dim \mathcal{C}(A) = \dim \mathcal{R}(A) = r$, $\dim \mathcal{N}(A) = n - r$ and $\dim \mathcal{N}(A') = m - r$ (if $r = 0$, then $A = 0$ and $\mathcal{C}(A) = \mathcal{R}(A) = \{0\}$).

Proof. The row operations multiply $[A \mid I_m]$ on the left by an invertible matrix E (Theorem 2.14(c)), and $E[A \mid I_m] = [EA \mid E]$, column by column (Theorem 1.4(a)). So $P = E$ is invertible and $PA = R$.

(a) $\mathcal{C}(A) = \text{span}\{a_1, \dots, a_n\}$; apply Theorem 4.3.

(b)

- Same row space: for $y \in \mathbb{R}^m$, $y'R = (y'P)A$ and $y'A = (y'P^{-1})R$, where $y'P = (P'y)'$ and $y'P^{-1} = ((P^{-1})'y)'$. So $\mathcal{R}(R) \subseteq \mathcal{R}(A)$ and $\mathcal{R}(A) \subseteq \mathcal{R}(R)$, that is, $\mathcal{R}(R) = \mathcal{R}(A)$.
- Spanning: the rows of R span $\mathcal{R}(R)$, and rows $r+1, \dots, m$ are zero, so $\rho_1, \dots, \rho_r$ span $\mathcal{R}(R) = \mathcal{R}(A)$.
- Independent: $\text{col}_{k_i}(R) = e_i$, so entry k_i of $c_1\rho_1 + \dots + c_r\rho_r$ is c_i ; if this combination is 0, every $c_i = 0$.

(c) The solutions of $Ax = 0$ are those of $Rx = 0$ (Theorem 2.3(b)).

- Independent: entry f_l of $\sum_j c_j s_j$ is c_l , so $\sum_j c_j s_j = 0$ forces every $c_l = 0$.
- Spanning: let $t \in \mathcal{N}(A)$ and $\tilde{t} = \sum_j t_{f_j} s_j$. Then $\tilde{t} \in \mathcal{N}(A)$, and $\tilde{t}$ has the same free unknowns as t . A solution is determined by its free unknowns (Proposition 2.7(b)), so $t = \tilde{t} \in \text{span}\{s_1, \dots, s_{n-r}\}$.
- If $r = n$, there are no free unknowns and $x = 0$ is the only solution.

(d)

- In $\mathcal{N}(A')$: for $i > r$, $\text{row}_i(P)A = \text{row}_i(PA) = \text{row}_i(R) = 0$ (Theorem 1.4(c)), so $A' \text{row}_i(P)' = (\text{row}_i(P)A)' = 0$.
- Independent: if $c'P = c_1 \text{row}_1(P) + \cdots + c_m \text{row}_m(P) = 0$, then $c' = c'PP^{-1} = 0$; so the rows of P are independent, and so are rows $r+1, \dots, m$ (Proposition 3.12(b)).
- Spanning: let $A'y = 0$, so $y'A = 0$, and put $z' = y'P^{-1}$, so that $y' = z'P = \sum_{i=1}^m z_i \text{row}_i(P)$. Then $0 = y'A = z'PA = z'R = z_1 \rho_1 + \cdots + z_r \rho_r$, and (b) gives $z_1 = \cdots = z_r = 0$. So $y' = \sum_{i>r} z_i \text{row}_i(P)$ (and $y = 0$ if $r = m$). $\square$

Remark 4.9. Row operations do not change $\mathcal{R}(A)$ or $\mathcal{N}(A)$, so R may be used for these two spaces. They do change $\mathcal{C}(A)$ in general: a basis of $\mathcal{C}(A)$ consists of the pivot columns of A itself, not of R (Example 4.10).

Example 4.10. Find bases of the four subspaces of

$$A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{pmatrix}$$

Solution. Row-reduce $[A \mid I_3]$:

$$\begin{aligned} & \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 1 & 3 & 0 & 5 & 0 & 1 & 0 \\ 2 & 5 & 1 & 8 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 1 & -1 & 2 & -2 & 0 & 1 \end{array} \right) \\ & \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 3 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cccc|ccc} 1 & 0 & 3 & -1 & 3 & -2 & 0 \\ 0 & 1 & -1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & -1 & 1 \end{array} \right) = [R \mid P] \end{aligned}$$

So $r = 2$, with pivot columns 1, 2 and free unknowns x_3, x_4 . (The pivots are already visible in the echelon form after the second arrow; Remark 4.2.)

- $\mathcal{C}(A)$: $a_1 = (1, 1, 2)'$ and $a_2 = (2, 3, 5)'$ form a basis; $\dim \mathcal{C}(A) = 2$. Columns 3, 4 of R give

$$a_3 = 3a_1 - a_2, \quad a_4 = -a_1 + 2a_2$$

Check: $3(1, 1, 2)' - (2, 3, 5)' = (1, 0, 1)'$ and $-(1, 1, 2)' + 2(2, 3, 5)' = (3, 5, 8)'$.

- $\mathcal{R}(A)$: $\rho_1 = (1, 0, 3, -1)$ and $\rho_2 = (0, 1, -1, 2)$ form a basis; $\dim \mathcal{R}(A) = 2$.
- $\mathcal{N}(A)$: $Rx = 0$ reads $x_1 + 3x_3 - x_4 = 0$, $x_2 - x_3 + 2x_4 = 0$. The special solutions

$$s_1 = \begin{pmatrix} -3 \\ 1 \\ 1 \\ 0 \end{pmatrix} \quad (x_3 = 1, x_4 = 0), \quad s_2 = \begin{pmatrix} 1 \\ -2 \\ 0 \\ 1 \end{pmatrix} \quad (x_3 = 0, x_4 = 1)$$

form a basis of $\mathcal{N}(A)$; $\dim \mathcal{N}(A) = 2 = 4 - 2$. Note $As_1 = -3a_1 + a_2 + a_3 = 0$ is the relation $a_3 = 3a_1 - a_2$: each special solution is a column relation.

- $\mathcal{N}(A')$: row 3 of P gives the basis $y = (-1, -1, 1)'$; $\dim \mathcal{N}(A') = 1 = 3 - 2$. Indeed $y'A = -\text{row}_1(A) - \text{row}_2(A) + \text{row}_3(A) = 0$, that is, $\text{row}_3(A) = \text{row}_1(A) + \text{row}_2(A)$: each vector of $\mathcal{N}(A')$ is a row relation.
- Warning: every column of R has third entry 0, so every vector of $\mathcal{C}(R)$ does; but a_1 has third entry 2. Thus $\mathcal{C}(R) \neq \mathcal{C}(A)$, and the pivot columns e_1, e_2 of R do not span $\mathcal{C}(A)$.

Rank

Definition 4.11. By Theorem 4.8, $\dim \mathcal{C}(A) = \dim \mathcal{R}(A) = r$, the number of pivots of the RREF of A (also when $r = 0$). This common number is the **rank** of A , written $\text{rank } A$; by Remark 4.2 it is also the number of nonzero rows of any echelon form of A . A has **full column rank** if $\text{rank } A = n$, and **full row rank** if $\text{rank } A = m$. The number $\dim \mathcal{N}(A)$ is the **nullity** of A .

Theorem 4.12. For every $A \in \mathbb{R}^{m \times n}$,

$$\text{rank } A + \dim \mathcal{N}(A) = n, \quad \text{rank } A + \dim \mathcal{N}(A') = m$$

Proof. Each of the n columns of R is either a pivot column (r of them) or a free column ($n - r$ of them), and by Theorem 4.8(c), $\dim \mathcal{N}(A) = n - r$, the number of free unknowns. Likewise $\dim \mathcal{N}(A') = m - r$, the number of zero rows of R (Theorem 4.8(d)). Since $\text{rank } A = r$, both equalities follow. $\square$

Theorem 4.13. Let $A \in \mathbb{R}^{m \times n}$.

- (a) $\text{rank } A' = \text{rank } A$, and $\text{rank } A \leq \min\{m, n\}$.
- (b) A has full column rank $\Leftrightarrow$ its columns are linearly independent $\Leftrightarrow \mathcal{N}(A) = \{0\}$.
- (c) A has full row rank $\Leftrightarrow$ its rows are linearly independent $\Leftrightarrow \mathcal{N}(A') = \{0\} \Leftrightarrow \mathcal{C}(A) = \mathbb{R}^m$.
- (d) If $m = n$, then A is invertible $\Leftrightarrow \text{rank } A = n$.

Proof. (a) For $y \in \mathbb{R}^m$, $(y'A)' = A'y$, so transposing maps $\mathcal{R}(A)$ onto $\mathcal{C}(A')$. It preserves linear combinations and is undone by transposing again, so it carries a basis of $\mathcal{R}(A)$ to a basis of $\mathcal{C}(A')$, and $\text{rank } A' = \dim \mathcal{C}(A') = \dim \mathcal{R}(A) = \text{rank } A$. Also $\text{rank } A = r \leq m$, as R has m rows, and $r \leq n$, as $k_1 < \dots < k_r$ lie in $\{1, \dots, n\}$.

(b) Since $\text{rank } A = r$, this is Theorem 4.1: (iii) $\Leftrightarrow$ (i) $\Leftrightarrow$ (ii).

- (c)
- $\text{rank } A = m$ if and only if $\dim \mathcal{N}(A') = m - \text{rank } A = 0$ (Theorem 4.12), that is, $\mathcal{N}(A') = \{0\}$.
 - $y'A = y_1 \text{row}_1(A) + \dots + y_m \text{row}_m(A)$ (Theorem 1.4(c)), and $y'A = 0$ if and only if $A'y = 0$. So $\mathcal{N}(A') = \{0\}$ says exactly that the rows of A are independent.
 - $\text{rank } A = \dim \mathcal{C}(A)$, and a subspace of $\mathbb{R}^m$ has dimension m only if it is $\mathbb{R}^m$ (Theorem 4.5(a)).
- (d) A square matrix A is invertible if and only if $Ax = 0$ has only the solution $x = 0$ (Theorem 2.15), that is, $\mathcal{N}(A) = \{0\}$; by (b), this means $\text{rank } A = n$. $\square$

Corollary 4.14. Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and let $[A \mid b] \in \mathbb{R}^{m \times (n+1)}$ be the augmented matrix.

- (a) $Ax = b$ is consistent $\Leftrightarrow b \in \mathcal{C}(A) \Leftrightarrow \text{rank}[A \mid b] = \text{rank } A$.
- (b) If $Ax = b$ is consistent, its solution is unique $\Leftrightarrow \text{rank } A = n$.

Proof. (a)

- $Ax = x_1 a_1 + \dots + x_n a_n$ (Theorem 1.4(b)), so $Ax = b$ has a solution if and only if b is a linear combination of the columns of A , that is, $b \in \mathcal{C}(A)$.
- $\mathcal{C}(A) \subseteq \mathcal{C}([A \mid b])$: a combination of $a_1, \dots, a_n$ is one of $a_1, \dots, a_n, b$, with coefficient 0 on b .
- If $b = Ac$, then $x_1 a_1 + \dots + x_n a_n + tb = A(x + tc) \in \mathcal{C}(A)$; so $\mathcal{C}([A \mid b]) = \mathcal{C}(A)$, and the ranks are equal.
- If the ranks are equal, the subspace $\mathcal{C}(A)$ of $\mathcal{C}([A \mid b])$ has the same dimension, so the two are equal (Theorem 4.5(a)), and $b \in \mathcal{C}([A \mid b]) = \mathcal{C}(A)$.

(b) By Theorem 2.20(a), the solutions are $x_p + z$, where x_p is one solution and z runs through the solutions of $Ax = 0$, that is, through $\mathcal{N}(A)$. So the solution is unique if and only if $\mathcal{N}(A) = \{0\}$, that is, $\text{rank } A = n$ (Theorem 4.13(b)). $\square$

In practice, consistency is read off from an echelon form of $[A \mid b]$ (Theorem 2.19).

Rank equalities and inequalities

Theorem 4.15. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$.

- (a) $\mathcal{C}(AB) \subseteq \mathcal{C}(A)$ and $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$; and $\text{rank}(AB) \leq \min\{\text{rank } A, \text{rank } B\}$.
- (b) If A has full column rank, then $\mathcal{N}(AB) = \mathcal{N}(B)$ and $\text{rank}(AB) = \text{rank } B$.
- (c) If B has full row rank, then $\mathcal{C}(AB) = \mathcal{C}(A)$ and $\text{rank}(AB) = \text{rank } A$.
- (d) $\text{rank}(PAQ) = \text{rank } A$ for all invertible $P \in \mathbb{R}^{m \times m}$ and $Q \in \mathbb{R}^{n \times n}$.

Proof. (a)

- $ABx = A(Bx) \in \mathcal{C}(A)$, and $Bx = 0$ gives $ABx = A0 = 0$.
- So $\text{rank}(AB) = \dim \mathcal{C}(AB) \leq \dim \mathcal{C}(A) = \text{rank } A$ (Theorem 4.5(a)).
- By this, $(AB)' = B'A'$ (Theorem 1.8) and Theorem 4.13(a), $\text{rank}(AB) = \text{rank}(B'A') \leq \text{rank } B' = \text{rank } B$.

(b)

- If $ABx = 0$, then $Bx \in \mathcal{N}(A) = \{0\}$ (Theorem 4.13(b)), so $x \in \mathcal{N}(B)$. With (a), $\mathcal{N}(AB) = \mathcal{N}(B)$.
- AB and B both have p columns, so by Theorem 4.12, $\text{rank}(AB) = p - \dim \mathcal{N}(AB) = p - \dim \mathcal{N}(B) = \text{rank } B$.

(c)

- $\mathcal{C}(B) = \mathbb{R}^n$ (Theorem 4.13(c)): every $x \in \mathbb{R}^n$ is Bw for some $w \in \mathbb{R}^p$, and then $Ax = (AB)w \in \mathcal{C}(AB)$.
- So $\mathcal{C}(A) \subseteq \mathcal{C}(AB)$; with (a), $\mathcal{C}(AB) = \mathcal{C}(A)$, and the ranks are equal.

(d) $Px = 0$ gives $x = P^{-1}Px = 0$, so P has full column rank (Theorem 4.13(b)); and $\text{rank } Q = n$ (Theorem 4.13(d)), so Q has full row rank. By (b) and (c), $\text{rank}(P(AQ)) = \text{rank}(AQ) = \text{rank } A$. $\square$

Theorem 4.16. (a) For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{m \times p}$, let $[A \ B] \in \mathbb{R}^{m \times (n+p)}$ have the columns of A followed by those of B . Then

$$\max\{\text{rank } A, \text{rank } B\} \leq \text{rank}[A \ B] \leq \text{rank } A + \text{rank } B$$

(b) For $A \in \mathbb{R}^{m \times n}$ and $C \in \mathbb{R}^{q \times n}$, let $\begin{pmatrix} A \\ C \end{pmatrix} \in \mathbb{R}^{(m+q) \times n}$ have the rows of A followed by those of C . Then

$$\max\{\text{rank } A, \text{rank } C\} \leq \text{rank} \begin{pmatrix} A \\ C \end{pmatrix} \leq \text{rank } A + \text{rank } C$$

(c) For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{q \times p}$, let $\text{diag}(A, B) = \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \in \mathbb{R}^{(m+q) \times (n+p)}$, with zero blocks of sizes $m \times p$ and $q \times n$. Then $\text{rank } \text{diag}(A, B) = \text{rank } A + \text{rank } B$.

Proof. (a)

- A combination of the columns of A , or of those of B , is a combination of the columns of $[A \ B]$ (the others with coefficient 0). So $\mathcal{C}(A)$ and $\mathcal{C}(B)$ are subspaces of $\mathcal{C}([A \ B])$, and Theorem 4.5(a) gives the lower bound.

- Let $r = \text{rank } A$, $s = \text{rank } B$, and take bases $v_1, \dots, v_r$ of $\mathcal{C}(A)$ and $w_1, \dots, w_s$ of $\mathcal{C}(B)$. Every column of $[A \ B]$ is a combination of these $r + s$ vectors of $\mathcal{C}([A \ B])$, so they span $\mathcal{C}([A \ B])$ (Theorem 3.7), and $\text{rank}[A \ B] \leq r + s$ (Theorem 4.5(c)).

(b) Transposing turns the rows of $\begin{pmatrix} A \\ C \end{pmatrix}$ into the columns of $[A' \ C']$, so $\begin{pmatrix} A \\ C \end{pmatrix}' = [A' \ C']$. Apply (a) to A' and C' , and use Theorem 4.13(a).

(c) Let $M = \text{diag}(A, B)$, $r = \text{rank } A$ and $s = \text{rank } B$.

- Upper bound: $M = [M_1 \ M_2]$ with $M_1 = \begin{pmatrix} A \\ 0 \end{pmatrix}$ and $M_2 = \begin{pmatrix} 0 \\ B \end{pmatrix}$. By (b), and as a zero matrix has rank 0, $r \leq \text{rank } M_1 \leq r + 0$; so $\text{rank } M_1 = r$, and likewise $\text{rank } M_2 = s$. By (a), $\text{rank } M \leq r + s$.
- Lower bound: let $a_{k_1}, \dots, a_{k_r}$ and $b_{l_1}, \dots, b_{l_s}$ be the pivot columns of A and of B , bases of $\mathcal{C}(A)$ and $\mathcal{C}(B)$ (Theorem 4.8(a)). The corresponding columns $\begin{pmatrix} a_{k_i} \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ b_{l_j} \end{pmatrix}$ of M are independent: a combination of them is $\begin{pmatrix} \sum_i \alpha_i a_{k_i} \\ \sum_j \beta_j b_{l_j} \end{pmatrix}$, which is 0 only if all α_i and β_j are 0. These $r + s$ vectors lie in $\mathcal{C}(M)$, so $\text{rank } M \geq r + s$ (Theorem 4.5(c)). $\square$

Corollary 4.17. For $A, B \in \mathbb{R}^{m \times n}$, $\text{rank}(A + B) \leq \text{rank } A + \text{rank } B$.

Proof. For $x \in \mathbb{R}^n$, $(A + B)x = Ax + Bx$ is a combination of the columns of A and of B , so it lies in $\mathcal{C}([A \ B])$. Hence $\mathcal{C}(A + B) \subseteq \mathcal{C}([A \ B])$, so $\text{rank}(A + B) \leq \text{rank}[A \ B]$ (Theorem 4.5(a)), and $\text{rank}[A \ B] \leq \text{rank } A + \text{rank } B$ (Theorem 4.16(a)). $\square$

Theorem 4.18. Let $A \in \mathbb{R}^{m \times n}$.

- (a) $\mathcal{N}(A'A) = \mathcal{N}(A)$.
- (b) $\text{rank}(A'A) = \text{rank } A = \text{rank}(AA')$.
- (c) $\mathcal{C}(A'A) = \mathcal{C}(A')$ and $\mathcal{C}(AA') = \mathcal{C}(A)$.

Proof. (a)

- If $Ax = 0$, then $A'Ax = A'0 = 0$.
- If $A'Ax = 0$, then $(Ax)'(Ax) = x'A'Ax = 0$. But $(Ax)'(Ax)$ is the sum of the squares of the entries of Ax (Theorem 1.8), so $Ax = 0$.

(b) $A'A$ and A both have n columns, so by (a) and Theorem 4.12, $\text{rank}(A'A) = n - \dim \mathcal{N}(A'A) = n - \dim \mathcal{N}(A) = \text{rank } A$. Applied to A' in place of A , this gives $\text{rank}(AA') = \text{rank } A' = \text{rank } A$ (Theorem 4.13(a)).

(c) $\mathcal{C}(A'A) \subseteq \mathcal{C}(A')$ (Theorem 4.15(a)), and both have dimension $\text{rank } A$, by (b) and Theorem 4.13(a); so they are equal (Theorem 4.5(a)). Likewise $\mathcal{C}(AA') \subseteq \mathcal{C}(A)$, both of dimension $\text{rank } A$. $\square$

Theorem 4.19 (Sylvester's inequality). For $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$,

$$\dim \mathcal{N}(AB) \leq \dim \mathcal{N}(A) + \dim \mathcal{N}(B), \quad \text{equivalently} \quad \text{rank}(AB) \geq \text{rank } A + \text{rank } B - n$$

In particular, if $AB = 0$, then $\text{rank } A + \text{rank } B \leq n$.

Proof. • Let $z_1, \dots, z_s$ be a basis of $\mathcal{N}(B)$. Since $\mathcal{N}(B) \subseteq \mathcal{N}(AB)$ (Theorem 4.15(a)), it extends to a basis $z_1, \dots, z_s, w_1, \dots, w_t$ of $\mathcal{N}(AB)$ (Theorem 4.5(b)); so $\dim \mathcal{N}(AB) = s + t$.

- The vectors $Bw_1, \dots, Bw_t$ lie in $\mathcal{N}(A)$, since $A(Bw_j) = (AB)w_j = 0$.
- They are independent: if $c_1Bw_1 + \dots + c_tBw_t = 0$, then $B(c_1w_1 + \dots + c_tw_t) = 0$, so $c_1w_1 + \dots + c_tw_t \in \mathcal{N}(B)$ equals some $d_1z_1 + \dots + d_s z_s$; as $z_1, \dots, z_s, w_1, \dots, w_t$ are independent, every $c_j = 0$.
- Hence $t \leq \dim \mathcal{N}(A)$ (Theorem 4.5(c)), and $\dim \mathcal{N}(AB) = s + t \leq \dim \mathcal{N}(B) + \dim \mathcal{N}(A)$.

- By Theorem 4.12, $\dim \mathcal{N}(AB) = p - \text{rank}(AB)$, $\dim \mathcal{N}(B) = p - \text{rank } B$ and $\dim \mathcal{N}(A) = n - \text{rank } A$. Substituting gives $p - \text{rank}(AB) \leq (p - \text{rank } B) + (n - \text{rank } A)$, which is the rank form. If $AB = 0$, then $\text{rank}(AB) = 0$. $\square$

Example 4.20. Let $A = \text{diag}(1, 1, 0)$ and $B = \text{diag}(0, 1, 1)$ in $\mathbb{R}^{3 \times 3}$. Their columns are $e_1, e_2, 0$ and $0, e_2, e_3$, so $\mathcal{C}(A) = \text{span}\{e_1, e_2\}$, $\mathcal{C}(B) = \text{span}\{e_2, e_3\}$ and $\text{rank } A = \text{rank } B = 2$.

- $AB = \text{diag}(0, 1, 0)$ (row i of AB is $a_{ii} \text{row}_i(B)$, Theorem 1.4(c)) has rank 1: less than $\min\{2, 2\}$ (Theorem 4.15(a)), and equal to $\text{rank } A + \text{rank } B - 3 = 1$, so Sylvester's bound is attained (Theorem 4.19).
- $A + B = \text{diag}(1, 2, 1)$ has rank $3 < 2 + 2$ (Corollary 4.17); and $A + (-A) = 0$ has rank 0.
- $[A \ B] \in \mathbb{R}^{3 \times 6}$ has the columns $e_1, e_2, 0, 0, e_2, e_3$, so $\mathcal{C}([A \ B]) = \mathbb{R}^3$ and $\text{rank}[A \ B] = 3$, strictly between $\max\{2, 2\}$ and $2 + 2$ (Theorem 4.16(a)).
- $\text{diag}(A, B) = \text{diag}(1, 1, 0, 0, 1, 1) \in \mathbb{R}^{6 \times 6}$ has rank $4 = 2 + 2$ (Theorem 4.16(c)).
- $A'A = A^2 = A$ has rank $2 = \text{rank } A$ (Theorem 4.18).

Theorem 4.21. Let $A \in \mathbb{R}^{m \times n}$ have rank r . If $r \geq 1$, a product $A = BC$ with $B \in \mathbb{R}^{m \times r}$ of full column rank and $C \in \mathbb{R}^{r \times n}$ of full row rank is a **full-rank factorization** of A .

- If $r \geq 1$, let R be the RREF of A , with pivot columns $k_1 < \dots < k_r$; let $B = [a_{k_1} \ \dots \ a_{k_r}] \in \mathbb{R}^{m \times r}$ consist of the pivot columns of A , and $C \in \mathbb{R}^{r \times n}$ of the r nonzero rows of R . Then $A = BC$ is a full-rank factorization. If $r = 0$, then $A = 0$, and there is nothing to factor.
- For every full-rank factorization $A = BC$, $\mathcal{C}(A) = \mathcal{C}(B)$ and $\mathcal{N}(A) = \mathcal{N}(C)$.
- For every full-rank factorization $A = BC$, the $r \times r$ matrices $B'B$ and CC' are invertible, and

$$L = (B'B)^{-1}B' \in \mathbb{R}^{r \times m}, \quad K = C'(CC')^{-1} \in \mathbb{R}^{n \times r}$$

satisfy $LB = I_r$ and $CK = I_r$: B has a left inverse and C has a right inverse.

Proof. (a)

- Column j of BC is $B \text{col}_j(C) = R_{1j}a_{k_1} + \dots + R_{rj}a_{k_r}$ (Theorem 1.4(a),(b)), which is a_j by Theorem 4.3(b). So $BC = A$.
- The columns of B form a basis of $\mathcal{C}(A)$, so they are independent: B has full column rank (Theorem 4.13(b)).
- The rows of C form a basis of $\mathcal{R}(A)$ (Theorem 4.8(b)), so they are independent: C has full row rank (Theorem 4.13(c)).
- If $r = 0$, then $A = 0$ (Theorem 4.8).

(b) Theorem 4.15(b) gives $\mathcal{N}(BC) = \mathcal{N}(C)$, as B has full column rank, and Theorem 4.15(c) gives $\mathcal{C}(BC) = \mathcal{C}(B)$, as C has full row rank.

(c)

- $\text{rank}(B'B) = \text{rank } B = r$ and $\text{rank}(CC') = \text{rank } C = r$ (Theorem 4.18(b)), so the $r \times r$ matrices $B'B$ and CC' are invertible (Theorem 4.13(d)).
- $LB = (B'B)^{-1}(B'B) = I_r$ and $CK = (CC')(CC')^{-1} = I_r$. $\square$

Example 4.22. For A of Example 4.10, Theorem 4.21(a) gives

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 1 & 3 \\ 2 & 5 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -1 & 2 \end{pmatrix}$$

- Columns 1, 2 of BC are a_1, a_2 , and columns 3, 4 are $3a_1 - a_2 = a_3$ and $-a_1 + 2a_2 = a_4$ (Example 4.10).
- By Theorem 4.21(b), $\mathcal{C}(A) = \mathcal{C}(B) = \text{span}\{a_1, a_2\}$ and $\mathcal{N}(A) = \mathcal{N}(C) = \text{span}\{s_1, s_2\}$: the system $Cx = 0$ is $Rx = 0$ without its zero row.
- With $N = [s_1 \ s_2] \in \mathbb{R}^{4 \times 2}$, $AN = 0$ and $\text{rank } N = 2$, and Sylvester's inequality $\text{rank } A + \text{rank } N \leq 4$ holds with equality: $2 + 2 = 4$.

Exercises and solutions

Exercise 4.1. Let $a_1, \dots, a_9$ be the columns of

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 & 5 \\ 0 & 1 & 1 & 2 & 2 & 3 & -1 & 3 & 3 \\ 1 & 1 & 2 & 3 & 5 & 5 & 1 & 0 & 4 \end{pmatrix}$$

Which of the following form a basis of $\mathbb{R}^3$? (a) a_1, a_2 ; (b) a_1, a_4, a_6, a_8 ; (c) a_2, a_4, a_7 ; (d) a_3, a_5, a_9 .

Solution. (a), (b) No: $\dim \mathbb{R}^3 = 3$, so two vectors cannot span $\mathbb{R}^3$, and four vectors of $\mathbb{R}^3$ are dependent (Theorem 4.5(c)).

(c) Yes. Three vectors of $\mathbb{R}^3$ form a basis if and only if they are independent (Theorem 4.5(d)). Row-reduce $[a_2 \ a_4 \ a_7]$:

$$\begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & -1 \\ 1 & 3 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 0 & 2 & -1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 5 \end{pmatrix} \\ \xrightarrow[R_2 \leftarrow R_2 + 3R_3]{R_3 \leftarrow \frac{1}{5}R_3, \ R_1 \leftarrow R_1 - 2R_3} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

All three columns are pivot columns, so a_2, a_4, a_7 are independent (Theorem 4.1).

(d) No: the three vectors are dependent, since

$$7a_3 - 2a_5 - a_9 = 7 \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 2 \\ 5 \end{pmatrix} - \begin{pmatrix} 5 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 7 - 2 - 5 \\ 7 - 4 - 3 \\ 14 - 10 - 4 \end{pmatrix} = 0$$

Exercise 4.2. Let $W = \mathcal{C}(A) \subseteq \mathbb{R}^4$ for

$$A = \begin{pmatrix} 1 & 2 & 3 \\ -2 & 3 & 8 \\ 5 & 1 & -3 \\ -3 & -4 & -5 \end{pmatrix}$$

(a) Find a basis of W . (b) What is $\dim W$?

Solution. Let a_1, a_2, a_3 be the columns of A ; they span W .

(a) a_1, a_2 are independent: otherwise, as $a_1 \neq 0$, $a_2 = ca_1$ for some c (Proposition 3.12(c)); the first entries give $c = 2$, and then the second entries give $3 = 2(-2)$ ✖. But

$$a_1 - 2a_2 + a_3 = \begin{pmatrix} 1 - 4 + 3 \\ -2 - 6 + 8 \\ 5 - 2 - 3 \\ -3 + 8 - 5 \end{pmatrix} = 0,$$

so $a_3 = 2a_2 - a_1$, and every combination $x_1a_1 + x_2a_2 + x_3a_3 = (x_1 - x_3)a_1 + (x_2 + 2x_3)a_2$ is a combination of a_1, a_2 . Thus a_1, a_2 span W and form a basis of W . (Check by Theorem 4.3: the RREF of A has rows $(1, 0, -1)$, $(0, 1, 2)$ and two zero rows; its pivot columns are 1, 2, and column 3 gives $a_3 = -a_1 + 2a_2$.)

(b) $\dim W = 2$.

Exercise 4.3. Let $u_1 = (1, -2, 1, -1)'$ and $u_2 = (0, 1, -1, 1)'$ in $\mathbb{R}^4$, and let $e_1, \dots, e_4$ be the unit vectors.

- (a) Row-reduce $[u_1 \ u_2 \ e_1 \ e_2 \ e_3 \ e_4] \in \mathbb{R}^{4 \times 6}$, and find a basis of $\mathbb{R}^4$ that consists of u_1, u_2 and two of the unit vectors.
- (b) Do u_1, u_2, e_1, e_2 form a basis of $\mathbb{R}^4$?

Solution. (a) Call the matrix $A = [u_1 \ u_2 \ e_1 \ e_2 \ e_3 \ e_4]$; its columns span $\mathbb{R}^4$, since they include $e_1, \dots, e_4$.

$$\begin{aligned} & \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_4 \leftarrow R_4 + R_1]{R_2 \leftarrow R_2 + 2R_1, \ R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array} \right) \\ & \xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & -1 & 0 & 1 \end{array} \right) \xrightarrow{R_4 \leftarrow R_4 + R_3} \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right) \end{aligned}$$

The last matrix is an echelon form, with leading 1s in columns 1, 2, 3, 5; these are the pivot columns of A (Remark 4.2). By Theorem 4.3(b), u_1, u_2, e_1, e_3 form a basis of the span of the columns, which is $\mathbb{R}^4$.

(b) No. Backward elimination finishes the RREF:

$$\xrightarrow{R_3 \leftarrow R_3 - R_4} \left(\begin{array}{cc|cccc} 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right) \xrightarrow[R_2 \leftarrow R_2 - 2R_3]{R_1 \leftarrow R_1 - R_3} \left(\begin{array}{cc|cccc} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & -1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right)$$

Column 4 of the RREF gives $e_2 = -u_1 - u_2 + e_1$ (Theorem 4.3(b)). So u_1, u_2, e_1, e_2 are linearly dependent and do not form a basis. Check: $-(1, -2, 1, -1)' - (0, 1, -1, 1)' + (1, 0, 0, 0)' = (0, 1, 0, 0)' = e_2$.

Exercise 4.4. Find the dimension of the space of solutions $(x, y, z)' \in \mathbb{R}^3$ of each system, and a basis of it: (a) $x - y + z = 0$; (b) $x + y + z = 0$, $x - y = 0$, $y + z = 0$.

Solution. The space of solutions is $\mathcal{N}(A)$ for the coefficient matrix A , of dimension $3 - \text{rank } A$ (Theorem 4.12).

(a) $A = (1, -1, 1)$ is a nonzero RREF with one row, so $\text{rank } A = 1$ and the dimension is 2. The vectors $(1, 1, 0)'$ and $(0, 1, 1)'$ are solutions ($1 - 1 + 0 = 0$, $0 - 1 + 1 = 0$) and are independent ($c_1(1, 1, 0)' + c_2(0, 1, 1)' = (c_1, c_1 + c_2, c_2)' = 0$ forces $c_1 = c_2 = 0$); so they form a basis (Theorem 4.5(d)).

(b) Row-reduce the coefficient matrix:

$$\left(\begin{array}{ccc} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & 1 \end{array} \right) \xrightarrow{R_2 \leftarrow R_2 - R_1} \left(\begin{array}{ccc} 1 & 1 & 1 \\ 0 & -2 & -1 \\ 0 & 1 & 1 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_3} \left(\begin{array}{ccc} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & -2 & -1 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 + 2R_2} \left(\begin{array}{ccc} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array} \right)$$

This echelon form has three nonzero rows, so $\text{rank } A = 3$ (Definition 4.11): the space of solutions is $\{0\}$, of dimension 0, with the empty basis.

Exercise 4.5. Find rank A and bases of $\mathcal{C}(A)$, $\mathcal{R}(A)$, $\mathcal{N}(A)$ and $\mathcal{N}(A')$ for

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ -1 & -2 & 0 & 1 \\ 1 & 2 & 2 & 1 \end{pmatrix},$$

and write the non-pivot columns of A as combinations of the pivot columns.

Solution. Row-reduce $[A \mid I_3]$ (Theorem 4.8):

$$\begin{aligned} & \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 \\ -1 & -2 & 0 & 1 & 0 & 1 & 0 \\ 1 & 2 & 2 & 1 & 0 & 0 & 1 \end{array} \right) \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 + R_1} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 0 & 1 \end{array} \right) \\ & \xrightarrow{R_3 \leftarrow R_3 - R_2} \left(\begin{array}{cccc|ccc} 1 & 2 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -2 & -1 & 1 \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 - R_2} \left(\begin{array}{cccc|ccc} 1 & 2 & 0 & -1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & -2 & -1 & 1 \end{array} \right) = [R \mid P] \end{aligned}$$

So rank $A = 2$, with pivot columns 1, 3 and free unknowns x_2, x_4 .

- $\mathcal{C}(A)$: $a_1 = (1, -1, 1)'$ and $a_3 = (1, 0, 2)'$. Columns 2 and 4 of R give $a_2 = 2a_1$ and $a_4 = -a_1 + a_3$ (Theorem 4.3(b)). Check: $2(1, -1, 1)' = (2, -2, 2)'$ and $-(1, -1, 1)' + (1, 0, 2)' = (0, 1, 1)'$.
- $\mathcal{R}(A)$: $\rho_1 = (1, 2, 0, -1)$ and $\rho_2 = (0, 0, 1, 1)$.
- $\mathcal{N}(A)$: $Rx = 0$ reads $x_1 + 2x_2 - x_4 = 0$, $x_3 + x_4 = 0$. The special solutions $s_1 = (-2, 1, 0, 0)'$ ($x_2 = 1, x_4 = 0$) and $s_2 = (1, 0, -1, 1)'$ ($x_2 = 0, x_4 = 1$) form a basis; $\dim \mathcal{N}(A) = 2 = 4 - 2$.
- $\mathcal{N}(A')$: row 3 of P gives the basis $y = (-2, -1, 1)'$; $\dim \mathcal{N}(A') = 1 = 3 - 2$. Indeed $y'A = -2\text{row}_1(A) - \text{row}_2(A) + \text{row}_3(A) = 0$, that is, $\text{row}_3(A) = 2\text{row}_1(A) + \text{row}_2(A)$.

Exercise 4.6. Find rank A , bases of $\mathcal{C}(A)$, $\mathcal{R}(A)$ and $\mathcal{N}(A)$, and $\dim \mathcal{N}(A')$ for

$$A = \begin{pmatrix} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 1 & 1 & 0 & 2 \\ 0 & -2 & -2 & 3 \end{pmatrix},$$

and write the non-pivot column of A as a combination of the pivot columns.

Solution. Row-reduce A :

$$\begin{aligned} A & \xrightarrow{R_3 \leftarrow R_3 - R_1} \left(\begin{array}{cccc} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & -1 & -1 & 1 \\ 0 & -2 & -2 & 3 \end{array} \right) \xrightarrow[R_4 \leftarrow R_4 + 2R_2]{R_3 \leftarrow R_3 + R_2} \left(\begin{array}{cccc} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right) \xrightarrow{R_3 \leftrightarrow R_4} \left(\begin{array}{cccc} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right) \\ & \xrightarrow[R_1 \leftarrow R_1 - R_3]{R_2 \leftarrow R_2 + R_3} \left(\begin{array}{cccc} 1 & 2 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{R_1 \leftarrow R_1 - 2R_2} \left(\begin{array}{cccc} 1 & 0 & -1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right) = R \end{aligned}$$

After the second step column 3 has no nonzero entry in rows 3, 4, and the next pivot column, 4, has its nonzero entry in row 4; so rows 3 and 4 are interchanged. The echelon form after the third arrow has three nonzero rows: rank $A = 3$, with pivot columns 1, 2, 4.

- $\mathcal{C}(A)$: $a_1 = (1, 0, 1, 0)'$, $a_2 = (2, 1, 1, -2)'$, $a_4 = (1, -1, 2, 3)'$. Column 3 of R gives $a_3 = -a_1 + a_2$; check: $-(1, 0, 1, 0)' + (2, 1, 1, -2)' = (1, 1, 0, -2)' = a_3$.

- $\mathcal{R}(A)$: $(1, 0, -1, 0)$, $(0, 1, 1, 0)$, $(0, 0, 0, 1)$.
- $\mathcal{N}(A)$: $Rx = 0$ reads $x_1 - x_3 = 0$, $x_2 + x_3 = 0$, $x_4 = 0$; the special solution $s = (1, -1, 1, 0)'$ ($x_3 = 1$) is a basis, and $\dim \mathcal{N}(A) = 1 = 4 - 3$.
- $\dim \mathcal{N}(A') = 4 - \text{rank } A = 1$ (Theorem 4.12).

Exercise 4.7. Let $A \in \mathbb{R}^{m \times n}$ have rank r . Show that A is the sum of r matrices of rank 1.

Solution. If $r = 0$, then $A = 0$, the empty sum. Let $r \geq 1$, and let $A = BC$ be the full-rank factorization of Theorem 4.21(a), where $B \in \mathbb{R}^{m \times r}$ has columns $b_1, \dots, b_r$ and $C \in \mathbb{R}^{r \times n}$. By the outer product expansion (Theorem 1.4(d)),

$$A = \sum_{j=1}^r A_j, \quad A_j = b_j \text{row}_j(C) \in \mathbb{R}^{m \times n}$$

- The columns of B are independent, and so are the rows of C (Theorem 4.13(b),(c)); hence every b_j and every $\text{row}_j(C)$ is nonzero, since a list containing 0 is dependent (Proposition 3.12(a)).
- If the l th entry c_{jl} of $\text{row}_j(C)$ is nonzero, then $\text{col}_l(A_j) = c_{jl}b_j \neq 0$; so $A_j \neq 0$, and $\text{rank } A_j \geq 1$.
- $\text{rank } A_j \leq \text{rank } b_j \leq 1$ (Theorems 4.15(a) and 4.13(a)), as b_j has one column.

So each A_j has rank 1.

Exercise 4.8. Show that $\text{rank}(A'AB) = \text{rank}(AB) = \text{rank}(ABB')$ for all $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$.

Solution. Apply Theorem 4.18(b) to AB , with $(AB)' = B'A'$ (Theorem 1.8), and Theorem 4.15(a) twice:

$$\begin{aligned} \text{rank}(AB) &= \text{rank}((AB)'AB) = \text{rank}(B'(A'AB)) \leq \text{rank}(A'AB) = \text{rank}(A'(AB)) \leq \text{rank}(AB), \\ \text{rank}(AB) &= \text{rank}(AB(AB)') = \text{rank}((ABB')A') \leq \text{rank}(ABB') = \text{rank}((AB)B') \leq \text{rank}(AB) \end{aligned}$$

So equality holds throughout.

Exercise 4.9. Let $A \in \mathbb{R}^{m \times n}$ with $m < n$. Show that there is no $B \in \mathbb{R}^{m \times n}$ with $B'A = I_n$.

Solution. If $B'A = I_n$, then by Theorems 4.13(a),(d) and 4.15(a),

$$n = \text{rank } I_n = \text{rank}(B'A) \leq \text{rank } A \leq m < n \quad \times$$

Exercise 4.10. Let $A \in \mathbb{R}^{n \times n}$ satisfy $A^2 = I$. Show that $\text{rank}(I - A) + \text{rank}(I + A) = n$.

Solution.

- $2I = (I - A) + (I + A)$, and $2I = \text{diag}(2, \dots, 2)$ is invertible (Definition 2.11). By Theorem 4.13(d) and Corollary 4.17, $n = \text{rank}(2I) \leq \text{rank}(I - A) + \text{rank}(I + A)$.
- $(I - A)(I + A) = I + A - A - A^2 = I - A^2 = 0$ (Theorem 1.3), so $\text{rank}(I - A) + \text{rank}(I + A) \leq n$ by Theorem 4.19.

Together, $\text{rank}(I - A) + \text{rank}(I + A) = n$.

Chapter 5. Block matrices and determinants

Block matrices

Definition 5.1. A **partition** of $A \in \mathbb{R}^{m \times n}$ splits the rows into consecutive groups of $m_1, \dots, m_r$ rows ($m_1 + \dots + m_r = m$) and the columns into consecutive groups of $n_1, \dots, n_c$ columns ($n_1 + \dots + n_c = n$). The **block** $A_{st} \in \mathbb{R}^{m_s \times n_t}$ consists of the entries in row group s and column group t , and we write

$$A = \begin{pmatrix} A_{11} & \cdots & A_{1c} \\ \vdots & & \vdots \\ A_{r1} & \cdots & A_{rc} \end{pmatrix}$$

Lines between rows and between columns may mark the partition. Two matrices are equal if and only if their corresponding blocks are equal. Partitions of $A, B \in \mathbb{R}^{m \times n}$ are **conformal for addition** if they use the same row and column groups; partitions of $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times p}$ are **conformal for multiplication** if the column groups of A are the row groups of B .

Theorem 5.2. Let the partitions below be conformal.

- (a) $A + B = (A_{st} + B_{st})$ and $cA = (cA_{st})$ ($c \in \mathbb{R}$).
- (b) A' has c block rows and r block columns, and its (t, s) block is A'_{st} : transpose the array of blocks and also each block.
- (c) If $A \in \mathbb{R}^{m \times n}$ has blocks A_{st} ($s \leq r, t \leq c$) and $B \in \mathbb{R}^{n \times p}$ has blocks B_{tu} ($t \leq c, u \leq v$), then AB has blocks

$$(AB)_{su} = \sum_{t=1}^c A_{st} B_{tu} = A_{s1} B_{1u} + A_{s2} B_{2u} + \cdots + A_{sc} B_{cu}$$

So conformal block matrices are added and multiplied as if the blocks were numbers, but the order of the factors in each $A_{st} B_{tu}$ must be kept. In particular, for 2×2 blocks,

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix} = \begin{pmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{pmatrix},$$

$$\text{and } [A_1 \ A_2] \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = A_1 B_1 + A_2 B_2, \quad \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} [B_1 \ B_2] = \begin{pmatrix} A_1 B_1 & A_1 B_2 \\ A_2 B_1 & A_2 B_2 \end{pmatrix}, \quad A[B_1 \ B_2] = [AB_1 \ AB_2].$$

Proof. (a) and (b) hold entry by entry. (c) Let i lie in row group s of A and j in column group u of B . In $(AB)_{ij} = \sum_{h=1}^n a_{ih} b_{hj}$, group the terms according to the column group t of A that contains h : $(AB)_{ij} = \sum_{t=1}^c \sum_{h \text{ in group } t} a_{ih} b_{hj}$. The inner sum runs over a row of A_{st} and a column of B_{tu} , so it is the corresponding entry of $A_{st} B_{tu}$. $\square$

Example 5.3. Partition

$$A = \left(\begin{array}{cc|c} 1 & 0 & 3 \\ 2 & -1 & 0 \\ 0 & 4 & 1 \end{array} \right), \quad B = \left(\begin{array}{cc|c} 3 & 3 & 1 \\ -4 & 2 & -1 \\ -2 & 0 & 0 \end{array} \right)$$

into blocks of sizes $2 \times 2, 2 \times 1, 1 \times 2, 1 \times 1$. By Theorem 5.2(c),

$$A_{11}B_{11} + A_{12}B_{21} = \begin{pmatrix} 3 & 3 \\ 10 & 4 \end{pmatrix} + \begin{pmatrix} -6 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} -3 & 3 \\ 10 & 4 \end{pmatrix}, \quad A_{11}B_{12} + A_{12}B_{22} = \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

$$A_{21}B_{11} + A_{22}B_{21} = (-16, 8) + (-2, 0) = (-18, 8), \quad A_{21}B_{12} + A_{22}B_{22} = -4 + 0$$

so $AB = \left(\begin{array}{cc|c} -3 & 3 & 1 \\ 10 & 4 & 3 \\ -18 & 8 & -4 \end{array} \right)$, the same matrix as the ordinary product. For instance $A_{12}B_{21} =$

$$\begin{pmatrix} 3 \\ 0 \end{pmatrix} (-2, 0) = \begin{pmatrix} -6 & 0 \\ 0 & 0 \end{pmatrix}. \text{ By Theorem 5.2(b), } B' = \begin{pmatrix} B'_{11} & B'_{21} \\ B'_{12} & B'_{22} \end{pmatrix}.$$

Definition 5.4. Let $A \in \mathbb{R}^{n \times n}$ be partitioned with the same groups $n_1, \dots, n_r$ for rows and columns, so that the **diagonal blocks** A_{ss} are square. A is **block diagonal**, written $A = \text{diag}(A_{11}, \dots, A_{rr})$, if $A_{st} = 0$ for $s \neq t$; **upper block triangular** if $A_{st} = 0$ for $s > t$; and **lower block triangular** if $A_{st} = 0$ for $s < t$.

Proposition 5.5. Let $A, B \in \mathbb{R}^{n \times n}$ be partitioned as in Definition 5.4, with the same groups.

- (a) If A and B are upper (lower) block triangular, so is AB , and its diagonal blocks are $A_{ss}B_{ss}$.
- (b) $\text{diag}(A_{11}, \dots, A_{rr}) \text{diag}(B_{11}, \dots, B_{rr}) = \text{diag}(A_{11}B_{11}, \dots, A_{rr}B_{rr})$. If every A_{ss} is invertible, then $\text{diag}(A_{11}, \dots, A_{rr})^{-1} = \text{diag}(A_{11}^{-1}, \dots, A_{rr}^{-1})$.
- (c) If $A_{11} \in \mathbb{R}^{p \times p}$ and $A_{22} \in \mathbb{R}^{q \times q}$ are invertible and $A_{12} \in \mathbb{R}^{p \times q}$, then

$$\begin{pmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{pmatrix}^{-1} = \begin{pmatrix} A_{11}^{-1} & -A_{11}^{-1}A_{12}A_{22}^{-1} \\ 0 & A_{22}^{-1} \end{pmatrix}$$

Proof. (a) The (s, u) block of AB is $\sum_t A_{st}B_{tu}$. For upper block triangular A, B , the term t can be nonzero only if $s \leq t \leq u$: no term survives when $s > u$, and only $t = s$ survives when $s = u$. The lower case is the same with the inequalities reversed. (b) Block diagonal means upper and lower block triangular, so (a) gives the product. Since $I_n = \text{diag}(I_{n_1}, \dots, I_{n_r})$, the products of $\text{diag}(A_{ss})$ and $\text{diag}(A_{ss}^{-1})$ in either order are I_n (Definition 2.11). (c) Call the two matrices M and X . By the 2×2 block formula,

$$MX = \begin{pmatrix} I_p & -A_{12}A_{22}^{-1} + A_{12}A_{22}^{-1} \\ 0 & I_q \end{pmatrix} = I, \quad XM = \begin{pmatrix} I_p & A_{11}^{-1}A_{12} - A_{11}^{-1}A_{12}A_{22}^{-1}A_{22} \\ 0 & I_q \end{pmatrix} = I \quad \square$$

Example 5.6. Find the inverse of

$$M = \left(\begin{array}{cc|cc} 2 & 1 & 1 & 3 \\ 1 & 1 & 2 & 1 \\ \hline 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

Solution. Here $A_{11} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $A_{12} = \begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$, $A_{22} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, and by the 2×2 formula of Exercise 2.12, $A_{11}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ and $A_{22}^{-1} = \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix}$. Then

$$A_{11}^{-1}A_{12} = \begin{pmatrix} -1 & 2 \\ 3 & -1 \end{pmatrix}, \quad -A_{11}^{-1}A_{12}A_{22}^{-1} = -\begin{pmatrix} -1 & 2 \\ 3 & -1 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -4 \\ -3 & 7 \end{pmatrix},$$

and by Proposition 5.5(c)

$$M^{-1} = \left(\begin{array}{cc|cc} 1 & -1 & 1 & -4 \\ -1 & 2 & -3 & 7 \\ \hline 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{array} \right)$$

Block elimination and the Schur complement

Proposition 5.7. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{m \times p}$, $B \in \mathbb{R}^{m \times q}$, $C \in \mathbb{R}^{n \times p}$, $D \in \mathbb{R}^{n \times q}$, and let $V \in \mathbb{R}^{n \times m}$, $F \in \mathbb{R}^{m \times n}$, $H \in \mathbb{R}^{p \times q}$. Then

$$\begin{pmatrix} I_m & 0 \\ V & I_n \end{pmatrix} M = \begin{pmatrix} A & B \\ C + VA & D + VB \end{pmatrix}, \quad \begin{pmatrix} I_m & F \\ 0 & I_n \end{pmatrix} M = \begin{pmatrix} A + FC & B + FD \\ C & D \end{pmatrix},$$

$$M \begin{pmatrix} I_p & H \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & B + AH \\ C & D + CH \end{pmatrix}$$

These **block elementary matrices** are invertible: $\begin{pmatrix} I & 0 \\ V & I \end{pmatrix}^{-1} = \begin{pmatrix} I & 0 \\ -V & I \end{pmatrix}$ and $\begin{pmatrix} I & F \\ 0 & I \end{pmatrix}^{-1} = \begin{pmatrix} I & -F \\ 0 & I \end{pmatrix}$.

Each identity is the 2×2 block formula; for example, the second block row of the first product is $(VA + I_n C, VB + I_n D)$. Thus left multiplication adds V times the first block row to the second, the block analogue of the row operation $R_2 \leftarrow R_2 + cR_1$ of Definition 2.2. The inverses follow from the products $\begin{pmatrix} I_p & 0 \\ -V & I \end{pmatrix} \begin{pmatrix} I & 0 \\ V & I \end{pmatrix} = I$ and $\begin{pmatrix} I & 0 \\ V & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -V & I \end{pmatrix} = I$, and likewise for F .

Theorem 5.8. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p \times p}$ invertible, $B \in \mathbb{R}^{p \times q}$, $C \in \mathbb{R}^{n \times p}$, $D \in \mathbb{R}^{n \times q}$. Then

$$\begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_n \end{pmatrix} \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}, \quad \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & 0 \\ C & D - CA^{-1}B \end{pmatrix},$$

and hence $\begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_n \end{pmatrix} M \begin{pmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{pmatrix}$. The matrix $D - CA^{-1}B$ is the **Schur complement** of A in M .

Proof. In Proposition 5.7 choose V so that the $(2, 1)$ block $C + VA$ is 0: $V = -CA^{-1}$; then $D + VB = D - CA^{-1}B$. Likewise $H = -A^{-1}B$ makes $B + AH = 0$ and $D + CH = D - CA^{-1}B$. Applying the column step to the result of the row step, $\begin{pmatrix} A & B \\ 0 & S \end{pmatrix} \begin{pmatrix} I & -A^{-1}B \\ 0 & I \end{pmatrix} = \begin{pmatrix} A & -AA^{-1}B+B \\ 0 & S \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & S \end{pmatrix}$ with $S = D - CA^{-1}B$. $\square$

This is Gaussian elimination done one block at a time. When D is invertible, the matrix $A - BD^{-1}C$ is likewise called the **Schur complement** of D in M .

Example 5.9. With the data matrix $X_0 \in \mathbb{R}^{4 \times 2}$ of Example 1.14 ($n = 4, k = 2$), let $X = [\mathbf{1}_4 \ X_0]$ be the design matrix. By Theorem 5.2,

$$X'X = \begin{pmatrix} \mathbf{1}' \\ X_0' \end{pmatrix} [\mathbf{1} \ X_0] = \begin{pmatrix} \mathbf{1}'\mathbf{1} & \mathbf{1}'X_0 \\ X_0'\mathbf{1} & X_0'X_0 \end{pmatrix} = \left(\begin{array}{c|cc} 4 & 4 & 8 \\ \hline 4 & 8 & 8 \\ 8 & 8 & 20 \end{array} \right)$$

Here $A = (4)$, $C = X_0'\mathbf{1} = (4, 8)'$ and $-CA^{-1} = -(1, 2)' = -\bar{x}$, so Theorem 5.8 gives

$$\left(\begin{array}{c|cc} 1 & 0 & 0 \\ \hline -1 & 1 & 0 \\ -2 & 0 & 1 \end{array} \right) \left(\begin{array}{c|cc} 4 & 4 & 8 \\ \hline 4 & 8 & 8 \\ 8 & 8 & 20 \end{array} \right) = \left(\begin{array}{c|cc} 4 & 4 & 8 \\ \hline 0 & 4 & 0 \\ 0 & 0 & 4 \end{array} \right)$$

The Schur complement of $n = 4$ is $D - CA^{-1}B = X_0'X_0 - \frac{1}{4}(X_0'\mathbf{1})(\mathbf{1}'X_0) = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$, the matrix $X_0'C_4X_0$ of corrected sums of squares and products of Example 1.14 (compare the first step of Example 2.17). The same holds for any data matrix $X_0 \in \mathbb{R}^{n \times k}$: for the design matrix $X = [\mathbf{1}_n \ X_0] \in \mathbb{R}^{n \times (k+1)}$, the Schur complement of $\mathbf{1}'\mathbf{1} = n$ in $X'X$ is $X_0'X_0 - \frac{1}{n}X_0'\mathbf{1}\mathbf{1}'X_0 = X_0'C_nX_0$ (the centering matrix keeps its subscript here, since C is a block).

Determinants: definition and computation

Definition 5.10. Let $A = (a_{ij}) \in \mathbb{R}^{n \times n}$. For $n \geq 2$ and $1 \leq i, j \leq n$, let $A(i|j) \in \mathbb{R}^{(n-1) \times (n-1)}$ be the matrix obtained from A by deleting row i and column j . The **determinant** $\det A$ is the number defined by $\det(a_{11}) = a_{11}$ if $n = 1$ and, recursively, for $n \geq 2$ by

$$\det A = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det A(1|j) = a_{11} \det A(1|1) - a_{12} \det A(1|2) + \cdots + (-1)^{1+n} a_{1n} \det A(1|n)$$

The number $\det A(i|j)$ is the **minor** of a_{ij} , and $c_{ij} = (-1)^{i+j} \det A(i|j)$ is its **cofactor**. Thus $\det A = a_{11}c_{11} + a_{12}c_{12} + \cdots + a_{1n}c_{1n}$, the **expansion along the first row**. Many books write $|A|$ for $\det A$.

For $n = 2$, $A(1|1) = (a_{22})$ and $A(1|2) = (a_{21})$, so

$$\det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a_{11}a_{22} - a_{12}a_{21},$$

the number $ad - bc$ of the 2×2 inverse formula of Exercise 2.12. The signs $(-1)^{i+j}$ form a checkerboard with $+$ on the diagonal, for $n = 3$ $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$.

Theorem 5.11. Let $A \in \mathbb{R}^{n \times n}$ with $n \geq 2$, and let c_{ij} be the cofactors of A . For every row i and every column j ,

$$\det A = a_{i1}c_{i1} + a_{i2}c_{i2} + \cdots + a_{in}c_{in} = a_{1j}c_{1j} + a_{2j}c_{2j} + \cdots + a_{nj}c_{nj}$$

(expansion along row i and along column j).

We omit the proof: it is an induction on n in which the minors are expanded once more, and its only difficulty is the bookkeeping of the signs $(-1)^{i+j}$. In practice one expands along a row or column with many zeros.

Example 5.12. Compute $\det A$ for $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 2 & 3 & 5 \end{pmatrix}$ by expansion along the first row, and check the value by expansion along the second row.

Solution. Along the first row,

$$\det A = 1 \det \begin{pmatrix} 5 & 6 \\ 3 & 5 \end{pmatrix} - 2 \det \begin{pmatrix} 4 & 6 \\ 2 & 5 \end{pmatrix} + 3 \det \begin{pmatrix} 4 & 5 \\ 2 & 3 \end{pmatrix} = 1(25 - 18) - 2(20 - 12) + 3(12 - 10) = -3$$

so the cofactors of row 1 are $c_{11} = 7$, $c_{12} = -8$, $c_{13} = 2$. Along the second row (Theorem 5.11),

$$\det A = -4 \det \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix} + 5 \det \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} - 6 \det \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} = -4(10 - 9) + 5(5 - 6) - 6(3 - 4) = -3$$

Theorem 5.13. Let $A \in \mathbb{R}^{n \times n}$.

- (a) If A is upper or lower triangular, then $\det A = a_{11}a_{22} \cdots a_{nn}$. In particular $\det \text{diag}(d_1, \dots, d_n) = d_1d_2 \cdots d_n$ and $\det I_n = 1$.
- (b) $\det A' = \det A$.

Proof. Both parts are proved by induction on n ; for $n = 1$ there is nothing to prove. Let $n \geq 2$.

(a)

- If A is lower (upper) triangular, the entries of its first row (column) other than a_{11} are 0, so the definition (the expansion along column 1, Theorem 5.11) gives $\det A = a_{11} \det A(1|1)$.
- $A(1|1)$ is triangular of the same kind with diagonal entries $a_{22}, \dots, a_{nn}$, so $\det A(1|1) = a_{22} \cdots a_{nn}$ by induction.

(b)

- Row 1 of A' is $(a_{11}, a_{21}, \dots, a_{n1})$, and $A'(1|j) = A(j|1)'$: both consist of the entries a_{lk} with $l \neq j$ and $k \neq 1$, arranged as transposes of each other.
- By the definition and the induction hypothesis, $\det A' = \sum_{j=1}^n (-1)^{1+j} a_{j1} \det A(j|1)' = \sum_{j=1}^n (-1)^{j+1} a_{j1} \det A(j|1)$, which is the expansion of $\det A$ along column 1 (Theorem 5.11). $\square$

Theorem 5.14. Let $A \in \mathbb{R}^{n \times n}$ with $n \geq 2$, let $r \neq s$ be row indices, and let $c \in \mathbb{R}$.

- (a) Multiplying row r of A by c multiplies $\det A$ by c . In particular, if A has a zero row, then $\det A = 0$.
- (b) Interchanging rows r and s changes the sign of $\det A$.
- (c) If two rows of A are equal, then $\det A = 0$.
- (d) Adding c times row s to row r does not change $\det A$.
- (e) For the elementary matrices of Definition 2.13, $\det D_r(c) = c$, $\det T_{rs}(c) = 1$ and $\det P_{rs} = -1$; and $\det(EA) = \det E \cdot \det A$ for every elementary matrix $E \in \mathbb{R}^{n \times n}$.
- (f) Statements (a)–(d) hold with columns in place of rows.

Proof. (a) Let B be A with row r multiplied by c . Deleting row r gives $B(r|j) = A(r|j)$, so the expansion along row r (Theorem 5.11) gives $\det B = \sum_j (-1)^{r+j} c a_{rj} \det A(r|j) = c \det A$. A zero row is 0 times itself, so then $\det A = 0 \cdot \det A = 0$.

(b) Let B be A with rows r and s interchanged; induction on n .

- For $n = 2$, $\det \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} = a_{11}a_{22} - a_{12}a_{21} = -\det A$.
- For $n \geq 3$, expand $\det A$ and $\det B$ along a row $i \notin \{r, s\}$, which is the same in both. Each $B(i|j)$ is $A(i|j)$ with two rows interchanged, so $\det B(i|j) = -\det A(i|j)$ by induction; hence $\det B = -\det A$.

(c) Interchanging the two equal rows does not change A , but by (b) it changes the sign of $\det A$. So $\det A = -\det A$, that is, $\det A = 0$.

(d) Let B be A with row r replaced by $\text{row}_r(A) + c \text{row}_s(A)$, and $\tilde{A}$ be A with row r replaced by $\text{row}_s(A)$. Since $B(r|j) = \tilde{A}(r|j) = A(r|j)$, expanding along row r ,

$$\det B = \sum_{j=1}^n (-1)^{r+j} (a_{rj} + c a_{sj}) \det A(r|j) = \det A + c \det \tilde{A}$$

Rows r and s of $\tilde{A}$ are equal, so $\det \tilde{A} = 0$ by (c).

(e) $D_r(c)$ is diagonal and $T_{rs}(c)$ is triangular with diagonal entries 1, so $\det D_r(c) = c$ and $\det T_{rs}(c) = 1$ (Theorem 5.13(a)); P_{rs} is I_n with rows r, s interchanged, so $\det P_{rs} = -1$ by (b). EA is obtained from A by the row operation of E (Theorem 2.14(a)), so by (a), (d), (b) $\det(EA)$ is $c \det A$, $\det A$ or $-\det A$, that is, $\det E \cdot \det A$.

(f) The columns of A are the transposed rows of A' , and $\det A = \det A'$ (Theorem 5.13(b)). An operation on the columns of A is the same operation on the rows of A' ; so (a)–(d) for the rows of A' give (a)–(d) for the columns of A . $\square$

Computing $\det A$ by row reduction. Forward elimination (Steps 1, 2 and 4 of Definition 2.5, without making the pivots 1) turns $A \in \mathbb{R}^{n \times n}$ into an upper triangular matrix U (the first nonzero entry of row i lies in column i or later), whose determinant is the product of its diagonal entries (Theorem 5.13(a)). By Theorem 5.14, each interchange of two rows changes the sign and adding a multiple of a row to another row changes nothing, so $\det A = (-1)^t u_{11} u_{22} \cdots u_{nn}$, where t is the number of interchanges; if a row is multiplied by $c \neq 0$ on the way, divide by c . Column operations may be used in the same way, and a row or column with a single nonzero entry can be expanded at once.

Example 5.15. Compute $\det A$ for $A = \begin{pmatrix} 0 & 2 & -1 \\ 1 & 3 & 2 \\ 2 & 4 & 1 \end{pmatrix}$.

Solution. Since $a_{11} = 0$, begin with an interchange:

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 2 & -1 \\ 2 & 4 & 1 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - 2R_1} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 2 & -1 \\ 0 & -2 & -3 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 1 & 3 & 2 \\ 0 & 2 & -1 \\ 0 & 0 & -4 \end{pmatrix} = U$$

The interchange changes the sign and the two additions change nothing, so $\det A = -\det U = -(1 \cdot 2 \cdot (-4)) = 8$. Check by expansion along column 1, whose entries are 0, 1, 2: $\det A = -1 \det \begin{pmatrix} 2 & -1 \\ 4 & 1 \end{pmatrix} + 2 \det \begin{pmatrix} 2 & -1 \\ 3 & 2 \end{pmatrix} = -6 + 14 = 8$.

Example 5.16. Compute $\det A$ for $A = \begin{pmatrix} 1 & -1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 4 & 1 & -1 & -1 \\ 1 & 2 & 3 & 0 \end{pmatrix}$.

Solution.

$$A \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1, R_3 \leftarrow R_3 - 4R_1} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 4 & -4 & -4 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix} \xrightarrow{R_2 \leftarrow \frac{1}{4}R_2} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 5 & -9 & -13 \\ 0 & 3 & 1 & -3 \end{pmatrix}$$

$$\xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 - 5R_2} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -4 & -8 \\ 0 & 0 & 4 & 0 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 + R_3} \begin{pmatrix} 1 & -1 & 2 & 3 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & -4 & -8 \\ 0 & 0 & 0 & -8 \end{pmatrix} = U$$

Only the second step changes the determinant: it multiplies it by $\frac{1}{4}$ (Theorem 5.14(a)). Hence $\det A = 4 \det U = 4 \cdot 1 \cdot 1 \cdot (-4) \cdot (-8) = 128$.

Main properties

Theorem 5.17. Let $A, B \in \mathbb{R}^{n \times n}$.

- (a) A is invertible $\Leftrightarrow \det A \neq 0 \Leftrightarrow \text{rank } A = n$.
- (b) $\det(AB) = \det A \cdot \det B$.
- (c) If A is invertible, then $\det A^{-1} = 1/\det A$. Moreover $\det(cA) = c^n \det A$ for $c \in \mathbb{R}$, and $\det A^k = (\det A)^k$ for $k = 1, 2, \dots$

Proof. For $n = 1$, $\det(a) = a$ and all statements are clear; let $n \geq 2$.

- (a)
 - Let $R = PA$ be the RREF of A , where $P = E_k \cdots E_1$ is a product of elementary matrices (Theorem 2.14(c)). By Theorem 5.14(e), applied k times, $\det R = \det E_k \cdots \det E_1 \cdot \det A$, and each $\det E_i$ is $c \neq 0, 1$ or -1 .
 - If A is invertible, then $R = I_n$ (Theorem 2.15), so $\det A \neq 0$. If not, then $\text{rank } A < n$ (Theorem 4.13(d)), so R has fewer than n nonzero rows (Definition 4.11), that is, a zero row; then $\det R = 0$ (Theorem 5.14(a)) and $\det A = 0$. Finally, A is invertible if and only if $\text{rank } A = n$ (Theorem 4.13(d)).
- (b)
 - If A is not invertible, neither is AB : an inverse C of AB would give $A(BC) = I_n$, and A would be invertible (Theorem 2.15). By (a), $\det(AB) = 0 = \det A \cdot \det B$.

- If A is invertible, then $A = F_1 \cdots F_k$ with elementary matrices F_i (Theorem 2.15). By Theorem 5.14(e), applied k times, $\det(AB) = \det F_1 \cdots \det F_k \cdot \det B$; for $B = I_n$ this reads $\det A = \det F_1 \cdots \det F_k$, so $\det(AB) = \det A \cdot \det B$.

(c) By (b), $\det A \cdot \det A^{-1} = \det(AA^{-1}) = \det I_n = 1$. Next, $cA = (cI_n)A$ and cI_n is diagonal, so $\det(cA) = c^n \det A$ by (b) and Theorem 5.13(a). Finally $\det A^{k+1} = \det(A^k A) = \det A^k \cdot \det A$ by (b), which gives the powers by induction on k . $\square$

Theorem 5.18. Let $A \in \mathbb{R}^{n \times n}$ with $n \geq 2$, and let c_{ij} be the cofactors of A . The **adjugate** of A is the transpose of the matrix of cofactors: $\text{adj } A = (c_{ij})'$, that is, $(\text{adj } A)_{ij} = c_{ji}$. Then

$$A \text{adj } A = (\det A) I_n$$

Hence, if $\det A \neq 0$, then $A^{-1} = \frac{1}{\det A} \text{adj } A$.

Proof. • The (i, k) entry of $A \text{adj } A$ is $\sum_{j=1}^n a_{ij}(\text{adj } A)_{jk} = \sum_{j=1}^n a_{ij}c_{kj}$.

- For $k = i$, this is the expansion of $\det A$ along row i (Theorem 5.11).
- For $k \neq i$, let $\tilde{A}$ be A with row k replaced by row i . Since $\tilde{A}(k|j) = A(k|j)$, the cofactors of row k of $\tilde{A}$ are $c_{k1}, \dots, c_{kn}$, and the sum is the expansion of $\det \tilde{A}$ along row k . As $\tilde{A}$ has two equal rows, the sum is 0 (Theorem 5.14(c)).
- If $\det A \neq 0$, then $A\left(\frac{1}{\det A} \text{adj } A\right) = I_n$, so $\frac{1}{\det A} \text{adj } A$ is the inverse of A (Theorem 2.15). $\square$

For $n = 2$, $\text{adj} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, and the theorem gives again the formula of Exercise 2.12.

Example 5.19. Find A^{-1} for $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 3 & 4 & 3 \end{pmatrix}$ by Theorem 5.18.

Solution. The cofactors are

$$\begin{aligned} c_{11} &= \det \begin{pmatrix} 1 & 1 \\ 4 & 3 \end{pmatrix} = -1, & c_{12} &= -\det \begin{pmatrix} 1 & 1 \\ 3 & 3 \end{pmatrix} = 0, & c_{13} &= \det \begin{pmatrix} 1 & 1 \\ 3 & 4 \end{pmatrix} = 1, \\ c_{21} &= -\det \begin{pmatrix} 1 & 1 \\ 4 & 3 \end{pmatrix} = 1, & c_{22} &= \det \begin{pmatrix} 0 & 1 \\ 3 & 3 \end{pmatrix} = -3, & c_{23} &= -\det \begin{pmatrix} 0 & 1 \\ 3 & 4 \end{pmatrix} = 3, \\ c_{31} &= \det \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = 0, & c_{32} &= -\det \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = 1, & c_{33} &= \det \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} = -1 \end{aligned}$$

Along row 1, $\det A = 0 \cdot (-1) + 1 \cdot 0 + 1 \cdot 1 = 1$. Hence

$$A^{-1} = \frac{1}{1} \text{adj } A = \begin{pmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{pmatrix} = \begin{pmatrix} -1 & 1 & 0 \\ 0 & -3 & 1 \\ 1 & 3 & -1 \end{pmatrix},$$

the inverse found by row reduction in Exercise 2.8. Check: $\text{row}_3(A) \text{adj } A = (3, 4, 3) \text{adj } A = (-3 + 0 + 3, 3 - 12 + 9, 0 + 4 - 3) = (0, 0, 1)$.

Determinants of block matrices and the Schur complement

Theorem 5.20. Let $A \in \mathbb{R}^{p \times p}$, $B \in \mathbb{R}^{p \times q}$, $C \in \mathbb{R}^{q \times p}$ and $D \in \mathbb{R}^{q \times q}$. Then

$$\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det A \cdot \det D = \det \begin{pmatrix} A & 0 \\ C & D \end{pmatrix}$$

In particular $\det \text{diag}(A, D) = \det A \cdot \det D$.

- Proof.* • The first row of $\begin{pmatrix} I_p & 0 \\ 0 & D \end{pmatrix}$ is $(1, 0, \dots, 0)$, and deleting row 1 and column 1 leaves $\begin{pmatrix} I_{p-1} & 0 \\ 0 & D \end{pmatrix}$; expanding along the first row p times gives $\det \begin{pmatrix} I_p & 0 \\ 0 & D \end{pmatrix} = \det D$. Likewise, expanding $\begin{pmatrix} A & 0 \\ 0 & I_q \end{pmatrix}$ q times along its last row $(0, \dots, 0, 1)$, whose entry 1 is a diagonal entry (sign $+1$), gives $\det A$.
- $\begin{pmatrix} I_p & B \\ 0 & I_q \end{pmatrix}$ is upper triangular with diagonal entries 1, so its determinant is 1 (Theorem 5.13(a)).
 - By the block product formula (Theorem 5.2),

$$\begin{pmatrix} I_p & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} I_p & B \\ 0 & I_q \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} I_p & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} A & B \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}$$

and by Theorem 5.17(b) its determinant is $\det D \cdot 1 \cdot \det A$.

- The transpose of $\begin{pmatrix} A & 0 \\ C & D \end{pmatrix}$ is $\begin{pmatrix} A' & C' \\ 0 & D' \end{pmatrix}$ (Theorem 5.2(b)); by Theorem 5.13(b) and the case just proved, its determinant is $\det A' \cdot \det D' = \det A \cdot \det D$. Finally, $\text{diag}(A, D)$ is the case $B = 0$. $\square$

Theorem 5.21. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p \times p}$, $B \in \mathbb{R}^{p \times q}$, $C \in \mathbb{R}^{q \times p}$ and $D \in \mathbb{R}^{q \times q}$.

- (a) If A is invertible, then $\det M = \det A \cdot \det(D - CA^{-1}B)$.
- (b) If D is invertible, then $\det M = \det D \cdot \det(A - BD^{-1}C)$.

In particular, if A is invertible, then M is invertible if and only if its Schur complement $D - CA^{-1}B$ is invertible.

Proof. (a) By Theorem 5.8, $\begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{pmatrix} M = \begin{pmatrix} A & B \\ 0 & D - CA^{-1}B \end{pmatrix}$. The left factor is lower triangular with diagonal entries 1, so its determinant is 1. By Theorems 5.17(b) and 5.20, $\det M = \det A \cdot \det(D - CA^{-1}B)$.

(b) By the block product formula, $\begin{pmatrix} I_p & -BD^{-1} \\ 0 & I_q \end{pmatrix} M = \begin{pmatrix} A - BD^{-1}C & B - BD^{-1}D \\ C & D \end{pmatrix} = \begin{pmatrix} A - BD^{-1}C & 0 \\ C & D \end{pmatrix}$. The left factor is upper triangular with diagonal entries 1; as in (a), $\det M = \det(A - BD^{-1}C) \cdot \det D$.

If A is invertible, then $\det A \neq 0$, so by (a) $\det M \neq 0$ if and only if $\det(D - CA^{-1}B) \neq 0$; apply Theorem 5.17(a) to M and to $D - CA^{-1}B$. $\square$

For $p = q = 1$ and $a \neq 0$, (a) reads $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = a(d - \frac{cb}{a}) = ad - bc$.

Theorem 5.22. Let $A \in \mathbb{R}^{n \times n}$ be invertible and $u, v \in \mathbb{R}^n$. Then

$$\det(A + uv') = \det A \cdot (1 + v'A^{-1}u)$$

In particular $\det(I_n + uv') = 1 + v'u$. (This is the **matrix determinant lemma**.)

Proof. Apply Theorem 5.21 to $M = \begin{pmatrix} A & -u \\ v' & 1 \end{pmatrix} \in \mathbb{R}^{(n+1) \times (n+1)}$ ($p = n$, $q = 1$).

- By (a), with the invertible block A , whose Schur complement is the 1×1 matrix $1 - v'A^{-1}(-u)$: $\det M = \det A \cdot (1 + v'A^{-1}u)$.
- By (b), with the invertible block $D = (1)$: $\det M = \det(1) \cdot \det(A - (-u)(1)^{-1}v') = \det(A + uv')$.
- Comparing gives the formula; for $A = I_n$, $A^{-1} = I_n$. $\square$

Example 5.23. Let $n \geq 2$ and $\rho \in \mathbb{R}$. The **equicorrelation matrix** $R = (1 - \rho)I_n + \rho J_n \in \mathbb{R}^{n \times n}$, with diagonal entries 1 and all other entries ρ , is the correlation matrix of n variables any two of which have correlation ρ . Find $\det R$, and the values of ρ for which R is invertible.

Solution.

- If $\rho = 1$, then $R = J_n$ has equal rows ($n \geq 2$), so $\det R = 0$ (Theorem 5.14(c)).

- Let $\rho \neq 1$. As $J_n = \mathbf{1}_n \mathbf{1}'_n$, $R = A + uv'$ with $A = (1-\rho)I_n$, $u = \rho \mathbf{1}_n$ and $v = \mathbf{1}_n$. Here $\det A = (1-\rho)^n$ (Theorem 5.13(a)), $A^{-1} = (1-\rho)^{-1}I_n$ and $v'A^{-1}u = \frac{\rho}{1-\rho} \mathbf{1}'\mathbf{1} = \frac{n\rho}{1-\rho}$ (Proposition 1.12). By Theorem 5.22,

$$\det R = (1-\rho)^n \left(1 + \frac{n\rho}{1-\rho}\right) = (1-\rho)^{n-1}(1-\rho+n\rho) = (1-\rho)^{n-1}(1+(n-1)\rho)$$

This formula also gives 0 for $\rho = 1$.

- By Theorem 5.17(a), R is invertible if and only if $\det R \neq 0$, that is, $\rho \neq 1$ and $\rho \neq -1/(n-1)$. For instance, for $n = 3$ and $\rho = -\frac{1}{2}$ every row of R has sum 0, so $R\mathbf{1}_3 = 0$.

Rank and the Schur complement

Theorem 5.24. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$.

- (a) If $A \in \mathbb{R}^{p \times p}$ is invertible, and $B \in \mathbb{R}^{p \times q}$, $C \in \mathbb{R}^{n \times p}$, $D \in \mathbb{R}^{n \times q}$, then $\text{rank } M = p + \text{rank}(D - CA^{-1}B)$.
- (b) If $D \in \mathbb{R}^{q \times q}$ is invertible, and $A \in \mathbb{R}^{m \times p}$, $B \in \mathbb{R}^{m \times q}$, $C \in \mathbb{R}^{q \times p}$, then $\text{rank } M = q + \text{rank}(A - BD^{-1}C)$.

Proof. (a)

- By Theorem 5.8, $\begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_n \end{pmatrix} M \begin{pmatrix} I_p & -A^{-1}B \\ 0 & I_q \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & D - CA^{-1}B \end{pmatrix}$, and the two outer factors are invertible (Proposition 5.7).
- Invertible factors do not change the rank (Theorem 4.15(d)), and the rank of a block-diagonal matrix is the sum of the ranks of its blocks (Theorem 4.16(c)); so $\text{rank } M = \text{rank } A + \text{rank}(D - CA^{-1}B)$, where $\text{rank } A = p$ (Theorem 4.13(d)).

(b)

- By the block product formula, $\begin{pmatrix} I_m & -BD^{-1} \\ 0 & I_q \end{pmatrix} M = \begin{pmatrix} A - BD^{-1}C & 0 \\ C & D \end{pmatrix}$ (as in the proof of Theorem 5.21(b)), and multiplying on the right by $\begin{pmatrix} I_p & 0 \\ -D^{-1}C & I_q \end{pmatrix}$ gives $\begin{pmatrix} A - BD^{-1}C & 0 \\ C - DD^{-1}C & D \end{pmatrix} = \begin{pmatrix} A - BD^{-1}C & 0 \\ 0 & D \end{pmatrix}$.
- Both outer factors are invertible (Proposition 5.7); as in (a), $\text{rank } M = \text{rank}(A - BD^{-1}C) + \text{rank } D = \text{rank}(A - BD^{-1}C) + q$. $\square$

Example 5.25. Let $d_1 = (1, 1, 1, 0, 0, 0)'$, $d_2 = (0, 0, 0, 1, 1, 0)'$ and $d_3 = (0, 0, 0, 0, 0, 1)'$ be the indicators of three groups of 3, 2 and 1 observations, and let $X = [\mathbf{1}_6 \ d_1 \ d_2 \ d_3] \in \mathbb{R}^{6 \times 4}$. Find $\text{rank}(X'X)$ and $\det(X'X)$ with a Schur complement, and $\text{rank } X$.

Solution. The (j, l) entry of $X'X$ is $\text{col}_j(X)' \text{col}_l(X)$ (Example 1.14): $\mathbf{1}'\mathbf{1} = 6$, $\mathbf{1}'d_l = d'_l d_l = n_l$, the size of group l ($n_1 = 3$, $n_2 = 2$, $n_3 = 1$), and $d'_l d_m = 0$ for $l \neq m$, as no observation lies in two groups. Partition after the first row and column:

$$X'X = \left(\begin{array}{c|ccc} 6 & 3 & 2 & 1 \\ \hline 3 & 3 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{array} \right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad A = (6), \quad B = (3, 2, 1), \quad C = B', \quad D = \text{diag}(3, 2, 1)$$

D is invertible with $D^{-1} = \text{diag}(\frac{1}{3}, \frac{1}{2}, 1)$, so $BD^{-1} = (1, 1, 1)$, and the Schur complement of D is $A - BD^{-1}C = 6 - (1, 1, 1)(3, 2, 1)' = 6 - 6 = 0$. Hence $\text{rank}(X'X) = 3 + \text{rank}(0) = 3$ (Theorem 5.24(b)) and $\det(X'X) = \det D \cdot 0 = 0$ (Theorem 5.21(b)): $X'X$ is not invertible. Also $\text{rank } X = \text{rank}(X'X) = 3$ (Theorem 4.18(b)), so the four columns of X are linearly dependent (Theorem 4.13(b)); indeed $d_1 + d_2 + d_3 = \mathbf{1}_6$, that is, $X(-1, 1, 1, 1)' = 0$.

Exercises and solutions

Exercise 5.1. Let A and B be the 3×5 matrices and C the 5×4 matrix partitioned as

$$A = \left(\begin{array}{ccc|cc} 1 & 3 & -2 & 1 & 2 \\ 2 & 1 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & 1 \end{array} \right), \quad B = \left(\begin{array}{ccc|cc} 1 & -3 & -2 & 2 & 1 \\ 1 & 2 & 3 & 2 & 0 \\ 1 & 0 & 2 & 0 & 1 \end{array} \right), \quad C = \left(\begin{array}{cc|cc} 1 & 0 & 2 & 1 \\ 0 & 2 & 0 & 0 \\ -1 & 0 & 3 & 1 \\ \hline 3 & 1 & 0 & 2 \\ 2 & -1 & 1 & 1 \end{array} \right)$$

With blocks $A = (A_{st})$, $B = (B_{st})$, $C = (C_{st})$ ($s, t = 1, 2$), show that

$$A + B = (A_{st} + B_{st}), \quad A' = \begin{pmatrix} A'_{11} & A'_{21} \\ A'_{12} & A'_{22} \end{pmatrix}, \quad AC = \begin{pmatrix} A_{11}C_{11} + A_{12}C_{21} & A_{11}C_{12} + A_{12}C_{22} \\ A_{21}C_{11} + A_{22}C_{21} & A_{21}C_{12} + A_{22}C_{22} \end{pmatrix}$$

Solution. Adding entrywise, $A + B = \left(\begin{array}{ccc|cc} 2 & 0 & -4 & 3 & 3 \\ 3 & 3 & 3 & 1 & 3 \\ 1 & 0 & 3 & 2 & 2 \end{array} \right)$, and each block is the sum of the corresponding blocks of A and B . Transposing,

$$A' = \left(\begin{array}{cc|c} 1 & 2 & 0 \\ 3 & 1 & 0 \\ -2 & 0 & 1 \\ \hline 1 & -1 & 2 \\ 2 & 3 & 1 \end{array} \right) = \begin{pmatrix} A'_{11} & A'_{21} \\ A'_{12} & A'_{22} \end{pmatrix}$$

Multiplying in the ordinary way, $AC = \left(\begin{array}{cc|cc} 10 & 5 & -2 & 3 \\ 5 & -2 & 7 & 3 \\ \hline 7 & 1 & 4 & 6 \end{array} \right)$. The blockwise products give the same four blocks:

$$A_{11}C_{11} + A_{12}C_{21} = \begin{pmatrix} 3 & 6 \\ 2 & 2 \end{pmatrix} + \begin{pmatrix} 7 & -1 \\ 3 & -4 \end{pmatrix} = \begin{pmatrix} 10 & 5 \\ 5 & -2 \end{pmatrix},$$

$$A_{11}C_{12} + A_{12}C_{22} = \begin{pmatrix} -4 & -1 \\ 4 & 2 \end{pmatrix} + \begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 3 \\ 7 & 3 \end{pmatrix},$$

$$A_{21}C_{11} + A_{22}C_{21} = (-1, 0) + (8, 1) = (7, 1),$$

$$A_{21}C_{12} + A_{22}C_{22} = (3, 1) + (1, 5) = (4, 6)$$

For instance, $A_{12}C_{21} = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ 2 & -1 \end{pmatrix} = \begin{pmatrix} 7 & -1 \\ 3 & -4 \end{pmatrix}$. This agrees with Theorem 5.2.

Exercise 5.2. Let $A \in \mathbb{R}^{n \times n}$ be invertible and partitioned by columns as $A = [A_1 \ A_2]$, where A_1 has k columns, and let A^{-1} be partitioned by rows into B_1 (k rows) and B_2 . Show that

$$B_1A_1 = I, \quad B_1A_2 = 0, \quad B_2A_1 = 0, \quad B_2A_2 = I, \quad A_1B_1 = I - A_2B_2, \quad A_2B_2 = I - A_1B_1$$

Solution. With the partition $(k, n - k)$ of I_n , Theorem 5.2(c) gives

$$\begin{pmatrix} B_1A_1 & B_1A_2 \\ B_2A_1 & B_2A_2 \end{pmatrix} = \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} [A_1 \ A_2] = A^{-1}A = I_n = \begin{pmatrix} I_k & 0 \\ 0 & I_{n-k} \end{pmatrix},$$

and comparing blocks gives the first four identities. Also $A_1B_1 + A_2B_2 = [A_1 \ A_2] \begin{pmatrix} B_1 \\ B_2 \end{pmatrix} = AA^{-1} = I_n$, which gives the last two.

Exercise 5.3. Show that the matrix $Z = \begin{pmatrix} -I_m & B \\ C & I_n \end{pmatrix}$ ($B \in \mathbb{R}^{m \times n}$, $C \in \mathbb{R}^{n \times m}$) satisfies $Z^2 = I_{m+n}$ if $B = 0$ or $C = 0$. Is this condition necessary?

Solution. By the 2×2 block formula (Theorem 5.2),

$$Z^2 = \begin{pmatrix} (-I)(-I) + BC & -B + B \\ -C + C & CB + I \end{pmatrix} = \begin{pmatrix} I_m + BC & 0 \\ 0 & I_n + CB \end{pmatrix},$$

so $Z^2 = I_{m+n}$ if and only if $BC = 0$ and $CB = 0$. This holds if $B = 0$ or $C = 0$. The condition is not necessary when $m, n \geq 2$: for $m = n = 2$ and $B = C = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq 0$ we have $BC = CB = 0$ (Exercise 1.3(b)); for larger m, n put this 2×2 matrix in the top left corner of zero matrices. If $m = 1$, then $CB \in \mathbb{R}^{n \times n}$ has entries $c_i b_j$, so $CB = 0$ forces $B = 0$ or $C = 0$; the case $n = 1$ is the same with BC .

Exercise 5.4. Show that $Z_1 = \begin{pmatrix} I_m & B_1 \\ 0 & I_n \end{pmatrix}$ and $Z_2 = \begin{pmatrix} I_m & B_2 \\ 0 & I_n \end{pmatrix}$ ($B_1, B_2 \in \mathbb{R}^{m \times n}$) commute.

Solution. Write $Z_1 = I_{m+n} + X_1$ and $Z_2 = I_{m+n} + X_2$ with $X_i = \begin{pmatrix} 0 & B_i \\ 0 & 0 \end{pmatrix}$. By the block formula, $X_1 X_2 = \begin{pmatrix} 0 \cdot 0 + B_1 \cdot 0 & 0 \cdot B_2 + B_1 \cdot 0 \\ 0 & 0 \end{pmatrix} = 0$, and likewise $X_2 X_1 = 0$. Hence

$$Z_1 Z_2 = I + X_1 + X_2 + X_1 X_2 = I + X_1 + X_2 = I + X_2 + X_1 + X_2 X_1 = Z_2 Z_1$$

Exercise 5.5. If $A \in \mathbb{R}^{m \times m}$ and $D \in \mathbb{R}^{n \times n}$ are invertible and $C \in \mathbb{R}^{n \times m}$, show that

$$\begin{pmatrix} A & 0 \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} A^{-1} & 0 \\ -D^{-1}CA^{-1} & D^{-1} \end{pmatrix}$$

Solution. Look for the inverse in the form $\begin{pmatrix} A^{-1} & 0 \\ R & D^{-1} \end{pmatrix}$. By the block formula,

$$\begin{pmatrix} A & 0 \\ C & D \end{pmatrix} \begin{pmatrix} A^{-1} & 0 \\ R & D^{-1} \end{pmatrix} = \begin{pmatrix} I_m & 0 \\ CA^{-1} + DR & I_n \end{pmatrix},$$

which is I if and only if $CA^{-1} + DR = 0$, that is, $R = -D^{-1}CA^{-1}$. With this R , the product in the other order is $\begin{pmatrix} A^{-1}A & 0 \\ RA + D^{-1}C & D^{-1}D \end{pmatrix} = \begin{pmatrix} I & 0 \\ -D^{-1}CA^{-1} + D^{-1}C & I \end{pmatrix} = I$.

Exercise 5.6. Use Exercise 5.5 to find the inverse of

$$N = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ \hline 1 & 0 & 3 & 1 \\ 4 & 1 & 2 & 1 \end{array} \right),$$

and check the answer by multiplication.

Solution. Here $A = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$. By the 2×2 formula of Exercise 2.12, $A^{-1} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}$ and $D^{-1} = \begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix}$. Then

$$CA^{-1} = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}, \quad -D^{-1}CA^{-1} = -\begin{pmatrix} 1 & -1 \\ -2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ -4 & -3 \end{pmatrix},$$

so

$$N^{-1} = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ \hline 1 & 1 & 1 & -1 \\ -4 & -3 & -2 & 3 \end{array} \right)$$

Check: the $(2, 1)$ block of NN^{-1} is $CA^{-1} + D \begin{pmatrix} 1 & 1 \\ -4 & -3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} + \begin{pmatrix} -1 & 0 \\ -2 & -1 \end{pmatrix} = 0$, the $(1, 2)$ block is $A \cdot 0 + 0 \cdot D^{-1} = 0$, and the diagonal blocks are $AA^{-1} = I$ and $DD^{-1} = I$; so $NN^{-1} = I_4$, and N^{-1} is the inverse by Theorem 2.15.

Exercise 5.7. Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ with $A \in \mathbb{R}^{p \times p}$, $B \in \mathbb{R}^{p \times q}$, $C \in \mathbb{R}^{q \times p}$ and $D \in \mathbb{R}^{q \times q}$.

(a) If A and its Schur complement $E = D - CA^{-1}B$ are invertible, show that M is invertible and

$$M^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BE^{-1}CA^{-1} & -A^{-1}BE^{-1} \\ -E^{-1}CA^{-1} & E^{-1} \end{pmatrix}$$

(b) If D and $F = A - BD^{-1}C$ are invertible, show that M is invertible and

$$M^{-1} = \begin{pmatrix} F^{-1} & -F^{-1}BD^{-1} \\ -D^{-1}CF^{-1} & D^{-1} + D^{-1}CF^{-1}BD^{-1} \end{pmatrix}$$

(c) Let $X = [\mathbf{1}_n \ X_0] \in \mathbb{R}^{n \times (k+1)}$ be a design matrix with $X'X$ invertible, let $\bar{x} = \frac{1}{n}X_0'\mathbf{1}$ be the mean vector, and let $S = X_0'C_nX_0$. Show that S is invertible and

$$(X'X)^{-1} = \begin{pmatrix} \frac{1}{n} + \bar{x}'S^{-1}\bar{x} & -\bar{x}'S^{-1} \\ -S^{-1}\bar{x} & S^{-1} \end{pmatrix},$$

and check this for the data of Example 1.14 against Example 2.17.

Solution. (a)

- By Theorem 5.8, $LM = U$ with $L = \begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{pmatrix}$ and $U = \begin{pmatrix} A & B \\ 0 & E \end{pmatrix}$.
- L is invertible with inverse $\begin{pmatrix} I_p & 0 \\ CA^{-1} & I_q \end{pmatrix}$ (Proposition 5.7), so $M = L^{-1}U$; and U is invertible with $U^{-1} = \begin{pmatrix} A^{-1} & -A^{-1}BE^{-1} \\ 0 & E^{-1} \end{pmatrix}$ (Proposition 5.5(c)).
- Hence M is invertible and $M^{-1} = U^{-1}(L^{-1})^{-1} = U^{-1}L$ (Theorem 2.12(a),(b)). By the block product formula (Theorem 5.2(c)),

$$U^{-1}L = \begin{pmatrix} A^{-1} & -A^{-1}BE^{-1} \\ 0 & E^{-1} \end{pmatrix} \begin{pmatrix} I_p & 0 \\ -CA^{-1} & I_q \end{pmatrix} = \begin{pmatrix} A^{-1} + A^{-1}BE^{-1}CA^{-1} & -A^{-1}BE^{-1} \\ -E^{-1}CA^{-1} & E^{-1} \end{pmatrix}$$

(b) Look for $X = \begin{pmatrix} P & Q \\ R & T \end{pmatrix}$ with $MX = I_{p+q}$; by the block product formula this means

$$AP + BR = I_p, \quad CP + DR = 0, \quad AQ + BT = 0, \quad CQ + DT = I_q$$

- The second equation gives $R = -D^{-1}CP$; then the first becomes $(A - BD^{-1}C)P = FP = I_p$, so $P = F^{-1}$ and $R = -D^{-1}CF^{-1}$.
- The fourth equation gives $T = D^{-1} - D^{-1}CQ$; then the third becomes $AQ + BD^{-1} - BD^{-1}CQ = FQ + BD^{-1} = 0$, so $Q = -F^{-1}BD^{-1}$ and $T = D^{-1} + D^{-1}CF^{-1}BD^{-1}$.
- Conversely, these blocks satisfy the four equations: $AP + BR = (A - BD^{-1}C)F^{-1} = I_p$, $CP + DR = CF^{-1} - CF^{-1} = 0$, $AQ + BT = -(A - BD^{-1}C)F^{-1}BD^{-1} + BD^{-1} = 0$ and $CQ + DT = -CF^{-1}BD^{-1} + I_q + CF^{-1}BD^{-1} = I_q$.
- So $MX = I_{p+q}$, and since M is square, M is invertible with $M^{-1} = X$ (Theorem 2.15).

(c)

- By Example 5.9, $X'X$ has the blocks $A = \mathbf{1}'\mathbf{1} = (n)$, $B = \mathbf{1}'X_0$, $C = X_0'\mathbf{1}$, $D = X_0'X_0$, and the Schur complement of A is $X_0'C_nX_0 = S$. As A and $X'X$ are invertible, so is S (Theorem 5.21).
- Here $A^{-1}B = \frac{1}{n}\mathbf{1}'X_0 = \bar{x}'$ and $CA^{-1} = \frac{1}{n}X_0'\mathbf{1} = \bar{x}$, so (a) gives the displayed formula: $A^{-1} + A^{-1}BS^{-1}CA^{-1} = \frac{1}{n} + \bar{x}'S^{-1}\bar{x}$, $-A^{-1}BS^{-1} = -\bar{x}'S^{-1}$ and $-S^{-1}CA^{-1} = -S^{-1}\bar{x}$. For $k = 1$ and $X_0 = x$, $S = x'C_nx = \sum_i (x_i - \bar{x})^2$ (Example 1.13), so the last diagonal entry of $(X'X)^{-1}$ is $1/\sum_i (x_i - \bar{x})^2$.

- In Example 1.14, $n = 4$, $\bar{x} = (1, 2)'$ and $S = 4I_2$, so $S^{-1} = \frac{1}{4}I_2$, $\frac{1}{4} + \bar{x}'S^{-1}\bar{x} = \frac{1}{4} + \frac{1+4}{4} = \frac{3}{2}$ and $-\bar{x}'S^{-1} = (-\frac{1}{4}, -\frac{1}{2})$: these are the entries of $(X'X)^{-1} = \frac{1}{4} \begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$ found in Example 2.17.

Exercise 5.8. Let

$$M = \left(\begin{array}{cc|cc} 2 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ \hline 1 & 1 & 2 & -1 \\ 2 & 1 & 1 & 2 \end{array} \right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

Find the Schur complement E of A and $\det M$, and use Exercise 5.7(a) to find M^{-1} .

Solution. Here $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}$; $\det A = 1$, and $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ by the 2×2 formula of Exercise 2.12. Then

$$A^{-1}B = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}, \quad CA^{-1} = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$$E = D - CA^{-1}B = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

So $\det M = \det A \cdot \det E = 1 \cdot 1 = 1$ (Theorem 5.21(a)), and $E^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. Next

$$A^{-1}BE^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad E^{-1}CA^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix},$$

and $A^{-1}BE^{-1}CA^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = I_2$. By Exercise 5.7(a),

$$M^{-1} = \begin{pmatrix} A^{-1} + I_2 & -A^{-1}BE^{-1} \\ -E^{-1}CA^{-1} & E^{-1} \end{pmatrix} = \begin{pmatrix} 2 & -1 & 0 & -1 \\ -1 & 3 & -1 & 0 \\ \hline -1 & -1 & 1 & 1 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

Check: $\text{row}_1(M)M^{-1} = 2(2, -1, 0, -1) + (-1, 3, -1, 0) + (-1, -1, 1, 1) + (-1, 0, 0, 1) = (1, 0, 0, 0)$ and $\text{row}_4(M)M^{-1} = 2(2, -1, 0, -1) + (-1, 3, -1, 0) + (-1, -1, 1, 1) + 2(-1, 0, 0, 1) = (0, 0, 0, 1)$; rows 2 and 3 likewise give $(0, 1, 0, 0)$ and $(0, 0, 1, 0)$.

Exercise 5.9. Show that

$$\det \begin{pmatrix} 1 & 5 & -5 \\ 3 & 2 & -5 \\ 6 & -2 & -5 \end{pmatrix} = -5 \quad \text{and} \quad \det \begin{pmatrix} 2 & 6 & 5 \\ -2 & -3 & -5 \\ 2 & 3 & 9 \end{pmatrix} = 24,$$

in each case (i) by expansion along the first row and (ii) by row and column operations.

Solution. Call the two matrices A and B .

(A)

- (i) $\det A = 1 \det \begin{pmatrix} 2 & -5 \\ -2 & -5 \end{pmatrix} - 5 \det \begin{pmatrix} 3 & -5 \\ 6 & -5 \end{pmatrix} - 5 \det \begin{pmatrix} 3 & 2 \\ 6 & -2 \end{pmatrix} = (-10 - 10) - 5(-15 + 30) - 5(-6 - 12) = -20 - 75 + 90 = -5$.
- (ii) Column 3 is -5 times $(1, 1, 1)'$. Taking the factor -5 out of column 3 (Theorem 5.14(a),(f)) and then subtracting row 1 from rows 2 and 3,

$$\det A = -5 \det \begin{pmatrix} 1 & 5 & 1 \\ 3 & 2 & 1 \\ 6 & -2 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 5 & 1 \\ 3 & 2 & 1 \\ 6 & -2 & 1 \end{pmatrix} \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 5 & 1 \\ 2 & -3 & 0 \\ 5 & -7 & 0 \end{pmatrix}$$

The last matrix has the single nonzero entry 1 in column 3, at position $(1, 3)$; expanding along column 3, its determinant is $(-1)^{1+3} \det \begin{pmatrix} 2 & -3 \\ 5 & -7 \end{pmatrix} = -14 + 15 = 1$. Hence $\det A = -5 \cdot 1 = -5$.

(B)

- (i) $\det B = 2 \det \begin{pmatrix} -3 & -5 \\ 3 & 9 \end{pmatrix} - 6 \det \begin{pmatrix} -2 & -5 \\ 2 & 9 \end{pmatrix} + 5 \det \begin{pmatrix} -2 & -3 \\ 2 & 3 \end{pmatrix} = 2(-27 + 15) - 6(-18 + 10) + 5(-6 + 6) = -24 + 48 + 0 = 24.$
- (ii)

$$B \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 + R_1} \begin{pmatrix} 2 & 6 & 5 \\ 0 & 3 & 0 \\ 0 & -3 & 4 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 + R_2} \begin{pmatrix} 2 & 6 & 5 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

No step changes the determinant, so $\det B = 2 \cdot 3 \cdot 4 = 24.$

Exercise 5.10. Compute the determinant of

$$A = \begin{pmatrix} 0 & 4 & 0 & 5 \\ 1 & 0 & -1 & 2 \\ 0 & 3 & 0 & -2 \\ 0 & 0 & -6 & 0 \end{pmatrix}$$

- (a) by repeated cofactor expansion: expand $\det A$ in terms of determinants of 3×3 matrices, these in terms of determinants of 2×2 matrices, and finally these in terms of determinants of 1×1 matrices;
 (b) by row reduction to triangular form.

Solution. (a)

- Column 1 has the single nonzero entry $a_{21} = 1$; expanding along column 1, $\det A = 1 \cdot (-1)^{2+1} \det \begin{pmatrix} 4 & 0 & 5 \\ 3 & 0 & -2 \\ 0 & -6 & 0 \end{pmatrix}.$
- Row 3 of this 3×3 matrix has the single nonzero entry -6 , in column 2: $\det \begin{pmatrix} 4 & 0 & 5 \\ 3 & 0 & -2 \\ 0 & -6 & 0 \end{pmatrix} = (-6)(-1)^{3+2} \det \begin{pmatrix} 4 & 5 \\ 3 & -2 \end{pmatrix}.$
- Expanding the 2×2 determinant along its first row, $4(-1)^{1+1} \det(-2) + 5(-1)^{1+2} \det(3) = -8 - 15 = -23.$
- Hence $\det A = (-1) \cdot 6 \cdot (-23) = 138.$

(b)

$$\begin{aligned} A &\xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 4 & 0 & 5 \\ 0 & 3 & 0 & -2 \\ 0 & 0 & -6 & 0 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_3} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 3 & 0 & -2 \\ 0 & 0 & -6 & 0 \end{pmatrix} \\ &\xrightarrow{R_3 \leftarrow R_3 - 3R_2} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 0 & -23 \\ 0 & 0 & -6 & 0 \end{pmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{pmatrix} 1 & 0 & -1 & 2 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & -6 & 0 \\ 0 & 0 & 0 & -23 \end{pmatrix} \end{aligned}$$

There are two interchanges, so $\det A = (-1)^2 \cdot 1 \cdot 1 \cdot (-6) \cdot (-23) = 138.$

Exercise 5.11. Let $\alpha \in \mathbb{R}$ and $A = \begin{pmatrix} 1 & \alpha & 0 \\ 4 & 5 & 3 \\ 1 & 0 & 2 \end{pmatrix}$. For which α is $\det A = 0$? Show that the columns of A are linearly dependent in that case.

Solution. Expanding along the first row, $\det A = 1 \det \begin{pmatrix} 5 & 3 \\ 1 & 2 \end{pmatrix} - \alpha \det \begin{pmatrix} 4 & 3 \\ 1 & 2 \end{pmatrix} = 10 - 5\alpha$, so $\det A = 0$ exactly when $\alpha = 2$ (and A is invertible exactly when $\alpha \neq 2$, Theorem 5.17(a)). For $\alpha = 2$ and $x = (2, -1, -1)'$, $Ax = (2 - 2 + 0, 8 - 5 - 3, 2 - 0 - 2)' = 0$; that is, $2a_1 - a_2 - a_3 = 0$, a nontrivial relation among the columns.

Exercise 5.12. Let $K \in \mathbb{R}^{12 \times 12}$ satisfy $K^5 = 3K$. What is the numerical value of $\det K$? Is there more than one possibility? If so, give all possible values.

Solution.

- By Theorem 5.17(c), $\det K^5 = (\det K)^5$ and $\det(3K) = 3^{12} \det K$. So $(\det K)^5 = 3^{12} \det K$, that is, $\det K \cdot ((\det K)^4 - 3^{12}) = 0$.
- Hence $\det K = 0$ or $(\det K)^4 = 3^{12}$, that is, $((\det K)^2 - 3^6)((\det K)^2 + 3^6) = 0$. The second factor is positive, so $(\det K)^2 = 3^6$ and $\det K = \pm 3^3 = \pm 27$.
- All three values occur. Let t be the positive number with $t^4 = 3$, so $t^5 = 3t$. For $K = 0$, $\det K = 0$. For $K = tI_{12}$, $K^5 = t^5 I_{12} = 3K$ and $\det K = t^{12} = 3^3 = 27$. For $K = \text{diag}(-t, t, \dots, t)$, $K^5 = \text{diag}(-t^5, t^5, \dots, t^5) = 3K$ and $\det K = -t^{12} = -27$.

So $\det K$ is 0, 27 or -27 .

Exercise 5.13. Let $A \in \mathbb{R}^{n \times n}$ be invertible. Show that if all entries of A and of A^{-1} are integers, then $\det A = \pm 1$.

Solution.

- A matrix with integer entries has an integer determinant: for $n = 1$ it is the entry, and for $n \geq 2$ Definition 5.10 gives $\det A$ as a sum of products of entries and of determinants of $(n-1) \times (n-1)$ matrices with integer entries; use induction on n . So $d = \det A$ and $\det A^{-1}$ are integers.
- By Theorem 5.17(a),(c), $d \neq 0$ and $\det A^{-1} = 1/d$. So d and $1/d$ are nonzero integers.
- A nonzero integer has absolute value at least 1: $|d| \geq 1$ and $|1/d| \geq 1$, that is, $|d| \leq 1$. Hence $|d| = 1$, and $\det A = \pm 1$.

Exercise 5.14. Let $A = \begin{pmatrix} 1 & 0 & -1 \\ -1 & 3 & -3 \\ 0 & -4 & 5 \end{pmatrix}$.

- Compute the cofactor c_{ij} of each entry of A .
- Compute $\det A$ by expansion along the second row, and check the answer by expansion along the second column.
- Use Theorem 5.18 to compute A^{-1} .

Solution. (a)

$$\begin{aligned} c_{11} &= \det \begin{pmatrix} 3 & -3 \\ -4 & 5 \end{pmatrix} = 3, & c_{12} &= -\det \begin{pmatrix} -1 & -3 \\ 0 & 5 \end{pmatrix} = 5, & c_{13} &= \det \begin{pmatrix} -1 & 3 \\ 0 & -4 \end{pmatrix} = 4, \\ c_{21} &= -\det \begin{pmatrix} 0 & -1 \\ -4 & 5 \end{pmatrix} = 4, & c_{22} &= \det \begin{pmatrix} 1 & -1 \\ 0 & 5 \end{pmatrix} = 5, & c_{23} &= -\det \begin{pmatrix} 1 & 0 \\ 0 & -4 \end{pmatrix} = 4, \\ c_{31} &= \det \begin{pmatrix} 0 & -1 \\ 3 & -3 \end{pmatrix} = 3, & c_{32} &= -\det \begin{pmatrix} 1 & -1 \\ -1 & -3 \end{pmatrix} = 4, & c_{33} &= \det \begin{pmatrix} 1 & 0 \\ -1 & 3 \end{pmatrix} = 3 \end{aligned}$$

(b) Along row 2: $\det A = (-1) \cdot 4 + 3 \cdot 5 + (-3) \cdot 4 = -1$. Along column 2: $\det A = 0 \cdot 5 + 3 \cdot 5 + (-4) \cdot 4 = -1$ (Theorem 5.11).

(c) Since $\det A = -1 \neq 0$,

$$A^{-1} = \frac{1}{-1} \text{adj } A = - \begin{pmatrix} 3 & 4 & 3 \\ 5 & 5 & 4 \\ 4 & 4 & 3 \end{pmatrix} = \begin{pmatrix} -3 & -4 & -3 \\ -5 & -5 & -4 \\ -4 & -4 & -3 \end{pmatrix}$$

Check: $\text{row}_1(A)A^{-1} = (1, 0, -1)A^{-1} = (-3 + 4, -4 + 4, -3 + 3) = (1, 0, 0)$; likewise $\text{row}_2(A)A^{-1} = (3 - 15 + 12, 4 - 15 + 12, 3 - 12 + 9) = (0, 1, 0)$ and $\text{row}_3(A)A^{-1} = (20 - 20, 20 - 20, 16 - 15) = (0, 0, 1)$.

Exercise 5.15. (a) Let $A \in \mathbb{R}^{n \times n}$ ($n \geq 2$) with $\det A \neq 0$, and let $b \in \mathbb{R}^n$. Show that the solution x of $Ax = b$ has the entries

$$x_j = \frac{\det A_j}{\det A} \quad (1 \leq j \leq n),$$

where A_j is A with column j replaced by b (**Cramer's rule**).

(b) Use (a) to solve the first system of Exercise 2.4, $\begin{pmatrix} 1 & -3 & -2 \\ -1 & 2 & 1 \\ -1 & 2 & 2 \end{pmatrix} x = (1, -1, -2)'$.

Solution. (a)

- By Theorem 5.18, A is invertible, so $x = A^{-1}b = \frac{1}{\det A}(\text{adj } A)b$ is the only solution.
- Entry j of $(\text{adj } A)b$ is $\sum_{i=1}^n (\text{adj } A)_{ji}b_i = \sum_{i=1}^n c_{ij}b_i$.
- A_j differs from A only in column j , so $A_j(i|j) = A(i|j)$, and the (i, j) entry of A_j is b_i . Expanding $\det A_j$ along column j (Theorem 5.11) gives $\sum_{i=1}^n b_i(-1)^{i+j} \det A(i|j) = \sum_{i=1}^n c_{ij}b_i$. Hence $x_j = \det A_j / \det A$.

(b) Expanding along the first row,

$$\det A = 1(4 - 2) + 3(-2 + 1) - 2(-2 + 2) = 2 - 3 + 0 = -1,$$

$$\det A_1 = \det \begin{pmatrix} 1 & -3 & -2 \\ -1 & 2 & 1 \\ -2 & 2 & 2 \end{pmatrix} = 1(4 - 2) + 3(-2 + 2) - 2(-2 + 4) = 2 + 0 - 4 = -2,$$

$$\det A_2 = \det \begin{pmatrix} 1 & 1 & -2 \\ -1 & -1 & 1 \\ -1 & -2 & 2 \end{pmatrix} = 1(-2 + 2) - 1(-2 + 1) - 2(2 - 1) = 0 + 1 - 2 = -1,$$

$$\det A_3 = \det \begin{pmatrix} 1 & -3 & 1 \\ -1 & 2 & -1 \\ -1 & 2 & -2 \end{pmatrix} = 1(-4 + 2) + 3(2 - 1) + 1(-2 + 2) = -2 + 3 + 0 = 1$$

So $x = \left(\frac{-2}{-1}, \frac{-1}{-1}, \frac{1}{-1} \right)' = (2, 1, -1)'$, the solution found by elimination in Exercise 2.4.

Exercise 5.16. Let $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times m}$. Show that $\det(I_m - AB) = \det(I_n - BA)$.

Solution. By the block product formula (Theorem 5.2),

$$\begin{pmatrix} I_m - AB & A \\ 0 & I_n \end{pmatrix} \begin{pmatrix} I_m & 0 \\ B & I_n \end{pmatrix} = \begin{pmatrix} I_m - AB + AB & A \\ B & I_n \end{pmatrix} = \begin{pmatrix} I_m & A \\ B & I_n \end{pmatrix},$$

$$\begin{pmatrix} I_m & 0 \\ B & I_n \end{pmatrix} \begin{pmatrix} I_m & A \\ 0 & I_n - BA \end{pmatrix} = \begin{pmatrix} I_m & A \\ B & BA + I_n - BA \end{pmatrix} = \begin{pmatrix} I_m & A \\ B & I_n \end{pmatrix}$$

So the two products are equal. By Theorems 5.17(b) and 5.20, the first has determinant $\det(I_m - AB) \det I_n \cdot \det I_m \det I_n = \det(I_m - AB)$, and the second $\det I_m \det I_n \cdot \det I_m \det(I_n - BA) = \det(I_n - BA)$.

Exercise 5.17. For $t \in \mathbb{R}$ let

$$M = \left(\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ \hline 1 & 1 & t & 2 \\ 0 & 1 & 2 & t+1 \end{array} \right) = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

(a) Find $\det M$ with the Schur complement of A . (b) For which t is M invertible? (c) Find $\text{rank } M$ for these exceptional values of t .

Solution. (a) Here $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, $C = B'$ and $D = \begin{pmatrix} t & 2 \\ 2 & t+1 \end{pmatrix}$. Here $\det A = 2 \cdot 1 - 1 \cdot 1 = 1$, and $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ by the 2×2 formula of Exercise 2.12. Then

$$CA^{-1} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix}, \quad CA^{-1}B = \begin{pmatrix} 0 & 1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

so the Schur complement is $S = D - CA^{-1}B = \begin{pmatrix} t-1 & 1 \\ 1 & t-1 \end{pmatrix}$. By Theorem 5.21(a), $\det M = \det A \cdot \det S = (t-1)^2 - 1 = t(t-2)$.

(b) By Theorem 5.17(a), M is invertible if and only if $t \neq 0$ and $t \neq 2$.

(c) For $t = 0$, $S = \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}$; for $t = 2$, $S = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$. In both cases the second row of S is a multiple of the first and $S \neq 0$, so $\text{rank } S = 1$. By Theorem 5.24(a), $\text{rank } M = 2 + 1 = 3$.

Exercise 5.18. Let $R = (1 - \rho)I_n + \rho J_n$ be the equicorrelation matrix of Example 5.23, with $\rho \neq 1$ and $\rho \neq -1/(n-1)$. Show that

$$R^{-1} = \frac{1}{1 - \rho} \left(I_n - \frac{\rho}{1 + (n-1)\rho} J_n \right)$$

Solution. Let $\tau = \rho/(1 + (n-1)\rho)$, so that $\tau(1 - \rho + n\rho) = \rho$. Using $J_n^2 = nJ_n$ (Proposition 1.12),

$$\begin{aligned} ((1 - \rho)I_n + \rho J_n)(I_n - \tau J_n) &= (1 - \rho)I_n + (\rho - (1 - \rho)\tau - n\rho\tau)J_n \\ &= (1 - \rho)I_n + (\rho - \tau(1 - \rho + n\rho))J_n = (1 - \rho)I_n \end{aligned}$$

Dividing by $1 - \rho \neq 0$ gives $R \cdot \frac{1}{1 - \rho}(I_n - \tau J_n) = I_n$, so $\frac{1}{1 - \rho}(I_n - \tau J_n) = R^{-1}$ (Theorem 2.15).

Chapter 6. Orthogonality, projections, and least squares

Inner products, lengths, and orthogonality

Definition 6.1. For $x, y \in \mathbb{R}^n$, the **inner product** of x and y is the number $x'y = x_1y_1 + \cdots + x_ny_n$; it equals $y'x$ (Theorem 1.8). The **length** (or norm) of x is $\|x\| = \sqrt{x'x} = (x_1^2 + \cdots + x_n^2)^{1/2}$, and $\|x - y\|$ is the **distance** between x and y . The vectors x and y are **orthogonal**, written $x \perp y$, if $x'y = 0$.

Since $x'x$ is a sum of squares, $\|x\| \geq 0$, and $\|x\| = 0$ only for $x = 0$; also $\|cx\| = |c| \|x\|$ for $c \in \mathbb{R}$. A vector of length 1 is a **unit vector**; for $x \neq 0$, $x/\|x\|$ is one. The zero vector is orthogonal to every vector, and it is the only vector orthogonal to itself.

Proposition 6.2. For $x, y \in \mathbb{R}^n$, $\|x + y\|^2 = \|x\|^2 + 2x'y + \|y\|^2$. In particular, if $x \perp y$, then $\|x + y\|^2 = \|x\|^2 + \|y\|^2$ (**Pythagoras**).

Proof. $(x + y)'(x + y) = x'x + x'y + y'x + y'y$, and $y'x = x'y$. □

Theorem 6.3. Let $x, y \in \mathbb{R}^n$.

- (a) (**Cauchy–Schwarz inequality**) $|x'y| \leq \|x\| \|y\|$, with equality if and only if one of x, y is a multiple of the other.
- (b) (**Triangle inequality**) $\|x + y\| \leq \|x\| + \|y\|$.

Proof. (a) If $x = 0$, both sides are 0, and $x = 0y$. Let $x \neq 0$.

- Put $c = x'y/\|x\|^2$ and $w = y - cx$. Then $x'w = x'y - cx'x = 0$, so $cx \perp w$.
- Pythagoras for $y = cx + w$ gives $\|y\|^2 = c^2\|x\|^2 + \|w\|^2 \geq c^2\|x\|^2 = (x'y)^2/\|x\|^2$. Multiply by $\|x\|^2$ and take square roots.
- Equality holds if and only if $w = 0$, that is, $y = cx$. Conversely, if $y = ax$, then $c = a$ and $w = 0$; if $x = ay$ with $x \neq 0$, then $a \neq 0$ and $y = a^{-1}x$.

(b) By Proposition 6.2 and (a), $\|x + y\|^2 \leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 = (\|x\| + \|y\|)^2$, and both $\|x + y\|$ and $\|x\| + \|y\|$ are ≥ 0 . □

Definition 6.4. Let W be a subspace of $\mathbb{R}^n$. A vector $y \in \mathbb{R}^n$ is **orthogonal to W** , written $y \perp W$, if $y'w = 0$ for every $w \in W$. The **orthogonal complement** of W is $W^\perp = \{y \in \mathbb{R}^n : y \perp W\}$.

Theorem 6.5. Let W be a subspace of $\mathbb{R}^n$.

- (a) If $w_1, \dots, w_k$ span W and $A = [w_1 \cdots w_k] \in \mathbb{R}^{n \times k}$, then $y \perp W$ if and only if $y'w_1 = \cdots = y'w_k = 0$, that is, $A'y = 0$. So $W^\perp = \mathcal{N}(A')$, a subspace of $\mathbb{R}^n$.
- (b) $W \cap W^\perp = \{0\}$.
- (c) $\dim W + \dim W^\perp = n$ and $\mathbb{R}^n = W \oplus W^\perp$: every $y \in \mathbb{R}^n$ is $y = w + v$ with $w \in W$ and $v \in W^\perp$ in exactly one way.

Proof. (a) If $y \perp W$, then $y'w_j = 0$, as $w_j \in W$. Conversely, let all $y'w_j = 0$. Every $w \in W$ is $w = Ac$ for some $c \in \mathbb{R}^k$ (Definition 3.6 and Theorem 1.4(b)), and $y'w = (A'y)'c = 0$. The j th entry of $A'y$ is w'_jy (Theorem 1.4(d)).

(b) If $y \in W \cap W^\perp$, then $y \perp y$, so $y = 0$.

(c) Let $d = \dim W$. If $d = 0$, then $W = \{0\}$ and $W^\perp = \mathbb{R}^n$. Let $d \geq 1$.

- Let $A \in \mathbb{R}^{n \times d}$ have a basis of W as its columns, so $\text{rank } A = d$. By (a) and Theorem 4.12, $\dim W^\perp = \dim \mathcal{N}(A') = n - \text{rank } A = n - d$.

- By (b), the sum $W + W^\perp$ is direct (Definition 3.20), and a basis of W together with a basis of $W^\perp$, $d + (n - d) = n$ vectors, is a basis of $W \oplus W^\perp$ (Theorem 3.23). So $W \oplus W^\perp$ is a subspace of $\mathbb{R}^n$ (Theorem 3.21(a)) of dimension n , hence equal to $\mathbb{R}^n$ (Theorem 4.5(a)). The decomposition $y = w + v$ is unique by Theorem 3.22. $\square$

Corollary 6.6. For $A \in \mathbb{R}^{m \times n}$,

$$\mathcal{N}(A) = \mathcal{C}(A')^\perp, \quad \mathbb{R}^n = \mathcal{C}(A') \oplus \mathcal{N}(A); \quad \mathcal{N}(A') = \mathcal{C}(A)^\perp, \quad \mathbb{R}^m = \mathcal{C}(A) \oplus \mathcal{N}(A')$$

So $Ax = 0$ if and only if x is orthogonal to every (transposed) row of A , and the four subspaces of A (Definition 4.7) are orthogonal in pairs: the transposed row space $\mathcal{C}(A')$ and the null space in $\mathbb{R}^n$, the column space and the left null space in $\mathbb{R}^m$.

Proof. The columns of A' span $\mathcal{C}(A')$, so Theorem 6.5(a) gives $\mathcal{C}(A')^\perp = \mathcal{N}((A')') = \mathcal{N}(A)$; applied to the columns of A , it gives $\mathcal{C}(A)^\perp = \mathcal{N}(A')$. The direct sums follow from Theorem 6.5(c). $\square$

Example 6.7. For $A = \begin{pmatrix} 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 5 \\ 2 & 5 & 1 & 8 \end{pmatrix}$ of Example 4.10, the transposed nonzero rows $\rho'_1 = (1, 0, 3, -1)'$, $\rho'_2 = (0, 1, -1, 2)'$ of its RREF form a basis of $\mathcal{C}(A')$, and $s_1 = (-3, 1, 1, 0)'$, $s_2 = (1, -2, 0, 1)'$ a basis of $\mathcal{N}(A)$. Indeed

$$\rho_1 s_1 = -3 + 0 + 3 + 0 = 0, \quad \rho_1 s_2 = 1 + 0 + 0 - 1 = 0, \quad \rho_2 s_1 = 0 + 1 - 1 + 0 = 0, \quad \rho_2 s_2 = 0 - 2 + 0 + 2 = 0,$$

and $2 + 2 = 4$. In $\mathbb{R}^3$, $y = (-1, -1, 1)'$ is a basis of $\mathcal{N}(A')$, orthogonal to the basis $a_1 = (1, 1, 2)'$, $a_2 = (2, 3, 5)'$ of $\mathcal{C}(A)$: $-1 - 1 + 2 = 0$ and $-2 - 3 + 5 = 0$; and $2 + 1 = 3$.

Orthonormal bases, Gram–Schmidt, and QR

Proposition 6.8. Let $u_1, \dots, u_k \in \mathbb{R}^n$ be nonzero and pairwise orthogonal ($u'_i u_j = 0$ for $i \neq j$).

- If $y = c_1 u_1 + \dots + c_k u_k$, then $c_j = u'_j y / \|u_j\|^2$ for $j = 1, \dots, k$.
- $u_1, \dots, u_k$ are linearly independent.

Proof. (a) $u'_j y = \sum_i c_i u'_j u_i = c_j \|u_j\|^2$, and $\|u_j\|^2 > 0$. (b) If $\sum_i c_i u_i = 0$, then (a) with $y = 0$ gives every $c_j = 0$. $\square$

Definition 6.9. Vectors $q_1, \dots, q_p \in \mathbb{R}^n$ are **orthonormal** if $q'_i q_j = 0$ for $i \neq j$ and $\|q_j\| = 1$ for every j . A matrix $Q = [q_1 \ \dots \ q_p] \in \mathbb{R}^{n \times p}$ has **orthonormal columns** if $Q'Q = I_p$, since $(Q'Q)_{ij} = q'_i q_j$ (Theorem 1.4(d)). A square matrix $Q \in \mathbb{R}^{n \times n}$ with $Q'Q = I_n$ is an **orthogonal matrix**. A basis of a subspace consisting of orthonormal vectors is an **orthonormal basis**.

Proposition 6.10. Let $Q \in \mathbb{R}^{n \times p}$ satisfy $Q'Q = I_p$.

- $(Qx)'(Qy) = x'y$ and $\|Qx\| = \|x\|$ for all $x, y \in \mathbb{R}^p$: Q preserves inner products and lengths.
- If $p = n$, then Q is invertible, $Q^{-1} = Q'$ and $QQ' = I_n$: the rows of Q are orthonormal too.

Proof. (a) $(Qx)'(Qy) = x'Q'Qy = x'y$; take $y = x$. (b) Q' is a left inverse of the square matrix Q , so $Q^{-1} = Q'$ (Theorem 2.15), and $QQ' = QQ^{-1} = I_n$. $\square$

Theorem 6.11 (Gram–Schmidt). Let $a_1, \dots, a_p \in \mathbb{R}^n$ be linearly independent. Put $b_1 = a_1$ and, for $j = 2, \dots, p$,

$$b_j = a_j - t_{1j}b_1 - t_{2j}b_2 - \dots - t_{j-1,j}b_{j-1}, \quad t_{ij} = \frac{a'_j b_i}{\|b_i\|^2} \quad (i < j)$$

Then $b_1, \dots, b_p$ are nonzero and pairwise orthogonal, and $\text{span}\{b_1, \dots, b_j\} = \text{span}\{a_1, \dots, a_j\}$ for every j . Hence the vectors $q_j = b_j / \|b_j\|$ ($1 \leq j \leq p$) form an orthonormal basis of $\text{span}\{a_1, \dots, a_p\}$.

Proof. Induction on j : $b_1 = a_1 \neq 0$, since an independent list does not contain 0. Let $j \geq 2$, and let $b_1, \dots, b_{j-1}$ be nonzero, pairwise orthogonal, with $\text{span } W_{j-1} = \text{span}\{a_1, \dots, a_{j-1}\}$; so b_j is defined.

- Orthogonal: for $l < j$, $b'_l b_j = b'_l a_j - \sum_{i < j} t_{ij} b'_l b_i = b'_l a_j - t_{lj} \|b_l\|^2 = 0$.
- Nonzero: otherwise $a_j = \sum_{i < j} t_{ij} b_i \in W_{j-1}$, and $a_1, \dots, a_j$ would be dependent (Proposition 3.12(c))
✗
- Span: b_j is a combination of a_j and $b_1, \dots, b_{j-1} \in W_{j-1}$, and $a_j = b_j + \sum_{i < j} t_{ij} b_i$; so each of the lists $a_1, \dots, a_j$ and $b_1, \dots, b_j$ lies in the span of the other, and the two spans are equal (Theorem 3.7).

Finally, the q_j are pairwise orthogonal unit vectors, hence independent (Proposition 6.8), and they span $\text{span}\{b_1, \dots, b_p\} = \text{span}\{a_1, \dots, a_p\}$. $\square$

Theorem 6.12 (QR decomposition). Let $A = [a_1 \ \dots \ a_p] \in \mathbb{R}^{n \times p}$ have full column rank p . Then $A = QR$, where $Q \in \mathbb{R}^{n \times p}$ has orthonormal columns ($Q'Q = I_p$) and $R \in \mathbb{R}^{p \times p}$ is upper triangular with positive diagonal entries; moreover $R = Q'A$.

Proof. • The columns of A are independent (Theorem 4.13(b)). Apply Theorem 6.11 to them, and let $Q = [q_1 \ \dots \ q_p]$.

- The construction gives $a_j = b_j + \sum_{i < j} t_{ij} b_i = \sum_{i \leq j} r_{ij} q_i$, where $r_{jj} = \|b_j\| > 0$ and $r_{ij} = t_{ij} \|b_i\|$ for $i < j$. Let $R = (r_{ij})$, with $r_{ij} = 0$ for $i > j$. Then $\text{col}_j(QR) = Q \text{col}_j(R) = \sum_{i \leq j} r_{ij} q_i = a_j$ (Theorem 1.4(a),(b)), so $A = QR$.
- $Q'Q = I_p$, as the q_j are orthonormal; hence $Q'A = Q'QR = R$. $\square$

Example 6.13. Find the QR decomposition of $A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 1 & 3 \\ 0 & 2 & 4 \\ 2 & 2 & 5 \end{pmatrix}$.

Solution. Apply Gram–Schmidt (Theorem 6.11) to the columns a_1, a_2, a_3 .

- $b_1 = a_1 = (1, 2, 0, 2)'$, $\|b_1\|^2 = 1 + 4 + 0 + 4 = 9$.
- $a'_2 b_1 = 3 + 2 + 0 + 4 = 9$, so $t_{12} = 1$ and $b_2 = a_2 - b_1 = (3, 1, 2, 2)' - (1, 2, 0, 2)' = (2, -1, 2, 0)'$, with $\|b_2\|^2 = 4 + 1 + 4 + 0 = 9$.
- $a'_3 b_1 = 2 + 6 + 0 + 10 = 18$ and $a'_3 b_2 = 4 - 3 + 8 + 0 = 9$, so $t_{13} = 2$, $t_{23} = 1$ and

$$b_3 = a_3 - 2b_1 - b_2 = \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \\ 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 2 \\ 1 \end{pmatrix}, \quad \|b_3\|^2 = 4 + 0 + 4 + 1 = 9$$

- Check: $b'_1 b_2 = 2 - 2 + 0 + 0 = 0$, $b'_1 b_3 = -2 + 0 + 0 + 2 = 0$, $b'_2 b_3 = -4 + 0 + 4 + 0 = 0$.
- All three lengths are 3, so $q_j = \frac{1}{3} b_j$, $r_{jj} = 3$, $r_{12} = 3t_{12} = 3$, $r_{13} = 3t_{13} = 6$ and $r_{23} = 3t_{23} = 3$:

$$Q = \frac{1}{3} \begin{pmatrix} 1 & 2 & -2 \\ 2 & -1 & 0 \\ 0 & 2 & 2 \\ 2 & 0 & 1 \end{pmatrix}, \quad R = \begin{pmatrix} 3 & 3 & 6 \\ 0 & 3 & 3 \\ 0 & 0 & 3 \end{pmatrix}$$

As a check, $R = Q'A$; for instance $r_{23} = q'_2 a_3 = \frac{1}{3}(4 - 3 + 8 + 0) = 3$.

Orthogonal projections

Definition 6.14. Let W be a subspace of $\mathbb{R}^n$ and $y \in \mathbb{R}^n$. By Theorem 6.5(c), $y = \hat{w} + (y - \hat{w})$ with $\hat{w} \in W$ and $y - \hat{w} \in W^\perp$ for exactly one vector $\hat{w}$, the **orthogonal projection** of y onto W . Thus $\hat{w}$ is the only vector of W with $y - \hat{w} \perp W$.

Theorem 6.15. Let $\hat{w}$ be the orthogonal projection of $y \in \mathbb{R}^n$ onto a subspace W of $\mathbb{R}^n$. Then

$$\|y - w\|^2 = \|y - \hat{w}\|^2 + \|\hat{w} - w\|^2 \quad (w \in W)$$

Hence $\hat{w}$ is the vector of W closest to y : $\|y - w\| > \|y - \hat{w}\|$ for every $w \in W$ with $w \neq \hat{w}$.

Proof. $y - w = (y - \hat{w}) + (\hat{w} - w)$, where $\hat{w} - w \in W$ and $y - \hat{w} \perp W$; apply Pythagoras (Proposition 6.2). If $w \neq \hat{w}$, then $\|\hat{w} - w\|^2 > 0$. $\square$

Theorem 6.16. Let $X \in \mathbb{R}^{n \times p}$ have full column rank p . Then $X'X$ is invertible. Put

$$H = X(X'X)^{-1}X' \in \mathbb{R}^{n \times n} \quad (\text{the } \mathbf{hat} \text{ matrix}), \quad M = I_n - H \quad (\text{the } \mathbf{residual} \text{ maker})$$

- (a) For every $y \in \mathbb{R}^n$, Hy is the orthogonal projection of y onto $\mathcal{C}(X)$, and $y = Hy + My$ is the decomposition of y along $\mathbb{R}^n = \mathcal{C}(X) \oplus \mathcal{C}(X)^\perp$.
- (b) $H' = H = H^2$ and $M' = M = M^2$: both are symmetric and idempotent.
- (c) $HX = X$, $MX = 0$ and $HM = MH = 0$.

Proof. $\text{rank}(X'X) = \text{rank } X = p$ (Theorem 4.18(b)), so the $p \times p$ matrix $X'X$ is invertible (Theorem 4.13(d)).

- (a) $Hy = X((X'X)^{-1}X'y) \in \mathcal{C}(X)$, and $X'(y - Hy) = X'y - (X'X)(X'X)^{-1}X'y = 0$, so $My = y - Hy \in \mathcal{C}(X)^\perp$ (Theorem 6.5(a)). By Definition 6.14, Hy is the projection.
- (b) $X'X$ is symmetric, hence so is its inverse: $((X'X)^{-1})' = ((X'X)')^{-1} = (X'X)^{-1}$ (Theorem 2.12(c)). So $H' = X((X'X)^{-1})'X' = H$ and $H^2 = X(X'X)^{-1}(X'X)(X'X)^{-1}X' = H$. Then $M' = I - H' = M$ and $M^2 = I - 2H + H^2 = M$.
- (c) $HX = X(X'X)^{-1}X'X = X$, $MX = X - HX = 0$, and $HM = H - H^2 = 0 = MH$. $\square$

Example 6.17. For $X = \mathbf{1}_n$ ($p = 1$), $X'X = \mathbf{1}'\mathbf{1} = n$, so $H = \frac{1}{n}\mathbf{1}\mathbf{1}' = \bar{J}_n$ and $M = I_n - \bar{J}_n = C_n$, the centering matrix of Definition 1.11: $Hy = \bar{y}\mathbf{1}$ and $My = y - \bar{y}\mathbf{1}$. The centered vectors form $W_0 = \{x : \mathbf{1}'x = 0\} = \text{span}\{\mathbf{1}\}^\perp$ (Theorem 6.5(a)), so the decomposition $y = \bar{y}\mathbf{1} + C_n y$ along $\mathbb{R}^n = \text{span}\{\mathbf{1}\} \oplus W_0$ of Example 3.25 is the orthogonal one, and $\bar{y}\mathbf{1}$ is the constant vector closest to y (Theorem 6.15). Likewise, for any nonzero $a \in \mathbb{R}^n$ the projection of y onto $\text{span}\{a\}$ is $a(a'a)^{-1}a'y = \frac{a'y}{a'a}a$.

Example 6.18. Let $X = [\mathbf{1}_4 \ X_0]$ be the design matrix of Example 1.14 ($n = 4$, $p = 3$), and let $y = (9, 3, 5, 3)'$ be a response measured on the four units. With $(X'X)^{-1}$ of Example 2.17,

$$(X'X)^{-1}X' = \frac{1}{4} \begin{pmatrix} 6 & -1 & -2 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 0 & 2 & 0 \\ 3 & 3 & 1 & 1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} -2 & 0 & 2 & 4 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix},$$

$$H = X(X'X)^{-1}X' = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & 3 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{pmatrix} \frac{1}{4} \begin{pmatrix} -2 & 0 & 2 & 4 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 3 & 1 & 1 & -1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 3 & 1 \\ -1 & 1 & 1 & 3 \end{pmatrix},$$

and $M = I_4 - H = \frac{1}{4} \begin{pmatrix} 1 & -1 & -1 & 1 \\ -1 & 1 & 1 & -1 \\ -1 & 1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}$. Hence

$$Hy = \frac{1}{4} \begin{pmatrix} 27 + 3 + 5 - 3 \\ 9 + 9 - 5 + 3 \\ 9 - 3 + 15 + 3 \\ -9 + 3 + 5 + 9 \end{pmatrix} = \begin{pmatrix} 8 \\ 4 \\ 6 \\ 2 \end{pmatrix}, \quad My = y - Hy = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}$$

Check: $X'(My) = (1 - 1 - 1 + 1, 2 - 2, 3 - 3 - 1 + 1)' = 0$, and $(Hy)'(My) = 8 - 4 - 6 + 2 = 0$.

Idempotent matrices: rank equals trace

Theorem 6.19. Let $A \in \mathbb{R}^{n \times n}$ be idempotent, $A^2 = A$.

- (a) $Av = v$ for every $v \in \mathcal{C}(A)$, and $\mathbb{R}^n = \mathcal{N}(A) \oplus \mathcal{C}(A)$: every $v \in \mathbb{R}^n$ is $v = (v - Av) + Av$ with $v - Av \in \mathcal{N}(A)$ and $Av \in \mathcal{C}(A)$.
- (b) $I_n - A$ is idempotent.
- (c) $\text{rank } A = \text{tr } A$.
- (d) If moreover $A' = A$, then Av is the orthogonal projection of v onto $\mathcal{C}(A)$, for every $v \in \mathbb{R}^n$.

Proof. (a)

- If $v = Aw$, then $Av = A^2w = Aw = v$. For every v , $A(v - Av) = Av - A^2v = 0$.
- If $v \in \mathcal{N}(A) \cap \mathcal{C}(A)$, then $v = Av = 0$. So the sum $\mathcal{N}(A) + \mathcal{C}(A)$ is direct (Definition 3.20), and it is $\mathbb{R}^n$ by the displayed decomposition.

(b) $(I - A)^2 = I - 2A + A^2 = I - A$.

(c) If $A = 0$, then $\text{rank } A = 0 = \text{tr } A$. Let $r = \text{rank } A \geq 1$.

- Take a full-rank factorization $A = BC$, with $B \in \mathbb{R}^{n \times r}$, $C \in \mathbb{R}^{r \times n}$, and $L \in \mathbb{R}^{r \times n}$, $K \in \mathbb{R}^{n \times r}$ such that $LB = I_r$ and $CK = I_r$ (Theorem 4.21(a),(c)).
- $A^2 = A$ reads $BCBC = BC$. Multiplying on the left by L and on the right by K , $L(BCBC)K = (LB)(CB)(CK) = CB$ and $L(BC)K = (LB)(CK) = I_r$; so $CB = I_r$.
- Hence $\text{tr } A = \text{tr}(BC) = \text{tr}(CB) = \text{tr } I_r = r$ (Theorem 1.10(b)).

(d) By Corollary 6.6, $\mathcal{N}(A) = \mathcal{C}(A')^\perp = \mathcal{C}(A)^\perp$. So in (a), $Av \in \mathcal{C}(A)$ and $v - Av \in \mathcal{C}(A)^\perp$: by Definition 6.14, Av is the orthogonal projection of v onto $\mathcal{C}(A)$. $\square$

Corollary 6.20. Let $X \in \mathbb{R}^{n \times p}$ have full column rank, with hat matrix H and residual maker M . Then $\text{rank } H = \text{tr } H = p = \text{rank } X$ and $\text{rank } M = \text{tr } M = n - p$. In particular $\text{rank } C_n = \text{tr } C_n = n - 1$.

Proof. By Theorem 1.10(a),(b), $\text{tr } H = \text{tr}(X(X'X)^{-1}X') = \text{tr}((X'X)^{-1}X'X) = \text{tr } I_p = p$ and $\text{tr } M = \text{tr } I_n - \text{tr } H = n - p$. Both matrices are idempotent (Theorem 6.16(b)), so their ranks equal their traces (Theorem 6.19(c)). For $X = \mathbf{1}_n$, $M = C_n$ (Example 6.17). $\square$

In Example 6.18, $\text{tr } H = \frac{1}{4}(3 + 3 + 3 + 3) = 3$ and $\text{tr } M = \frac{1}{4}(1 + 1 + 1 + 1) = 1$; indeed every column of M is a multiple of $(1, -1, -1, 1)'$, so $\text{rank } M = 1$.

Least squares

In the **linear regression model**, the response vector $y = (y_1, \dots, y_n)'$ holds n observations, and x_{ij} is the value of the j th explanatory variable on observation i ($1 \leq j \leq k$):

$$y_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_k x_{ik} + \varepsilon_i \quad (1 \leq i \leq n), \quad \text{that is,} \quad y = X\beta + \varepsilon$$

The **design matrix** $X \in \mathbb{R}^{n \times (k+1)}$ has i th row $x'_i = (1, x_{i1}, \dots, x_{ik})$, so its first column is $\mathbf{1}_n$, the column of the **intercept** β_0 ; $\beta = (\beta_0, \dots, \beta_k)'$, and $\varepsilon = (\varepsilon_1, \dots, \varepsilon_n)'$ is the vector of errors. A **least-squares estimate** of β is a vector $\hat{\beta} \in \mathbb{R}^{k+1}$ that minimizes

$$S(\beta) = \|y - X\beta\|^2 = \sum_{i=1}^n (y_i - x'_i \beta)^2 \quad (\beta \in \mathbb{R}^{k+1}),$$

and the $k+1$ equations $X'X\beta = X'y$ are the **normal equations**. The results of this section and the next hold for every $X \in \mathbb{R}^{n \times (k+1)}$, with or without the column $\mathbf{1}_n$, unless an intercept is assumed.

Theorem 6.21. Let $X \in \mathbb{R}^{n \times (k+1)}$ and $y \in \mathbb{R}^n$.

- (a) The normal equations $X'X\beta = X'y$ (unknown $\beta \in \mathbb{R}^{k+1}$) are consistent.
- (b) $X'X$ is invertible if and only if X has full column rank, $\text{rank } X = k+1$. Then the normal equations have exactly one solution, $\hat{\beta} = (X'X)^{-1}X'y$.
- (c) If $\text{rank } X < k+1$ and $\hat{\beta}$ is one solution, the solutions are the vectors $\hat{\beta} + z$ with $z \in \mathcal{N}(X)$; there are infinitely many, and all of them give the same vector $X\hat{\beta}$.

Proof. (a) $X'y \in \mathcal{C}(X') = \mathcal{C}(X'X)$ (Theorem 4.18(c)), so the system is consistent (Corollary 4.14(a)).

(b) $X'X \in \mathbb{R}^{(k+1) \times (k+1)}$ is invertible if and only if $\text{rank}(X'X) = k+1$ (Theorem 4.13(d)), that is, $\text{rank } X = k+1$ (Theorem 4.18(b)). Then multiplying $X'X\beta = X'y$ by $(X'X)^{-1}$ gives $\beta = (X'X)^{-1}X'y$, and this vector is a solution.

(c) By Theorem 2.20(a), the solutions are $\hat{\beta} + z$ with $z \in \mathcal{N}(X'X) = \mathcal{N}(X)$ (Theorem 4.18(a)), and $X(\hat{\beta} + z) = X\hat{\beta} + Xz = X\hat{\beta}$. As $\text{rank } X < k+1$, some $z \neq 0$ lies in $\mathcal{N}(X)$ (Theorem 4.13(b)), and the solutions $\hat{\beta} + tz$ ($t \in \mathbb{R}$) are pairwise distinct, since $t_1 z = t_2 z$ forces $t_1 = t_2$. $\square$

Theorem 6.22. Let $X \in \mathbb{R}^{n \times (k+1)}$ and $y \in \mathbb{R}^n$.

- (a) $\hat{\beta}$ is a least-squares estimate if and only if $X'X\hat{\beta} = X'y$. For every solution $\hat{\beta}$ of the normal equations,

$$S(\beta) = S(\hat{\beta}) + \|X(\hat{\beta} - \beta)\|^2 \quad (\beta \in \mathbb{R}^{k+1})$$

- (b) All least-squares estimates give the same vector of **fitted values** $\hat{y} = X\hat{\beta}$, the orthogonal projection of y onto $\mathcal{C}(X)$; the vector of **residuals** $e = y - \hat{y}$ lies in $\mathcal{C}(X)^\perp$.
- (c) If $\text{rank } X = k+1$, the least-squares estimate is unique, $\hat{\beta} = (X'X)^{-1}X'y$, and $\hat{y} = Hy$, $e = My$.

Proof. (a)

- Let $X'X\hat{\beta} = X'y$. Then $X'(y - X\hat{\beta}) = 0$, so $y - X\hat{\beta} \perp \mathcal{C}(X)$ (Theorem 6.5(a)); as $X\hat{\beta} \in \mathcal{C}(X)$, it is the orthogonal projection of y onto $\mathcal{C}(X)$ (Definition 6.14).
- For $\beta \in \mathbb{R}^{k+1}$, $X\beta \in \mathcal{C}(X)$, and Theorem 6.15 gives $\|y - X\beta\|^2 = \|y - X\hat{\beta}\|^2 + \|X\hat{\beta} - X\beta\|^2$, the identity. Hence $S(\beta) \geq S(\hat{\beta})$: $\hat{\beta}$ is a least-squares estimate.
- Solutions exist (Theorem 6.21(a)). If β^* is a least-squares estimate and $\hat{\beta}$ a solution, then $S(\beta^*) \leq S(\hat{\beta})$, so the identity gives $\|X(\hat{\beta} - \beta^*)\|^2 = 0$, that is, $X\beta^* = X\hat{\beta}$; then $X'X\beta^* = X'X\hat{\beta} = X'y$.

- (b) By the last step, every least-squares estimate β^* gives $X\beta^* = X\hat{\beta}$, the projection found in the first step; and $e = y - X\hat{\beta} \perp \mathcal{C}(X)$.
- (c) By Theorem 6.21(b) and (a), $(X'X)^{-1}X'y$ is the only least-squares estimate, and then $\hat{y} = X(X'X)^{-1}X'y = Hy$ and $e = y - Hy = My$ (Theorem 6.16). $\square$

So least squares needs no calculus: $\hat{y}$ is the projection of y onto $\mathcal{C}(X)$. The formula $\hat{\beta} = (X'X)^{-1}X'y$ needs a design matrix of full column rank, $\text{rank } X = k + 1$; since $\text{rank } X \leq n$ (Theorem 4.13(a)), this requires at least $k + 1$ observations. Then Corollary 6.20 with $p = k + 1$ gives $\text{tr } H = k + 1 = \text{rank } X$ and $\text{tr } M = n - k - 1$.

Proposition 6.23. Let $\hat{\beta}$ be a least-squares estimate, $\hat{y} = X\hat{\beta}$ and $e = y - \hat{y}$.

- (a) $X'e = 0$: the residuals are orthogonal to every column of X .
- (b) If $\mathbf{1}_n \in \mathcal{C}(X)$ (for instance, if the model has an intercept), then $\mathbf{1}'e = e_1 + \cdots + e_n = 0$; hence $\hat{y}$ and y have the same mean $\bar{y}$.
- (c) $\hat{y}'e = 0$.

Proof. (a) This is $X'(y - X\hat{\beta}) = 0$, the normal equations (Theorem 6.22(a)). (b) If $\mathbf{1} = Xc$, then $\mathbf{1}'e = c'X'e = 0$ by (a), and $\mathbf{1}'\hat{y} = \mathbf{1}'y - \mathbf{1}'e = \mathbf{1}'y$. (c) $\hat{y}'e = \hat{\beta}'X'e = 0$. $\square$

Example 6.24. For X and y of Example 6.18, X has full column rank, as $X'X$ is invertible (Example 2.17, Theorem 6.21(b)); so the least-squares estimate is

$$\hat{\beta} = (X'X)^{-1}X'y = \frac{1}{4} \begin{pmatrix} -2 & 0 & 2 & 4 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} 9 \\ 3 \\ 5 \\ 3 \end{pmatrix} = \frac{1}{4} \begin{pmatrix} -18 + 0 + 10 + 12 \\ 9 - 3 + 5 - 3 \\ 9 + 3 - 5 - 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix},$$

and indeed, with $X'X$ of Example 1.14, $X'X\hat{\beta} = (4 + 8 + 8, 4 + 16 + 8, 8 + 16 + 20)' = (20, 28, 44)'$, which is $X'y = (9 + 3 + 5 + 3, 18 + 10, 27 + 9 + 5 + 3)'$. The fitted values and residuals are $\hat{y} = Hy = (8, 4, 6, 2)'$ and $e = My = (1, -1, -1, 1)'$, and the smallest value of S is $S(\hat{\beta}) = e'e = 4$. As it is positive, no β gives $X\beta = y$: the system $X\beta = y$ has no solution. As Proposition 6.23 says, $X'e = 0$ (Example 6.18), $\mathbf{1}'e = 1 - 1 - 1 + 1 = 0$ and $\hat{y}'e = 8 - 4 - 6 + 2 = 0$. By the identity of Theorem 6.22(a), for instance $\beta = (1, 2, 2)'$ gives $S(\beta) = 4 + \|X(0, 0, -1)'\|^2 = 4 + (9 + 9 + 1 + 1) = 24$.

Corollary 6.25. Let $k = 1$ (**simple regression**): $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, where $x_i = x_{i1}$ is a number, and $X = [\mathbf{1}_n \ x]$ with $x = (x_1, \dots, x_n)'$. Put

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2, \quad S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}$$

(expand, using $\sum_i x_i = n\bar{x}$ and $\sum_i y_i = n\bar{y}$, as in Example 1.13).

- (a) X has full column rank if and only if x is not constant, that is, $S_{xx} > 0$.
- (b) If $S_{xx} > 0$, then $\hat{\beta}_1 = S_{xy}/S_{xx}$ and $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$: the fitted line $\hat{\beta}_0 + \hat{\beta}_1 x$ passes through the point $(\bar{x}, \bar{y})$.

Proof. (a) As $\mathbf{1} \neq 0$, the columns $\mathbf{1}, x$ are dependent if and only if $x = c\mathbf{1}$ for some c (Proposition 3.12(c)), that is, x is constant; and S_{xx} , a sum of squares, is 0 exactly then.

(b) By (a), X has full column rank, so $\hat{\beta} = (X'X)^{-1}X'y$ (Theorem 6.21(b)). Here $X'X = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_i x_i^2 \end{pmatrix}$, with determinant $n \sum_i x_i^2 - n^2 \bar{x}^2 = nS_{xx}$, and $X'y = (n\bar{y}, \sum_i x_i y_i)'$. By Theorem 5.18 for 2×2 matrices,

$$\hat{\beta} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum_i x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix} \begin{pmatrix} n\bar{y} \\ \sum_i x_i y_i \end{pmatrix}, \quad \text{so} \quad \hat{\beta}_1 = \frac{n \sum_i x_i y_i - n^2 \bar{x}\bar{y}}{nS_{xx}} = \frac{S_{xy}}{S_{xx}}$$

The first normal equation, $n\hat{\beta}_0 + n\bar{x}\hat{\beta}_1 = n\bar{y}$, gives $\hat{\beta}_0 + \hat{\beta}_1\bar{x} = \bar{y}$. $\square$

Rank in regression

Proposition 6.26. Let $X \in \mathbb{R}^{n \times p}$ with $p \geq 2$. Then X has full column rank if and only if no column of X is a linear combination of the other columns. If some column is such a combination (**exact collinearity**), then $Xz = 0$ for some $z \neq 0$, the matrix $X'X$ is singular, and β and $\beta + tz$ give the same $X\beta$ for every $t \in \mathbb{R}$.

Proof. By Theorem 4.13(b), X has full column rank if and only if its columns are independent, and $p \geq 2$ vectors are dependent if and only if one of them is a linear combination of the others (Proposition 3.12(c)). If they are dependent, $Xz = 0$ for some $z \neq 0$ (Theorem 4.13(b)), $X'X$ is singular (Theorem 6.21(b) with $p = k + 1$), and $X(\beta + tz) = X\beta + tXz = X\beta$. $\square$

For example, if a regression uses the scores x_{i1} , x_{i2} of two tests and also their total $x_{i3} = x_{i1} + x_{i2}$, then column 4 of X is the sum of columns 2 and 3, and $X(0, 1, 1, -1)' = 0$.

Proposition 6.27. Let $X \in \mathbb{R}^{n \times p}$, $T \in \mathbb{R}^{p \times q}$ and $X_1 = XT \in \mathbb{R}^{n \times q}$, with $\text{rank } X_1 = \text{rank } X$. Then $\mathcal{C}(X_1) = \mathcal{C}(X)$, the least-squares fitted values (and residuals) of the designs X_1 and X are the same, and if $\hat{\beta}_1$ is a least-squares estimate for X_1 , then $T\hat{\beta}_1$ is one for X .

Proof. $\mathcal{C}(X_1) \subseteq \mathcal{C}(X)$ (Theorem 4.15(a)), and both have dimension $\text{rank } X$, so they are equal (Theorem 4.5(a)). For either design, the fitted values are the projection $\hat{y}$ of y onto this subspace (Theorem 6.22(b)). Finally $X(T\hat{\beta}_1) = X_1\hat{\beta}_1 = \hat{y}$, so $\|y - X(T\hat{\beta}_1)\|^2 = \|y - \hat{y}\|^2 \leq \|y - X\beta\|^2$ for every β (Theorem 6.15). $\square$

For instance, adding to the design X of Example 6.24 the total $X_{\cdot 1} + X_{\cdot 2}$ of its two variables gives $\tilde{X} = [X \ X_{\cdot 1} + X_{\cdot 2}] \in \mathbb{R}^{4 \times 4}$, with $X = \tilde{X}T$ for T the first three columns of I_4 , and $\text{rank } \tilde{X} = \text{rank } X = 3$. So $\tilde{X}$ has the same fitted values $(8, 4, 6, 2)'$, and $T(1, 2, 1)' = (1, 2, 1, 0)'$ is a least-squares estimate for $\tilde{X}$.

Theorem 6.28. Let $n_1, \dots, n_g \geq 1$ and $n = n_1 + \dots + n_g$. Arrange n observations in g groups (first the n_1 observations of group 1, then the n_2 of group 2, and so on), and let the **indicator** $d_l \in \mathbb{R}^n$ of group l have entry 1 in the rows of group l and 0 elsewhere. The model $y_i = \mu + \alpha_l + \varepsilon_i$ for an observation i of group l reads $y = X\beta + \varepsilon$ with $\beta = (\mu, \alpha_1, \dots, \alpha_g)'$ and

$$X = [\mathbf{1}_n \ d_1 \ \dots \ d_g] \in \mathbb{R}^{n \times (g+1)}$$

Let $X_1 = [d_1 \ \dots \ d_g]$ (intercept dropped) and $X_2 = [\mathbf{1}_n \ d_2 \ \dots \ d_g]$ (indicator d_1 dropped), both in $\mathbb{R}^{n \times g}$.

- (a) X_1 has full column rank, and $X = X_1S$ with $S = [\mathbf{1}_g \ I_g] \in \mathbb{R}^{g \times (g+1)}$ of full row rank.
- (b) $\text{rank } X = g$ and $\mathcal{N}(X) = \text{span}\{(-1, 1, \dots, 1)'\}$: X does not have full column rank.
- (c) $X_1 = XT_1$ and $X_2 = XT_2$, where $T_1, T_2 \in \mathbb{R}^{(g+1) \times g}$ are I_{g+1} with its first, respectively second, column deleted. Both X_1 and X_2 have full column rank g , and $\mathcal{C}(X_1) = \mathcal{C}(X_2) = \mathcal{C}(X)$.

Proof. (a)

- No observation lies in two groups, so $d'_l d_m = 0$ for $l \neq m$, and $d'_l d_l = n_l$. If $c_1 d_1 + \dots + c_g d_g = 0$, multiplying by d'_m gives $c_m n_m = 0$, so $c_m = 0$: X_1 has full column rank (Theorem 4.13(b)).
- Every observation lies in exactly one group, so $d_1 + \dots + d_g = \mathbf{1}_n$, that is, $X_1 \mathbf{1}_g = \mathbf{1}_n$. By the block product (Theorem 5.2), $X_1 S = [X_1 \mathbf{1}_g \ X_1 I_g] = [\mathbf{1}_n \ d_1 \ \dots \ d_g] = X$.
- The columns of S include those of I_g , so $\mathcal{C}(S) = \mathbb{R}^g$: S has full row rank (Theorem 4.13(c)).

(b) By (a) and Theorem 4.15(b), $\text{rank } X = \text{rank } S = g$ and $\mathcal{N}(X) = \mathcal{N}(S)$. Now $S(c_0, c_1, \dots, c_g)' = c_0 \mathbf{1}_g + (c_1, \dots, c_g)'$ is 0 if and only if $c_l = -c_0$ for every l , that is, $(c_0, c_1, \dots, c_g)' = -c_0(-1, 1, \dots, 1)'$.
(c)

- The columns of XT_1 are the columns of X other than the first (Theorem 1.4(a),(b)), so $XT_1 = X_1$; likewise $XT_2 = X_2$.
- X_1 has full column rank by (a). For X_2 : if $c_1 \mathbf{1}_n + c_2 d_2 + \dots + c_g d_g = 0$, then $\mathbf{1}_n = d_1 + \dots + d_g$ turns this into $c_1 d_1 + (c_1 + c_2) d_2 + \dots + (c_1 + c_g) d_g = 0$, and (a) gives $c_1 = 0$ and then $c_2 = \dots = c_g = 0$.
- So $\text{rank } X_1 = \text{rank } X_2 = g = \text{rank } X$, and Proposition 6.27 gives $\mathcal{C}(X_1) = \mathcal{C}(X_2) = \mathcal{C}(X)$. $\square$

Using the intercept together with all g indicators (the **dummy variable trap**) thus makes $X'X$ singular (Theorem 6.21(b)); dropping the intercept or one indicator gives a design of full column rank with the same column space, hence the same fitted values (Proposition 6.27).

Example 6.29. Six plants of three types have weights 5, 7, 6 (type 1), 3, 5 (type 2) and 2 (type 3). With $y = (5, 7, 6, 3, 5, 2)'$, the model of Theorem 6.28 ($g = 3$, $n_1 = 3$, $n_2 = 2$, $n_3 = 1$) has

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \quad X'X = \begin{pmatrix} 6 & 3 & 2 & 1 \\ 3 & 3 & 0 & 0 \\ 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \quad X'y = \begin{pmatrix} 28 \\ 18 \\ 8 \\ 2 \end{pmatrix}$$

The (j, l) entry of $X'X$ is $\text{col}_j(X)' \text{col}_l(X)$ (Theorems 1.4(d) and 1.8), and $X'y$ holds the total and the three group totals.

- $\text{rank } X = 3$ and $\mathcal{N}(X) = \text{span}\{(-1, 1, 1, 1)'\}$ (Theorem 6.28), so $X'X$ is singular (Theorem 6.21(b)).
- $\hat{\beta} = (0, 6, 4, 2)'$ solves the normal equations: $X'X\hat{\beta} = (18 + 8 + 2, 18, 8, 2)' = X'y$. By Theorem 6.21(c), all solutions are

$$\begin{pmatrix} 0 \\ 6 \\ 4 \\ 2 \end{pmatrix} + t \begin{pmatrix} -1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -t \\ 6+t \\ 4+t \\ 2+t \end{pmatrix} \quad (t \in \mathbb{R}),$$

and all give $X\hat{\beta} = (6, 6, 6, 4, 4, 2)'$: the fitted value $\hat{\mu} + \hat{\alpha}_l$ is the mean of group l ($18/3 = 6$, $8/2 = 4$, $2/1 = 2$).

- Intercept dropped: $X'_1 X_1 = \text{diag}(3, 2, 1)$ and $X'_1 y = (18, 8, 2)'$, so $\hat{\beta} = (X'_1 X_1)^{-1} X'_1 y = (6, 4, 2)'$, the three group means.
- Indicator d_1 dropped:

$$X'_2 X_2 = \begin{pmatrix} 6 & 2 & 1 \\ 2 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad X'_2 y = \begin{pmatrix} 28 \\ 8 \\ 2 \end{pmatrix}, \quad X'_2 X_2 \begin{pmatrix} 6 \\ -2 \\ -4 \end{pmatrix} = \begin{pmatrix} 36 - 4 - 4 \\ 12 - 4 \\ 6 - 4 \end{pmatrix} = X'_2 y$$

so $\hat{\beta} = (6, -2, -4)'$, the unique solution (Theorem 6.21(b)): the mean 6 of group 1 and the differences $4 - 6$, $2 - 6$ of the other group means from it.

- In all three designs $\hat{y} = (6, 6, 6, 4, 4, 2)'$, as Proposition 6.27 says; moreover $T_1(6, 4, 2)' = (0, 6, 4, 2)'$ and $T_2(6, -2, -4)' = (6, 0, -2, -4)'$ are the solutions above with $t = 0$ and $t = -6$.

Proposition 6.30. Let $x = (x_1, \dots, x_n)' \in \mathbb{R}^n$, and let $X \in \mathbb{R}^{n \times 3}$ have i th row $(1, x_i, x_i^2)$: the design matrix of the quadratic regression $y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \varepsilon_i$ ($k = 2$, with $x_{i1} = x_i$ and $x_{i2} = x_i^2$). Then X has full column rank if and only if $x_1, \dots, x_n$ take at least three different values.

Proof. • If every x_i is one of two numbers a, b (possibly $a = b$), then $(x_i - a)(x_i - b) = ab - (a + b)x_i + x_i^2 = 0$ for every i ; so $X(ab, -(a + b), 1)' = 0$, and X does not have full column rank (Theorem 4.13(b)).

- Let $x_i = a, x_j = b, x_l = c$ be three different values. If $Xv = 0$, rows i, j, l give $Mv = 0$ for

$$M = \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix}$$

The operations $R_2 \leftarrow R_2 - R_1, R_3 \leftarrow R_3 - R_1$ give the rows $(0, b - a, b^2 - a^2)$ and $(0, c - a, c^2 - a^2)$; dividing them by $b - a \neq 0$ and $c - a \neq 0$ gives $(0, 1, a + b)$ and $(0, 1, a + c)$; then $R_3 \leftarrow R_3 - R_2$ gives $(0, 0, c - b)$, and dividing by $c - b \neq 0$ gives an echelon form with a leading 1 in every column. So $Mv = 0$ only for $v = 0$ (Theorem 4.1 and Remark 4.2), and hence $Xv = 0$ only for $v = 0$. $\square$

Example 6.31. (a) For $x = (0, 1, 2, 3)'$, four different values, X has full column rank by Proposition 6.30. Directly:

$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - R_1, R_3 \leftarrow R_3 - R_1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 2 & 4 \\ 0 & 3 & 9 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 - 2R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 0 & 6 \end{pmatrix} \xrightarrow[R_3 \leftarrow \frac{1}{2}R_3]{R_4 \leftarrow R_4 - 3R_3} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

an echelon form with three nonzero rows, so $\text{rank } X = 3$.

(b) If a dose is given at only two levels, 1 and 3, say $x = (1, 3, 3, 1)'$, then $x_i^2 = 4x_i - 3$ for every i ($1 = 4 - 3, 9 = 12 - 3$). So column 3 of X is 4 times column 2 minus 3 times column 1, and $X(3, -4, 1)' = 0$. Columns 1 and 2, $(1, 1, 1, 1)'$ and $(1, 3, 3, 1)'$, are not multiples of each other, so $\text{rank } X = 2$. Thus β and $\beta + t(3, -4, 1)'$ give the same $X\beta$: two dose levels cannot determine the curvature β_2 .

Sums of squares and R^2

Definition 6.32. For a least-squares fit, with fitted values $\hat{y}$ and residuals $e = y - \hat{y}$, the **total, regression** and **error sums of squares** are

$$\text{SST} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad \text{SSR} = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2, \quad \text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = e'e$$

If $\text{SST} > 0$, the **coefficient of determination** is $R^2 = \text{SSR}/\text{SST}$.

Theorem 6.33. If $\mathbf{1}_n \in \mathcal{C}(X)$ (for instance, if the model has an intercept), then $\text{SST} = \text{SSR} + \text{SSE}$, where $\text{SST} = \|C_n y\|^2 = y' C_n y$ and $\text{SSR} = \|C_n \hat{y}\|^2$. Hence, if $\text{SST} > 0$, then $R^2 = 1 - \text{SSE}/\text{SST}$ and $0 \leq R^2 \leq 1$.

Proof. • For $v \in \mathbb{R}^n$, $C_n v = v - \bar{v}\mathbf{1}$, so $\|C_n v\|^2 = \sum_i (v_i - \bar{v})^2$, and $\|C_n v\|^2 = v' C_n' C_n v = v' C_n v$ (Proposition 1.12). With $v = y$ this is SST; with $v = \hat{y}$, whose mean is $\bar{y}$ (Proposition 6.23(b)), it is SSR.

- $C_n e = e - \bar{e}\mathbf{1} = e$, as $\bar{e} = 0$ (Proposition 6.23(b)). So $C_n y = C_n \hat{y} + C_n e = C_n \hat{y} + e$.
- $(C_n \hat{y})' e = \hat{y}' C_n e = \hat{y}' e = 0$ (Proposition 6.23(c)). By Pythagoras, $\|C_n y\|^2 = \|C_n \hat{y}\|^2 + \|e\|^2$, that is, $\text{SST} = \text{SSR} + \text{SSE}$.

- Dividing by $SST > 0$ gives $R^2 = 1 - SSE/SST$; and $0 \leq SSR \leq SST$, as $SSE \geq 0$. $\square$

In terms of projections: $y = \bar{y}\mathbf{1} + C_n y$ along $\text{span}\{\mathbf{1}\} \oplus \text{span}\{\mathbf{1}\}^\perp$ (Example 6.17), and $C_n y = C_n \hat{y} + e$ splits the centered response into $C_n \hat{y} = \hat{y} - \bar{y}\mathbf{1} \in \mathcal{C}(X)$ and $e \in \mathcal{C}(X)^\perp$; $SST = SSR + SSE$ is Pythagoras for this split. For the design $X = \mathbf{1}_n$ alone, $\hat{y} = \bar{y}\mathbf{1}$, so $SSR = 0$ and $SSE = SST$; R^2 is the fraction of SST that the other columns of X explain.

Example 6.34. In Example 6.24, $\bar{y} = \frac{1}{4}(9 + 3 + 5 + 3) = 5$, so

$$C_4 y = \begin{pmatrix} 4 \\ -2 \\ 0 \\ -2 \end{pmatrix}, \quad C_4 \hat{y} = \begin{pmatrix} 3 \\ -1 \\ 1 \\ -3 \end{pmatrix}, \quad e = \begin{pmatrix} 1 \\ -1 \\ -1 \\ 1 \end{pmatrix}, \quad C_4 y = C_4 \hat{y} + e$$

Hence $SST = 16 + 4 + 0 + 4 = 24$, $SSR = 9 + 1 + 1 + 9 = 20$ and $SSE = 1 + 1 + 1 + 1 = 4$; indeed $24 = 20 + 4$, and $R^2 = \frac{20}{24} = \frac{5}{6}$.

Exercises and solutions

Exercise 6.1. (a) Find all vectors of $\mathbb{R}^3$ that are orthogonal to $x = (2, 2, -1)'$.

(b) What is the dimension of the subspace of $\mathbb{R}^6$ consisting of the vectors orthogonal to both $(1, 1, -2, 3, 4, 5)'$ and $(0, 0, 1, 1, 0, 7)'$?

Solution. (a) $y = (y_1, y_2, y_3)'$ is orthogonal to x if and only if $x'y = 2y_1 + 2y_2 - y_3 = 0$, that is, $y_3 = 2y_1 + 2y_2$. So these vectors are

$$y = \begin{pmatrix} y_1 \\ y_2 \\ 2y_1 + 2y_2 \end{pmatrix} = y_1 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + y_2 \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \quad (y_1, y_2 \in \mathbb{R}) :$$

$\text{span}\{x\}^\perp = \mathcal{N}(x') = \text{span}\{(1, 0, 2)', (0, 1, 2)'\}$, of dimension $2 = 3 - 1$ (Theorem 6.5).

(b) Let a_1, a_2 be the two vectors and $A = [a_1 \ a_2] \in \mathbb{R}^{6 \times 2}$. The subspace is $\text{span}\{a_1, a_2\}^\perp = \mathcal{N}(A')$ (Theorem 6.5(a)). The vectors are independent: if $c_1 a_1 + c_2 a_2 = 0$, the first entry gives $c_1 = 0$, and then $c_2 a_2 = 0$ with $a_2 \neq 0$ gives $c_2 = 0$. So $\dim \text{span}\{a_1, a_2\} = 2$, and the answer is $6 - 2 = 4$ (Theorem 6.5(c)).

Exercise 6.2. Let X be the design matrix of Example 6.18.

(a) Find the QR decomposition of X by the Gram–Schmidt process.

(b) Show that $Q'Q = I_3$ but $QQ' \neq I_4$.

Solution. (a) The columns are $a_1 = \mathbf{1}_4$, $a_2 = (2, 0, 2, 0)'$ and $a_3 = (3, 3, 1, 1)'$.

- $b_1 = a_1$, $\|b_1\|^2 = 4$.
- $a'_2 b_1 = 4$, so $t_{12} = 1$ and $b_2 = a_2 - b_1 = (1, -1, 1, -1)'$, with $\|b_2\|^2 = 4$.
- $a'_3 b_1 = 8$ and $a'_3 b_2 = 3 - 3 + 1 - 1 = 0$, so $t_{13} = 2$, $t_{23} = 0$ and $b_3 = a_3 - 2b_1 = (1, 1, -1, -1)'$, with $\|b_3\|^2 = 4$.
- All three lengths are 2: $q_j = \frac{1}{2}b_j$, $r_{jj} = 2$, $r_{12} = 2t_{12} = 2$, $r_{13} = 2t_{13} = 4$ and $r_{23} = 0$:

$$Q = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 2 & 2 & 4 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}$$

Check: the third column of QR is $Q(4, 0, 2)' = \frac{1}{2}(4 + 2, 4 + 2, 4 - 2, 4 - 2)' = (3, 3, 1, 1)' = a_3$.

(b) The columns of Q are orthonormal, so $Q'Q = I_3$. But

$$QQ' = \frac{1}{4} \begin{pmatrix} 3 & 1 & 1 & -1 \\ 1 & 3 & -1 & 1 \\ 1 & -1 & 3 & 1 \\ -1 & 1 & 1 & 3 \end{pmatrix} \neq I_4;$$

for instance, its $(1, 1)$ entry is $\frac{1}{4}(1 + 1 + 1) = \frac{3}{4}$. The rows of Q are not orthonormal.

Exercise 6.3. Let $X = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$, $W = \mathcal{C}(X)$ and $y = (4, 2, 5)'$.

- (a) Find a basis of $W^\perp$.
- (b) Compute the hat matrix H and the residual maker M . Check that $H^2 = H$, $HX = X$ and $MX = 0$, and find $\text{tr } H$ and $\text{tr } M$.
- (c) Write $y = Hy + My$, the decomposition along $\mathbb{R}^3 = W \oplus W^\perp$, and check that the two parts are orthogonal.

Solution. (a) $W^\perp = \mathcal{N}(X')$ (Theorem 6.5(a)). Row-reduce X' :

$$\begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \xrightarrow{R_2 \leftarrow -R_2} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \end{pmatrix} \xrightarrow{R_1 \leftarrow R_1 - R_2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix}$$

With $x_3 = t$ free, $x_1 = -t$ and $x_2 = t$; so $(-1, 1, 1)'$ is a basis of $W^\perp$, of dimension $3 - 2 = 1$.

(b) $X'X = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, with determinant 3, so $(X'X)^{-1} = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ (Theorem 5.18),

$$(X'X)^{-1}X' = \frac{1}{3} \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix},$$

$$H = X(X'X)^{-1}X' = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix} \frac{1}{3} \begin{pmatrix} 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix},$$

and $M = I_3 - H = \frac{1}{3} \begin{pmatrix} 1 & -1 & -1 \\ -1 & 1 & 1 \\ -1 & 1 & 1 \end{pmatrix}$. Then

$$9H^2 = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}^2 = \begin{pmatrix} 6 & 3 & 3 \\ 3 & 6 & -3 \\ 3 & -3 & 6 \end{pmatrix} = 9H, \quad HX = \frac{1}{3} \begin{pmatrix} 3 & 3 \\ 3 & 0 \\ 0 & 3 \end{pmatrix} = X,$$

so $H^2 = H$, $HX = X$ and $MX = X - HX = 0$. Finally $\text{tr } H = \frac{1}{3}(2 + 2 + 2) = 2$ and $\text{tr } M = \frac{1}{3}(1 + 1 + 1) = 1$, the ranks of H and M (Corollary 6.20).

(c) $Hy = \frac{1}{3}(8+2+5, 4+4-5, 4-2+10)' = (5, 1, 4)'$, which is $X(1, 4)'$, and $My = y - Hy = (-1, 1, 1)'$, a vector of $W^\perp$ by (a). They are orthogonal: $(Hy)'(My) = -5 + 1 + 4 = 0$.

Exercise 6.4. Let $A = \begin{pmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}$ and $v = (2, 1, 0)'$.

- (a) Show that A is idempotent but not symmetric.
- (b) Find $\text{rank } A$, $\text{tr } A$, $\text{rank}(I_3 - A)$ and $\text{tr}(I_3 - A)$.
- (c) Decompose v along $\mathbb{R}^3 = \mathcal{N}(A) \oplus \mathcal{C}(A)$. Are the two parts orthogonal?

Solution. (a) Row by row, $\text{row}_1(A)A = (4 - 1 - 1, -2 + 0 + 1, -2 + 1 + 0) = (2, -1, -1)$, $\text{row}_2(A)A = (2 + 0 - 1, -1 + 0 + 1, -1 + 0 + 0) = (1, 0, -1)$ and $\text{row}_3(A)A = (2 - 1 + 0, -1 + 0 + 0, -1 + 1 + 0) = (1, -1, 0)$; so $A^2 = A$. It is not symmetric: $a_{12} = -1 \neq 1 = a_{21}$.

(b) Row-reduce A :

$$A \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & -1 \\ 2 & -1 & -1 \\ 1 & -1 & 0 \end{pmatrix} \xrightarrow[\substack{R_3 \leftarrow R_3 - R_1 \\ R_2 \leftarrow R_2 - 2R_1}]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 0 & -1 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{pmatrix} \xrightarrow[\substack{R_2 \leftarrow -R_2 \\ R_3 \leftarrow R_3 - R_2}]{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix},$$

an echelon form with two nonzero rows, so $\text{rank } A = 2 = 2 + 0 + 0 = \text{tr } A$. Next $I_3 - A = \begin{pmatrix} -1 & 1 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & 1 \end{pmatrix}$ has three equal nonzero rows, so $\text{rank}(I_3 - A) = 1 = -1 + 1 + 1 = \text{tr}(I_3 - A)$, as Theorem 6.19(b),(c) say.

(c) $Av = (4 - 1 - 0, 2 + 0 - 0, 2 - 1 + 0)' = (3, 2, 1)' \in \mathcal{C}(A)$ and $v - Av = (-1, -1, -1)'$, which lies in $\mathcal{N}(A)$: $A(-1, -1, -1)' = (-2 + 1 + 1, -1 + 0 + 1, -1 + 1 + 0)' = 0$. So $v = (-1, -1, -1)' + (3, 2, 1)'$ (Theorem 6.19(a)). The parts are not orthogonal: $(v - Av)'(Av) = -3 - 2 - 1 = -6$. (By Theorem 6.19(d), they would be if A were symmetric.)

Exercise 6.5. Show that $I_n - 2A$ is an orthogonal matrix for every symmetric idempotent $A \in \mathbb{R}^{n \times n}$. In particular, if $h \in \mathbb{R}^n$ and $h'h = 1$, then $I_n - 2hh'$ is orthogonal.

Solution. Using $A' = A$ and $A^2 = A$, $(I - 2A)'(I - 2A) = (I - 2A)^2 = I - 4A + 4A^2 = I$; so $I - 2A$ is orthogonal (Definition 6.9). The matrix hh' is symmetric and idempotent: $(hh')' = hh'$ and $(hh')(hh') = h(h'h)h' = hh'$.

Exercise 6.6. Five observations of a variable x and a response y are

$$\begin{array}{c|ccccc} x_i & 1 & 2 & 3 & 4 & 5 \\ \hline y_i & 3 & 2 & 4 & 4 & 7 \end{array}$$

Fit the simple regression $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$.

- (a) Compute $X'X$, $X'y$, $(X'X)^{-1}$ and $\hat{\beta}$.
- (b) Compute $\hat{y}$ and e , and verify that $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$.
- (c) Compute SSE.
- (d) Compute SST and SSR, verify $\text{SST} = \text{SSR} + \text{SSE}$, and compute R^2 .

Solution. (a) $X = [\mathbf{1}_5 \ x]$, $\sum_i x_i = 15$, $\sum_i x_i^2 = 55$, $\sum_i y_i = 20$ and $\sum_i x_i y_i = 3 + 4 + 12 + 16 + 35 = 70$:

$$X'X = \begin{pmatrix} 5 & 15 \\ 15 & 55 \end{pmatrix}, \quad X'y = \begin{pmatrix} 20 \\ 70 \end{pmatrix}, \quad (X'X)^{-1} = \frac{1}{50} \begin{pmatrix} 55 & -15 \\ -15 & 5 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 11 & -3 \\ -3 & 1 \end{pmatrix}$$

($\det X'X = 275 - 225 = 50$; Theorem 5.18), and $\hat{\beta} = \frac{1}{10}(220 - 210, -60 + 70)' = (1, 1)'$. Check with Corollary 6.25: $\bar{x} = 3$, $\bar{y} = 4$, $S_{xx} = 55 - 45 = 10$ and $S_{xy} = 70 - 60 = 10$, so $\hat{\beta}_1 = 1$ and $\hat{\beta}_0 = 4 - 3 = 1$.

(b) $\hat{y}_i = 1 + x_i$: $\hat{y} = (2, 3, 4, 5, 6)'$ and $e = y - \hat{y} = (1, -1, 0, -1, 1)'$. Then $\sum_i e_i = 1 - 1 + 0 - 1 + 1 = 0$ and $\sum_i x_i e_i = 1 - 2 + 0 - 4 + 5 = 0$ (Proposition 6.23).

(c) $\text{SSE} = 1 + 1 + 0 + 1 + 1 = 4$.

(d) $\text{SST} = \sum_i (y_i - 4)^2 = 1 + 4 + 0 + 0 + 9 = 14$ and $\text{SSR} = \sum_i (\hat{y}_i - 4)^2 = 4 + 1 + 0 + 1 + 4 = 10$; indeed $10 + 4 = 14$ (Theorem 6.33), and $R^2 = \frac{10}{14} = \frac{5}{7}$.

Exercise 6.7. Add to the design matrix X of Example 6.18 the sum of its two variables as a fourth column: $\tilde{X} = [\mathbf{1}_4 \ X_1 \ X_2 \ X_1 + X_2] \in \mathbb{R}^{4 \times 4}$, and let $y = (9, 3, 5, 3)'$.

(a) Find $\text{rank } \tilde{X}$ and a basis of $\mathcal{N}(\tilde{X})$.

(b) Solve the normal equations $\tilde{X}'\tilde{X}\beta = \tilde{X}'y$ by row reduction.

(c) Show that all least-squares estimates give the fitted values $\hat{y} = (8, 4, 6, 2)'$ of Example 6.24, and find SSE.

Solution. (a) The columns are $\mathbf{1}_4$, $(2, 0, 2, 0)'$, $(3, 3, 1, 1)'$ and $(5, 3, 3, 1)'$. The fourth is the sum of the second and the third, so $\tilde{X}(0, 1, 1, -1)' = 0$; the first three are the columns of X , which are independent (Example 6.24). So $\text{rank } \tilde{X} = 3$, $\dim \mathcal{N}(\tilde{X}) = 4 - 3 = 1$ (Theorem 4.12), and $(0, 1, 1, -1)'$ is a basis of $\mathcal{N}(\tilde{X})$.

(b) The entries of $\tilde{X}'\tilde{X}$ and $\tilde{X}'y$ are inner products of columns (Theorems 1.4(d) and 1.8):

$$\begin{aligned} & \left(\begin{array}{cccc|c} 4 & 4 & 8 & 12 & 20 \\ 4 & 8 & 8 & 16 & 28 \\ 8 & 8 & 20 & 28 & 44 \\ 12 & 16 & 28 & 44 & 72 \end{array} \right) \xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - R_1, \ R_3 \leftarrow R_3 - 2R_1} \left(\begin{array}{cccc|c} 4 & 4 & 8 & 12 & 20 \\ 0 & 4 & 0 & 4 & 8 \\ 0 & 0 & 4 & 4 & 4 \\ 0 & 4 & 4 & 8 & 12 \end{array} \right) \\ & \xrightarrow{R_4 \leftarrow R_4 - R_2} \left(\begin{array}{cccc|c} 4 & 4 & 8 & 12 & 20 \\ 0 & 4 & 0 & 4 & 8 \\ 0 & 0 & 4 & 4 & 4 \\ 0 & 0 & 4 & 4 & 4 \end{array} \right) \xrightarrow[R_1 \leftarrow \frac{1}{4}R_1, \ R_2 \leftarrow \frac{1}{4}R_2, \ R_3 \leftarrow \frac{1}{4}R_3]{R_4 \leftarrow R_4 - R_3} \left(\begin{array}{cccc|c} 1 & 1 & 2 & 3 & 5 \\ 0 & 1 & 0 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \end{aligned}$$

The last matrix is an echelon form with pivots in columns 1, 2, 3 and none in the last column, so the system is consistent and $\beta_3 = t$ is free (Theorem 2.19). Back substitution: $\beta_2 = 1 - t$, $\beta_1 = 2 - t$ and $\beta_0 = 5 - \beta_1 - 2\beta_2 - 3\beta_3 = 5 - (2 - t) - 2(1 - t) - 3t = 1$. So the solutions are

$$\beta = \begin{pmatrix} 1 \\ 2 - t \\ 1 - t \\ t \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 0 \\ -1 \\ -1 \\ 1 \end{pmatrix} \quad (t \in \mathbb{R}),$$

a particular solution plus $\mathcal{N}(\tilde{X}'\tilde{X}) = \mathcal{N}(\tilde{X})$, as Theorem 6.21(c) says.

(c) These solutions are the least-squares estimates (Theorem 6.22(a)), and each gives $\tilde{X}\beta = X(1, 2, 1)' + t\tilde{X}(0, -1, -1, 1)' = (8, 4, 6, 2)'$. So $\hat{y}$ and $e = (1, -1, -1, 1)'$ are those of Example 6.24, and $\text{SSE} = 4$.

Exercise 6.8. Let $X \in \mathbb{R}^{n \times (k+1)}$ have full column rank, with hat matrix H and residual maker M .

(a) In the model $y = X\beta + \varepsilon$, show that $e = M\varepsilon$ and $\text{SSE} = \varepsilon'M\varepsilon$.

(b) Let $\mathbf{1}_n \in \mathcal{C}(X)$. Show that $H\mathbf{1}_n = \mathbf{1}_n$, $H\bar{J}_n = \bar{J}_nH = \bar{J}_n$ and $MC_n = C_nM = M$.

(c) Under (b), show that $H - \bar{J}_n$ is symmetric and idempotent of rank k , that $(H - \bar{J}_n)M = 0$, and that $\text{SSR} = y'(H - \bar{J}_n)y$.

Solution. (a) $MX = 0$ (Theorem 6.16(c)), so $e = My = M(X\beta + \varepsilon) = MX\beta + M\varepsilon = M\varepsilon$ (Theorem 6.22(c)). As M is symmetric and idempotent, $\text{SSE} = e'e = \varepsilon'M'M\varepsilon = \varepsilon'M\varepsilon$.

(b)

- $\mathbf{1}_n = Xc$ for some c , so $H\mathbf{1}_n = HXc = Xc = \mathbf{1}_n$ (Theorem 6.16(c)).
- $H\bar{J}_n = \frac{1}{n}H\mathbf{1}\mathbf{1}' = \frac{1}{n}\mathbf{1}\mathbf{1}' = \bar{J}_n$, and, as H and $\bar{J}_n$ are symmetric, $\bar{J}_nH = (H\bar{J}_n)' = \bar{J}_n$.
- $M\bar{J}_n = \bar{J}_n - H\bar{J}_n = 0$, so $MC_n = M(I_n - \bar{J}_n) = M$, and $C_nM = (MC_n)' = M$.

(c)

- $H - \bar{J}_n$ is symmetric, and by (b) and $\bar{J}_n^2 = \bar{J}_n$ (Proposition 1.12), $(H - \bar{J}_n)^2 = H^2 - H\bar{J}_n - \bar{J}_nH + \bar{J}_n^2 = H - \bar{J}_n - \bar{J}_n + \bar{J}_n = H - \bar{J}_n$.
- Its rank is its trace (Theorem 6.19(c)): $\text{tr } H - \text{tr } \bar{J}_n = (k+1) - 1 = k$ (Corollary 6.20; the diagonal entries of $\bar{J}_n$ are $\frac{1}{n}$).
- $(H - \bar{J}_n)M = HM - \bar{J}_nM = 0 - (M\bar{J}_n)' = 0$, by Theorem 6.16(c) and (b).
- $\hat{y} - \bar{y}\mathbf{1} = Hy - \bar{J}_ny = (H - \bar{J}_n)y$ (Example 6.17), so $\text{SSR} = \|(H - \bar{J}_n)y\|^2 = y'(H - \bar{J}_n)'(H - \bar{J}_n)y = y'(H - \bar{J}_n)y$.

Exercise 6.9. Let $x, y \in \mathbb{R}^n$ with $S_{xx} > 0$ and $S_{yy} = \sum_i (y_i - \bar{y})^2 > 0$ (notation of Corollary 6.25), and put $r = S_{xy} / \sqrt{S_{xx}S_{yy}}$.

(a) Show that $|r| \leq 1$.

(b) In the simple regression of y on x , show that $\text{SSR} = \hat{\beta}_1^2 S_{xx}$ and $R^2 = r^2$.

Solution. (a) Let $u = C_n x = x - \bar{x}\mathbf{1}$ and $v = C_n y = y - \bar{y}\mathbf{1}$ (Example 1.13). Then $S_{xy} = u'v$, $S_{xx} = \|u\|^2$ and $S_{yy} = \|v\|^2$. By the Cauchy–Schwarz inequality (Theorem 6.3(a)), $|S_{xy}| \leq \sqrt{S_{xx}}\sqrt{S_{yy}}$, that is, $|r| \leq 1$.

(b) By Corollary 6.25(b), $\hat{y}_i - \bar{y} = \hat{\beta}_0 + \hat{\beta}_1 x_i - (\hat{\beta}_0 + \hat{\beta}_1 \bar{x}) = \hat{\beta}_1(x_i - \bar{x})$, so

$$\text{SSR} = \sum_i (\hat{y}_i - \bar{y})^2 = \hat{\beta}_1^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}^2} S_{xx} = \frac{S_{xy}^2}{S_{xx}}, \quad R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = r^2,$$

since $\text{SST} = S_{yy}$.

Exercise 6.10. Let $X \in \mathbb{R}^{n \times p}$ with $\mathbf{1}_n \in \mathcal{C}(X)$, let $x \in \mathbb{R}^n$, let $X_+ = [X \ x]$ be X with the column x added, and let $y \in \mathbb{R}^n$ with $\text{SST} > 0$. Let SSE and R^2 belong to the least-squares fit with X , and SSE_+ and R_+^2 to the fit with X_+ . Show that $\text{SSE}_+ \leq \text{SSE}$ and $R_+^2 \geq R^2$: adding a column cannot decrease R^2 .

Solution.

- Every column of X is a column of X_+ , so $\mathcal{C}(X) \subseteq \mathcal{C}(X_+)$, and both contain $\mathbf{1}_n$.
- The fitted values $\hat{y}_+$ of X_+ form the vector of $\mathcal{C}(X_+)$ closest to y (Theorems 6.22(b) and 6.15), and the fitted values $\hat{y}$ of X lie in $\mathcal{C}(X) \subseteq \mathcal{C}(X_+)$. Hence $\text{SSE}_+ = \|y - \hat{y}_+\|^2 \leq \|y - \hat{y}\|^2 = \text{SSE}$.
- SST does not depend on the design, so by Theorem 6.33, $R_+^2 = 1 - \text{SSE}_+/\text{SST} \geq 1 - \text{SSE}/\text{SST} = R^2$.

Exercise 6.11. Wages y_i of n workers are regressed on years of schooling x_i , an intercept, and the indicators $d_1, d_2, d_3, d_4 \in \mathbb{R}^n$ of all four regions of a country (each worker lives in exactly one region): $X = [\mathbf{1}_n \ x \ d_1 \ d_2 \ d_3 \ d_4] \in \mathbb{R}^{n \times 6}$. (a) Show that X does not have full column rank, so that $\hat{\beta} = (X'X)^{-1}X'y$ is not defined. (b) Let $X_2 = [\mathbf{1}_n \ x \ d_2 \ d_3 \ d_4]$ (indicator d_1 dropped). Show that $\mathcal{C}(X_2) = \mathcal{C}(X)$, so that $\text{rank } X_2 = \text{rank } X$.

Solution. (a) Each worker lives in exactly one region, so $d_1 + d_2 + d_3 + d_4 = \mathbf{1}_n$, that is, $X(-1, 0, 1, 1, 1, 1)' = 0$. So X does not have full column rank (Theorem 4.13(b)), and $X'X$ is singular (Theorem 6.21(b)): the dummy variable trap.

(b) Every column of X_2 is a column of X , so $\mathcal{C}(X_2) \subseteq \mathcal{C}(X)$. Conversely $d_1 = \mathbf{1}_n - d_2 - d_3 - d_4 \in \mathcal{C}(X_2)$, so every column of X lies in the subspace $\mathcal{C}(X_2)$, and therefore $\mathcal{C}(X) \subseteq \mathcal{C}(X_2)$ (Theorem 3.7). Hence $\mathcal{C}(X_2) = \mathcal{C}(X)$, and $\text{rank } X_2 = \text{rank } X$; by Proposition 6.27 (with $X_2 = XT$, T the columns 1, 2, 4, 5, 6 of I_6), the two designs give the same fitted values.

Exercise 6.12. Six plants of three varieties are grown under two fertilizer treatments, one plant for each combination. The model $y_{ij} = \mu + \alpha_i + \gamma_j + \varepsilon_{ij}$ (variety i , treatment j) has $\beta = (\mu, \alpha_1, \alpha_2, \alpha_3, \gamma_1, \gamma_2)'$ and, with the observations ordered $y_{11}, y_{12}, y_{21}, y_{22}, y_{31}, y_{32}$,

$$X = \begin{pmatrix} 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 \end{pmatrix}$$

Show that $\text{rank } X = 4$, and find a basis of $\mathcal{N}(X)$.

Solution. Let $c_1, \dots, c_6$ be the columns of X .

- Each plant has one variety and one treatment, so $c_2 + c_3 + c_4 = c_1$ and $c_5 + c_6 = c_1$; that is, $z_1 = (-1, 1, 1, 1, 0, 0)'$ and $z_2 = (-1, 0, 0, 0, 1, 1)'$ lie in $\mathcal{N}(X)$.
- c_2, c_3, c_4, c_5 are independent: if $a_2c_2 + a_3c_3 + a_4c_4 + a_5c_5 = 0$, rows 2, 4, 6 give $a_2 = a_3 = a_4 = 0$, and then row 1 gives $a_5 = 0$.
- $c_1 = c_2 + c_3 + c_4$ and $c_6 = c_1 - c_5$ are combinations of c_2, c_3, c_4, c_5 , so these four columns span $\mathcal{C}(X)$ and form a basis of it: $\text{rank } X = 4$.
- $\dim \mathcal{N}(X) = 6 - 4 = 2$ (Theorem 4.12), and z_1, z_2 are independent (entries 2 and 5 of $t_1z_1 + t_2z_2$ are t_1 and t_2); so they form a basis of $\mathcal{N}(X)$ (Theorem 4.5(d)).